COURSE SLIDES · WSQ

AI Vibe Coding for iOS Mobile Apps Development

Portable C++ core · Objective-C++ adapter · SwiftUI shell

TGS-2023037545 · Software Design 3 · ICT-DES-3005-1.1
Tertiary Infotech Academy Pte Ltd · UEN 201200696W
Trainer: Dr Alfred Ang



v1.2 · 12 September 2026

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MANDATORY · SSG DIGITAL ATTENDANCE

Digital Attendance · TRAQOM

1 01

Open the SSG TRAQOM attendance request shown by the trainer.

TRAQOM

2 02

Verify course, session and identity details before confirming.

TRAQOM

3 03

Complete attendance at the required start and end checkpoints.

TRAQOM

4 04

Tell the trainer immediately if the status is not confirmed.

TRAQOM

DESIGN JUDGEMENT

TRAQOM records SSG digital attendance. It is not a course survey.

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About the Trainer



General Trainer

Template profile

NAME _____

TITLE / DESIGNATION _____

QUALIFICATIONS _____

AREAS OF EXPERTISE _____

TRAINING EXPERIENCE _____

CONTACT _____

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About the Trainer

AA

Dr Alfred Ang

Lead Trainer · AI and Software Engineering

ROLE	Founder and principal trainer
CERTIFICATION	Adult educator and technology specialist
DELIVERS	AI, mobile, cloud and software design programmes
INDUSTRY FOCUS	Applied architecture and product delivery
TRAINING APPROACH	Mechanism-led, evidence-backed practice
CONTACT	enquiry@tertiaryinfotech.com

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COURSE ADMINISTRATION

Learner Introduction

1 Experience

Share your Swift, C++ and iOS development background.

LEARNER

2 Outcome

Name one iOS capability you want to design and validate.

LEARNER

3 Evidence

Describe how you currently decide that software works.

LEARNER

4 Boundary

Identify one area where AI assistance needs review.

LEARNER

DESIGN JUDGEMENT

Use the introductions to calibrate examples without changing the regulated learning outcomes.

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GROUND RULES

Working Agreements

1 Safe

Use mock data; never paste credentials or personal information.

GOVERNANCE

2 Scoped

Protect unrelated files and approved requirements.

GOVERNANCE

3 Observable

Run checks and retain evidence for material claims.

GOVERNANCE

4 Accountable

Human review owns consequential design and release decisions.

GOVERNANCE

DESIGN JUDGEMENT

Ask early when a requirement, permission or acceptance criterion is unclear.

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THE THROUGH-LINE

Course Architecture

1 SwiftUI shell

Apple-native controls, navigation, accessibility and state.

ARCH-CLEAN

2 Objective-C++ adapter

The single translation seam for types, ownership and errors.

ARCH-CLEAN

3 Portable C++ core

Business invariants, exact values, ports and domain tests.

ARCH-CLEAN

4 Evidence pack

Traceability, tests, ADRs, privacy and deployment checklist.

ARCH-CLEAN

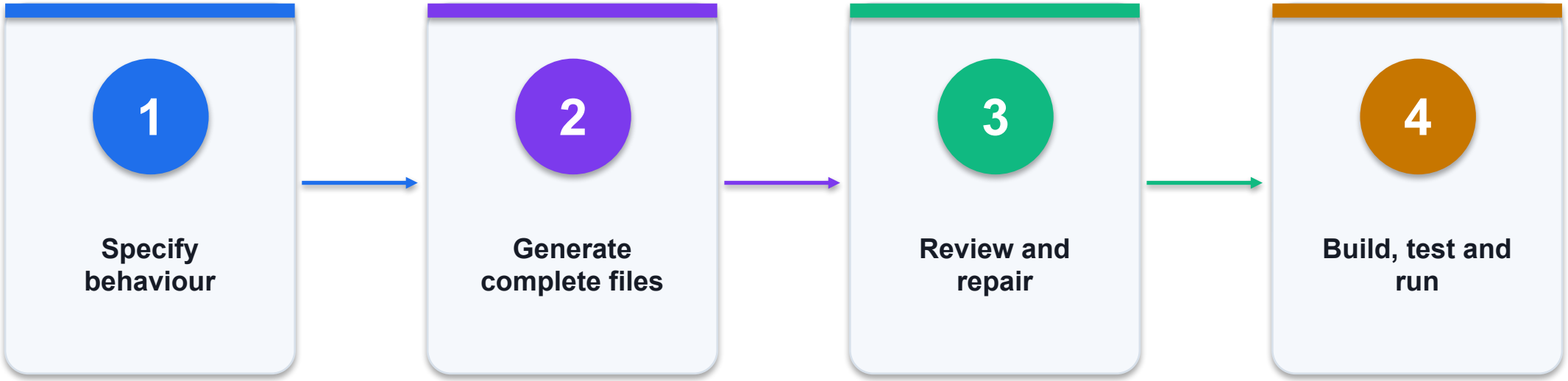
DESIGN JUDGEMENT

Swift supplements the regulated C++ scope; it does not replace it.

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THE COURSE CONTRACT

You Specify · AI Generates · Xcode Proves



MECHANISM Swift appears as review vocabulary for generated iOS code. Learners own the requirement, protected interfaces, diagnostic and acceptance evidence; Xcode and the Simulator decide whether the proposal works.

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XCODEGEN · XCODE 26 · IOS 17+

Three Runnable Apps · One Evidence-Driven Journey



BudgetBuddy

Regulated portable C++ policy, Objective-C++ seam, Observation state and accessible SwiftUI.

THREE-APP-JOURNEY



FocusCards

Modern SwiftUI flashcard interaction inspired by the legacy course project, generated and tested as a complete app.

THREE-APP-JOURNEY



HabitPulse

Stable-identity habit actions that make red → repair → green practice visible in the Simulator.

THREE-APP-JOURNEY

4

Release gate

Every app regenerates from project.yml, builds, tests and launches before any credentialled App Store gate.

THREE-APP-JOURNEY

DESIGN JUDGEMENT

Legacy UIKit examples contribute app concepts only. The learner package uses current SwiftUI, Observation, Swift Testing and XCTest UI automation.

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Vibe Coding Loop · From Context to Evidence

1 BASELINE

Capture the current build or failing test before generation.

2 PROMPT

Name the exact files, protected interfaces and observable finish condition.

3 REVIEW

Inspect complete-file output and the diff for boundary or contract drift.

4 REPAIR

Feed back the first causal diagnostic. Request the smallest patch.

5 GREEN

Run focused, layer and cross-layer checks. Retain the evidence.

Prompt output contract: FILE: relative/path, then complete code. No ellipses, TODOs, placeholders or unverified run claims.

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COPY / ADAPT / VERIFY

Prompt Anatomy · A Bounded Code-Generation Contract

1 Context

BudgetBuddy slice, current files and first failing evidence.

AI-PROMPT-CONTRACT

2 Contract

Exact paths, complete files, no omitted imports or placeholder code.

AI-PROMPT-CONTRACT

3 Guardrails

C++ stays portable; Objective-C++ owns translation; Swift uses @Observable.

AI-PROMPT-CONTRACT

4 Proof

CMake/CTest, Swift Testing, XCTest/XCUIAutomation and retained observations.

AI-PROMPT-CONTRACT

DESIGN JUDGEMENT

A useful prompt constrains the change and defines evidence. The assistant proposes code; the learner verifies behaviour and owns acceptance.

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TWO-DAY ROADMAP

Course Outline

1 Topic 1

Requirements and the Portable Component Boundary

COURSE-MAP

2 Topic 2

Vibe-Generated Portable Logic and Quality Gates

COURSE-MAP

3 Topic 3

AI-Generated SwiftUI App Shell and Controls

COURSE-MAP

4 Topic 4

iOS Interoperability, Persistence and Validation

COURSE-MAP

DESIGN JUDGEMENT

Topic 5 completes the evidence and deployment thread; ten Activities progressively build BudgetBuddy.

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8 CONTACT HOURS · LUNCH 13:00–14:00

Lesson Plan · Day 1

1 **09:30–11:00**

Requirements, component boundaries and Activity 1.

SCHEDULE

2 **11:00–13:00**

Portable C++ design and Activity 2.

SCHEDULE

3 **14:00–16:00**

C++ methods, tools and AI-assisted TDD.

SCHEDULE

4 **16:00–18:30**

Activities 3–4, reflection and evidence review.

SCHEDULE

DESIGN JUDGEMENT

Detailed timings, methods and slide references are in the Lesson Plan.

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8 CONTACT HOURS · LUNCH 13:00–14:00

Lesson Plan · Day 2

1

09:30–11:15

SwiftUI controls and Objective-C++ interoperability.

SCHEDULE

2

11:15–13:00

Activities 5–6 and state validation.

SCHEDULE

3

14:00–16:30

Activities 7–10, documentation and release evidence.

SCHEDULE

4

16:30–18:30

Written Assessment 1 h · Practical Performance 1 h.

SCHEDULE

DESIGN JUDGEMENT

Both assessments retain the approved one-hour duration.

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Regulated Learning Outcomes

1 LO1

Determine basic software components using C++ methodologies to meet functional specifications.

LO1-LO4

2 LO2

Apply C++ methodologies and tools for software creation.

LO1-LO4

3 LO3

Select essential C++ controls and features to meet software design requirements.

LO1-LO4

4 LO4

Examine the interoperability and functionality of C++ software components.

LO1-LO4

DESIGN JUDGEMENT

LO5: Generate C++ design documentation aligned with user specifications.

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TSC TRACEABILITY

Knowledge and Ability Pairing

1

K1 / A1

Design requirements for simple, basic software components. Apply through the corresponding Activity evidence.

K/A-MAP

2

K2 / A2

Basic software design tools and techniques. Apply through the corresponding Activity evidence.

K/A-MAP

3

K3 / A3

Types of controls, elements and features in software. Apply through the corresponding Activity evidence.

K/A-MAP

4

K4 / A4

Indicators of software functionality and interoperability. Apply through the corresponding Activity evidence.

K/A-MAP

DESIGN JUDGEMENT

K5 / A5: Documentation of design details. Produce detailed design documentation mapped to user specifications.

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READ BEFORE ASSESSMENT

Briefing for Assessment

1 Written Assessment

Five open-ended questions · K1–K5 · 1 hour.

ASSESSMENT

2 Practical Performance

Three coherent tasks · A1–A5 · 1 hour.

ASSESSMENT

3 Submission

Complete the provided document and upload it to the LMS.

ASSESSMENT

4 Decision

Competent or Not Yet Competent, based on stated criteria.

ASSESSMENT

DESIGN JUDGEMENT

Candidate papers contain instructions and grading on page 2; questions and scenario begin on page 3.

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COURSE ADMINISTRATION

Assessment and Funding

1 Attendance

Meet the required attendance threshold and complete TRAQOM.

ASSESSMENT

2 Evidence

Submit both assessment instruments through the LMS.

ASSESSMENT

3 Authenticity

Complete individual work and explain your implementation.

ASSESSMENT

4 Support

Ask the assessor to clarify instructions, not supply answers.

ASSESSMENT

DESIGN JUDGEMENT

Assessment answer keys are trainer-only and never linked from learner channels.

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Access the Course Activities

1 Clone

Use Git when you want version history and can run a terminal command.

ACTIVITY-ACCESS

2 Download ZIP

Use Code → Download ZIP when Git is not available on the learner device.

ACTIVITY-ACCESS

3 Verify identity

Confirm the repository name and course code TGS-2023037545 before using files.

ACTIVITY-ACCESS

4 Use the guides

Open activities/README.md, then enter the matching numbered Activity folder.

ACTIVITY-ACCESS

DESIGN JUDGEMENT

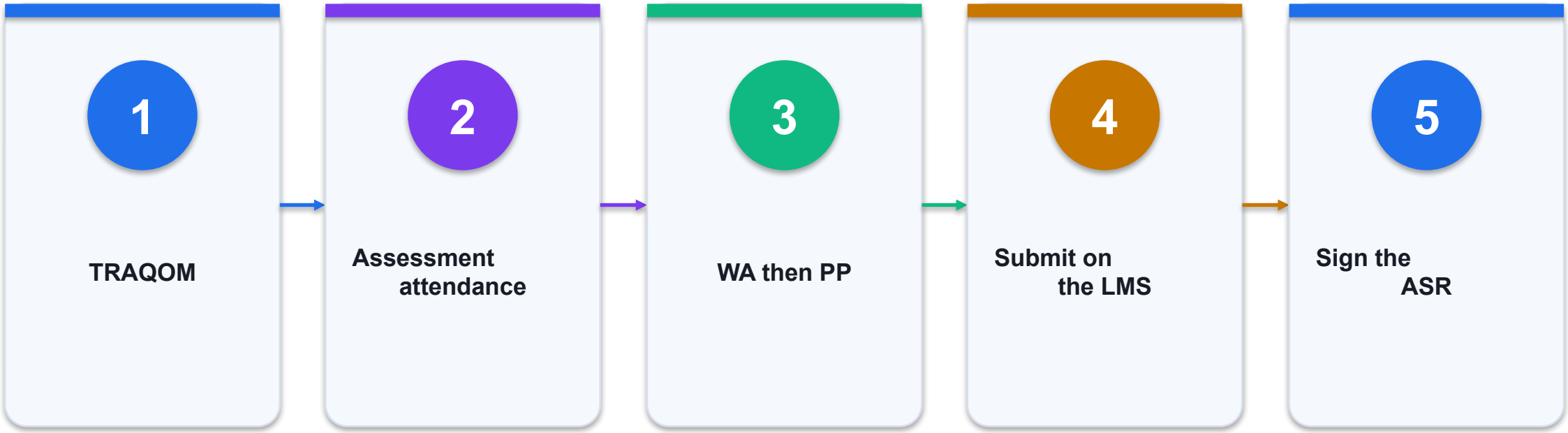
Clone: git clone https://github.com/tertiarycourses/TGS-2023037545-AI-Vibe-Coding-for-iOS-Mobile-Apps-Development.git

https://github.com/tertiarycourses/TGS-2023037545-AI-Vibe-Coding-for-iOS-Mobile-Apps-Development

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ASSESSMENT ADMINISTRATION

Assessment Flow



ASR = Assessment Summary Record · complete the sign-off only after both instruments are submitted.

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01

TOPIC 1 · LO1 · K1 · A1

Requirements and the Portable Component Boundary

LO1: Determine basic software components using C++ methodologies to meet functional specifications.

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Functional requirements become contracts

1 Elicit need

actor

TSC-K1

2 State invariant

input

TSC-K1

3 Define interface

rule

TSC-K1

4 Write acceptance

output

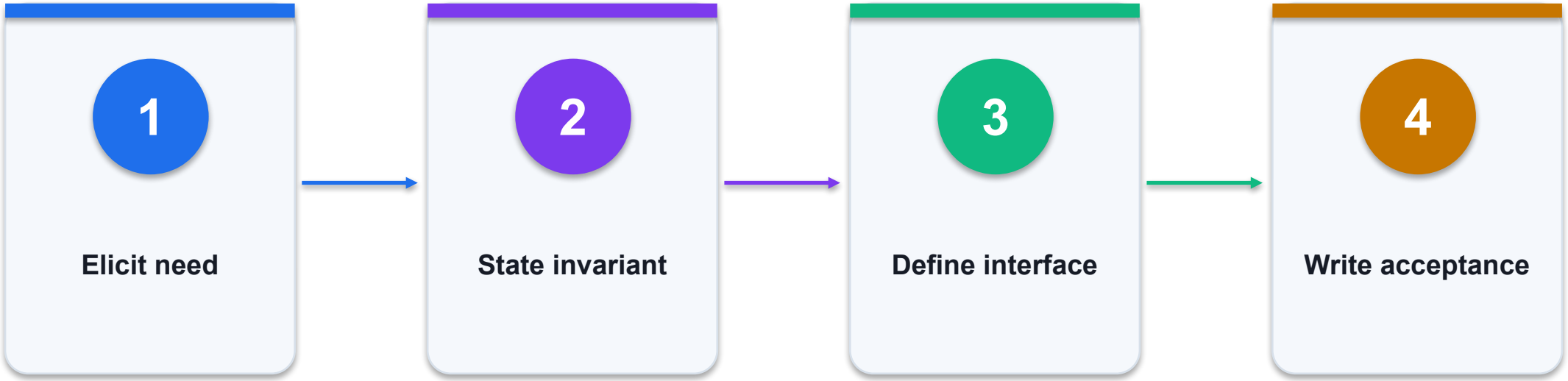
TSC-K1

DESIGN JUDGEMENT

Translate each user story into an invariant and testable acceptance criterion.

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Functional requirements become contracts · Mechanism



MECHANISM

Failure mode: A prompt says build an app but never defines observable success. Control: Translate each user story into an invariant and testable acceptance criterion.

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WORKED EXAMPLE

Functional requirements become contracts · Technical Anchor

```
struct Requirement { std::string id; bool testable; };
```

1. actor
2. input
3. rule
4. output

EVIDENCE METRIC Accepted behaviours / specified behaviours

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Separate business rules from the iOS shell

1 **SwiftUI event**

View

CPP-BOUNDARY

2 **Adapter call**

Bridge

CPP-BOUNDARY

3 **C++ rule**

Domain

CPP-BOUNDARY

4 **Value result**

Result

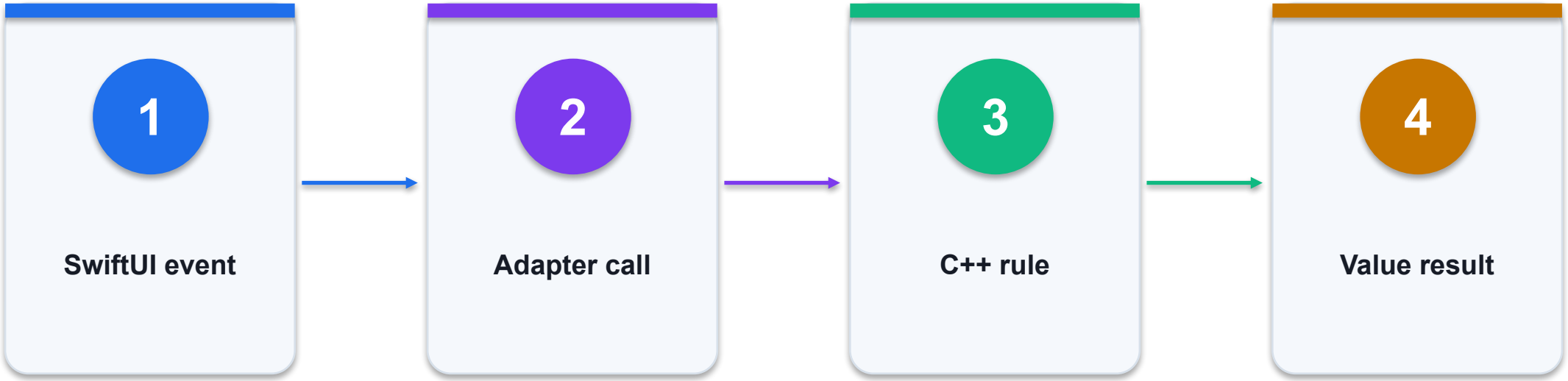
CPP-BOUNDARY

DESIGN JUDGEMENT

Keep portable policy in C++; keep presentation and Apple frameworks in Swift.

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Separate business rules from the iOS shell · Mechanism



MECHANISM

Failure mode: Rules leak into View bodies and cannot be reused or isolated. Control: Keep portable policy in C++; keep presentation and Apple frameworks in Swift.

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WORKED EXAMPLE

Separate business rules from the iOS shell · Technical Anchor

```
class BudgetPolicy { public: Decision evaluate(const
Input&) const; };
```

1. View
2. Bridge
3. Domain
4. Result

EVIDENCE METRIC Domain tests runnable without Simulator

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Define a narrow C ABI seam

1 Choose values

int64_t

OBJCXX-ABI

2 Stabilise types

const char*

OBJCXX-ABI

3 Expose function

opaque handle

OBJCXX-ABI

4 Wrap in Swift

status code

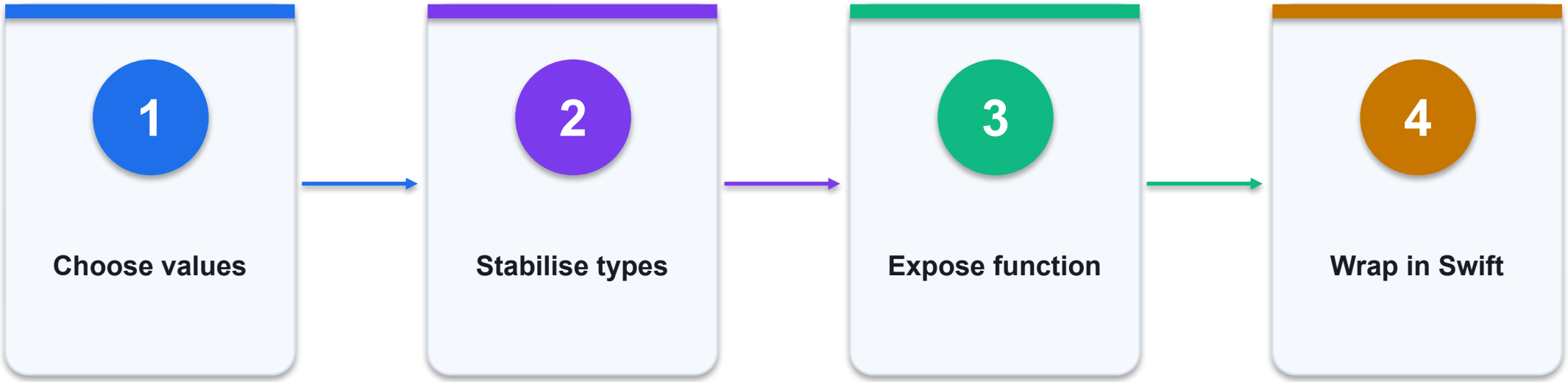
OBJCXX-ABI

DESIGN JUDGEMENT

Expose C-compatible data at the boundary; hide implementation behind the adapter.

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Define a narrow C ABI seam · Mechanism



MECHANISM

Failure mode: C++ templates or STL containers cross directly into Swift. Control: Expose C-compatible data at the boundary; hide implementation behind the adapter.

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WORKED EXAMPLE

Define a narrow C ABI seam · Technical Anchor

```
extern "C" BudgetResult budget_evaluate(BudgetInput
input);
```

1. int64_t
2. const char*
3. opaque handle
4. status code

EVIDENCE METRIC Boundary surface measured in stable operations

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Model money without floating-point drift

1 **Accept decimal**

minor units

CPP-VALUE

2 **Convert to cents**

currency

CPP-VALUE

3 **Compute in integer**

rounding

CPP-VALUE

4 **Format for UI**

display

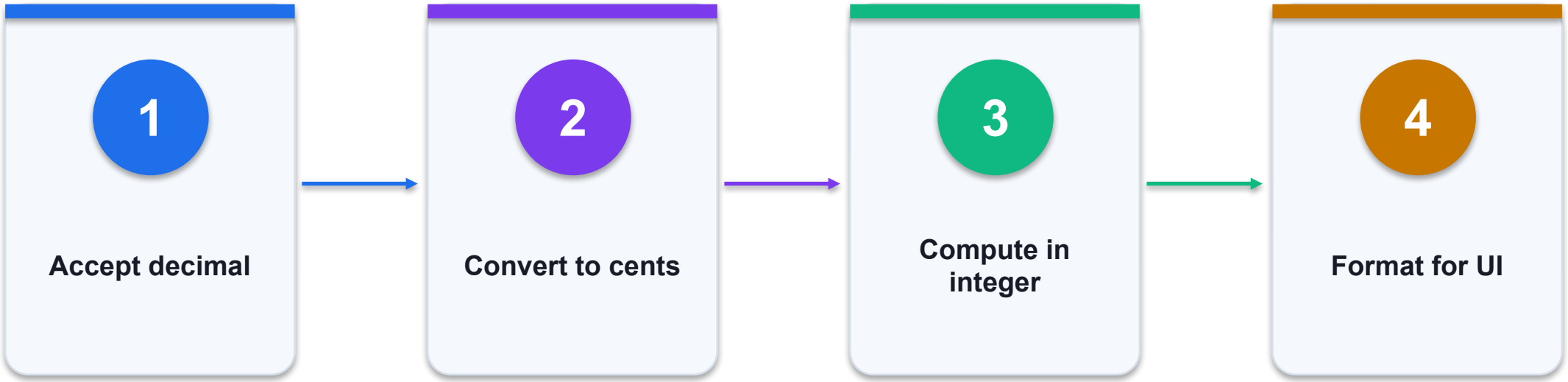
CPP-VALUE

DESIGN JUDGEMENT

Represent monetary amounts as signed 64-bit minor units in the domain.

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Model money without floating-point drift · Mechanism



MECHANISM Failure mode: Binary floating point creates a one-cent mismatch. Control: Represent monetary amounts as signed 64-bit minor units in the domain.

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WORKED EXAMPLE

Model money without floating-point drift · Technical Anchor

```
struct Money { std::int64_t cents; };
```

1. minor units
2. currency
3. rounding
4. display

EVIDENCE METRIC Exact equality for ledger totals

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Design immutable value objects

1 **Validate input**

invariant

CPP-OOP

2 **Construct value**

constructor

CPP-OOP

3 **Pass by const ref**

const

CPP-OOP

4 **Return new value**

equality

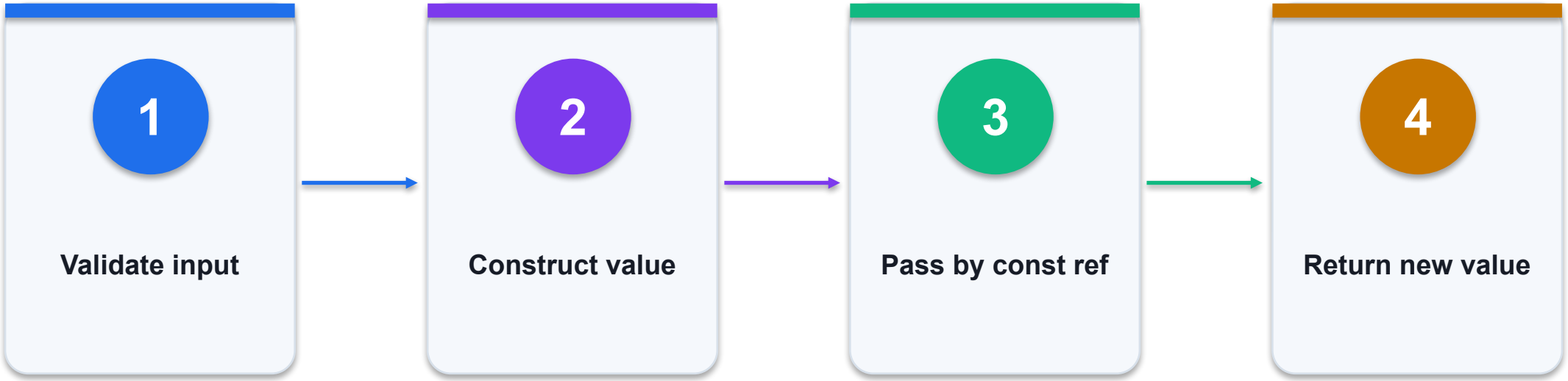
CPP-OOP

DESIGN JUDGEMENT

Make invariants constructor-level and expose values through read-only accessors.

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Design immutable value objects · Mechanism



MECHANISM

Failure mode: Partially initialised objects travel through the system. Control: Make invariants constructor-level and expose values through read-only accessors.

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WORKED EXAMPLE

Design immutable value objects · Technical Anchor

```
class Category { std::string value_; public: explicit
Category(std::string); };
```

1. **invariant**
2. **constructor**
3. **const**
4. **equality**

EVIDENCE METRIC Invalid states rejected at construction

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Map a user story to an interface

1 **Read story**

story

TSC-A1

2 **Find verb**

command

TSC-A1

3 **Name operation**

query

TSC-A1

4 **Specify result**

error

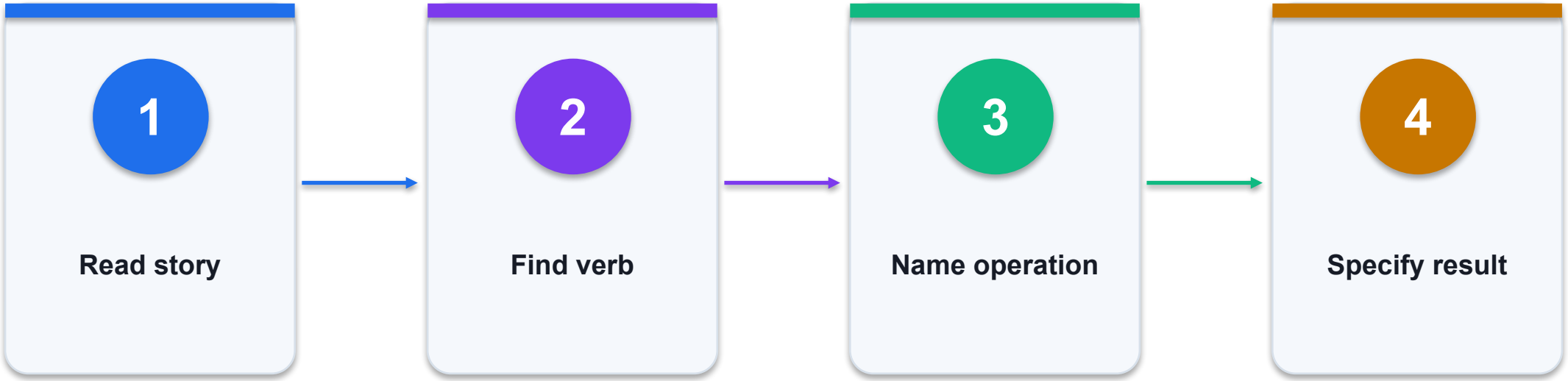
TSC-A1

DESIGN JUDGEMENT

Name operations after domain decisions, then let SwiftUI adapt them.

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Map a user story to an interface · Mechanism



MECHANISM **Failure mode: The API mirrors screen widgets instead of business intent. Control: Name operations after domain decisions, then let SwiftUI adapt them.**

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WORKED EXAMPLE

Map a user story to an interface · Technical Anchor

```
virtual AddResult addExpense(const ExpenseDraft&) = 0;
```

1. story
2. command
3. query
4. error

EVIDENCE METRIC Traceable story-to-operation mapping

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Use acceptance criteria as design pressure

1 **Given state**

precondition

REQ-TRACE

2 **When action**

trigger

REQ-TRACE

3 **Then outcome**

postcondition

REQ-TRACE

4 **And evidence**

artifact

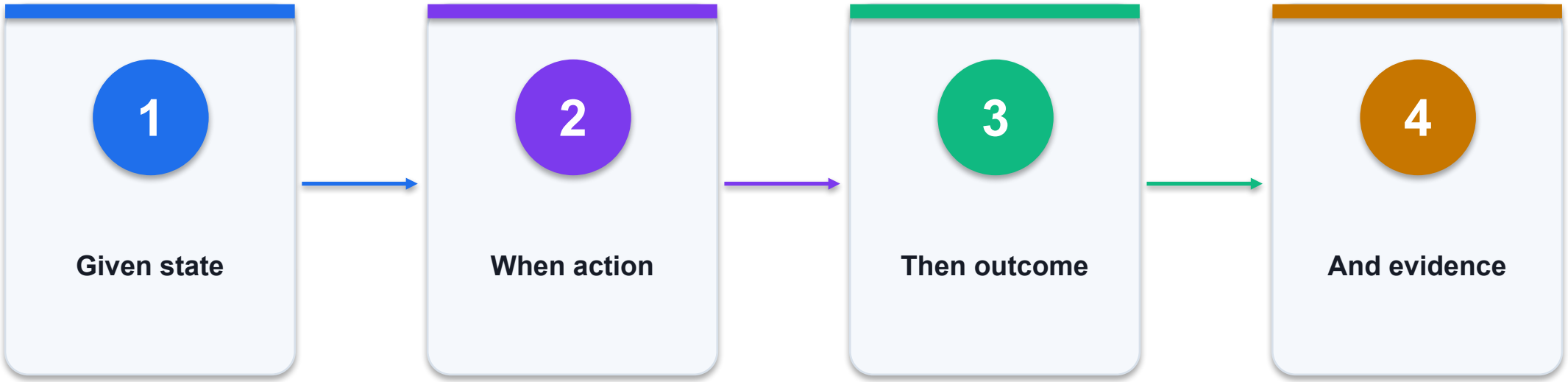
REQ-TRACE

DESIGN JUDGEMENT

Write externally observable outcomes before selecting controls or tools.

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Use acceptance criteria as design pressure · Mechanism



MECHANISM

Failure mode: Acceptance criteria describe implementation steps rather than outcomes. Control: Write externally observable outcomes before selecting controls or tools.

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WORKED EXAMPLE

Use acceptance criteria as design pressure · Technical Anchor

```
// Given valid draft, when added, then total increases
by exact cents.
```

1. precondition
2. trigger
3. postcondition
4. artifact

EVIDENCE METRIC Criteria with deterministic evidence

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Choose component granularity

1 Find responsibility

responsibility

ARCH-CLEAN

2 Measure cohesion

cohesion

ARCH-CLEAN

3 Inspect coupling

dependency

ARCH-CLEAN

4 Cut boundary

contract

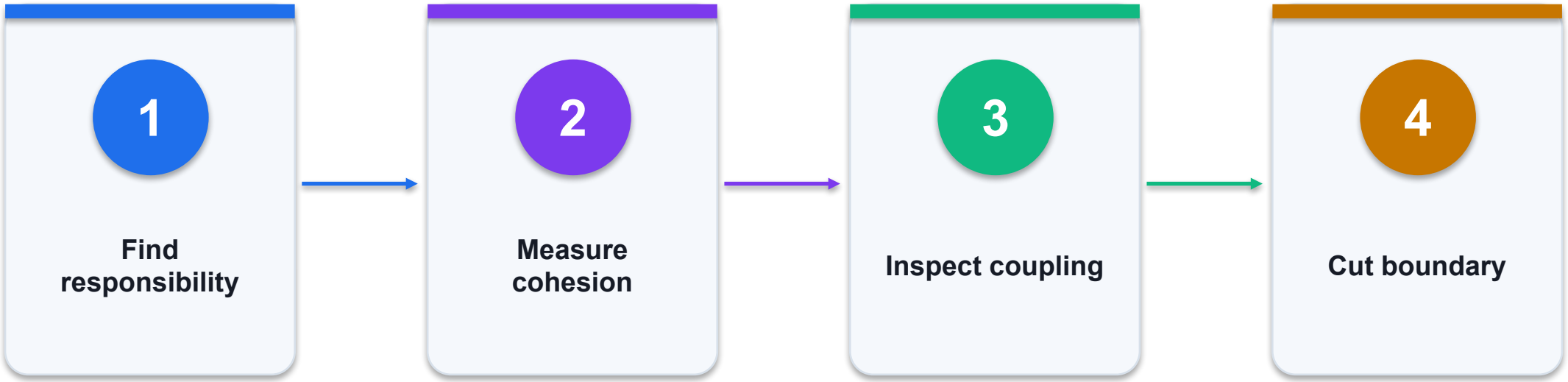
ARCH-CLEAN

DESIGN JUDGEMENT

Partition by responsibility and dependency direction, not by file-count targets.

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Choose component granularity · Mechanism



MECHANISM

Failure mode: A single Manager object owns UI, persistence, validation and policy. Control: Partition by responsibility and dependency direction, not by file-count targets.

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WORKED EXAMPLE

Choose component granularity · Technical Anchor

```
ExpenseService -> ExpenseRepository; ViewModel ->
ExpenseService;
```

1. responsibility
2. cohesion
3. dependency
4. contract

EVIDENCE METRIC Reasons-to-change per component

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Express failure in the contract

1 **Enumerate faults**

validation

CPP-ERROR

2 **Classify recoverability**

storage

CPP-ERROR

3 **Create result type**

bridge

CPP-ERROR

4 **Map to message**

unexpected

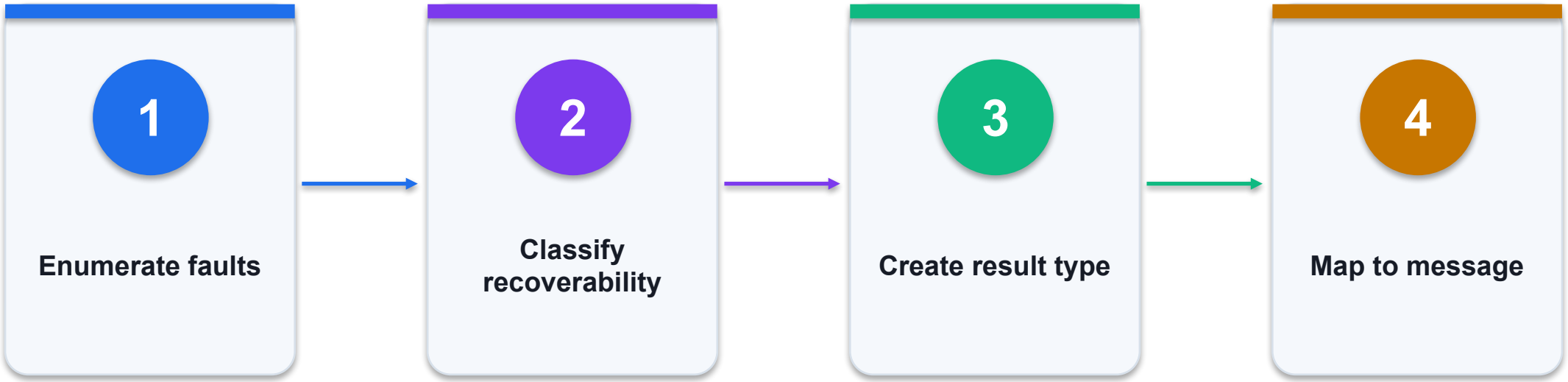
CPP-ERROR

DESIGN JUDGEMENT

Use typed errors in C++; translate them once at the Objective-C++ edge.

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Express failure in the contract · Mechanism



MECHANISM

Failure mode: A Boolean failure discards the cause and repair path. Control: Use typed errors in C++; translate them once at the Objective-C++ edge.

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WORKED EXAMPLE

Express failure in the contract · Technical Anchor

```
using AddResult = std::variant<ExpenseId, DomainError>;
```

1. validation
2. storage
3. bridge
4. unexpected

EVIDENCE METRIC — Failure variants handled explicitly

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Estimate before implementation

1 Slice capability

must

DESIGN-TOOL

2 Expose unknowns

should

DESIGN-TOOL

3 Prototype seam

risk

DESIGN-TOOL

4 Commit scope

spike

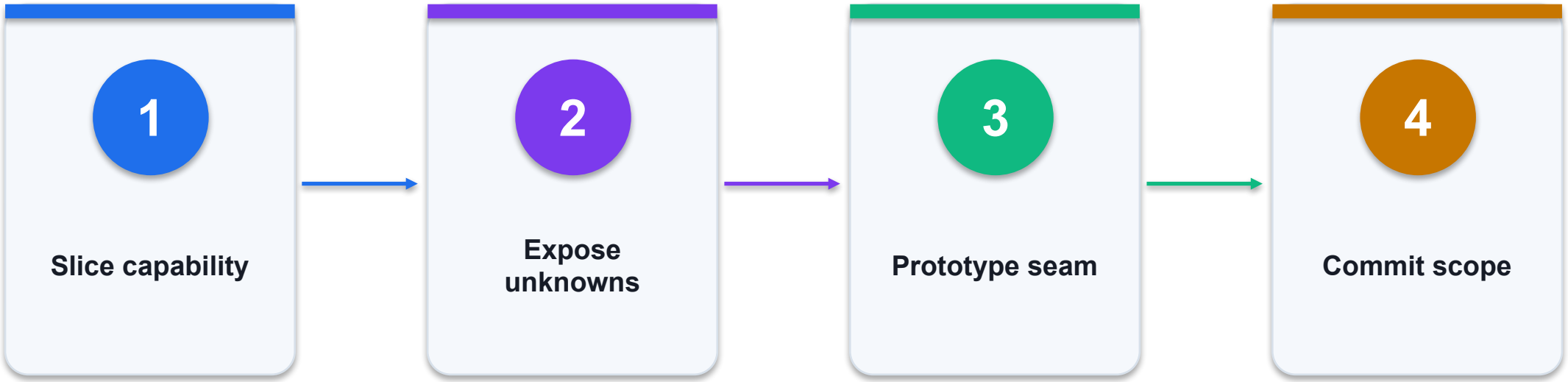
DESIGN-TOOL

DESIGN JUDGEMENT

Time-box an Objective-C++ seam spike before committing the architecture.

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Estimate before implementation · Mechanism



MECHANISM

Failure mode: The hardest bridge uncertainty appears late in the capstone. Control: Time-box an Objective-C++ seam spike before committing the architecture.

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WORKED EXAMPLE

Estimate before implementation · Technical Anchor

```
// Spike: Swift -> ObjC++ -> C++ -> immutable result
```

1. must
2. should
3. risk
4. spike

EVIDENCE METRIC Unknowns retired before full build

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Requirements baseline and change control

1 **Version baseline**

requirement id

GOV-AI

2 **Record decision**

version

GOV-AI

3 **Link change**

rationale

GOV-AI

4 **Re-run impact**

affected tests

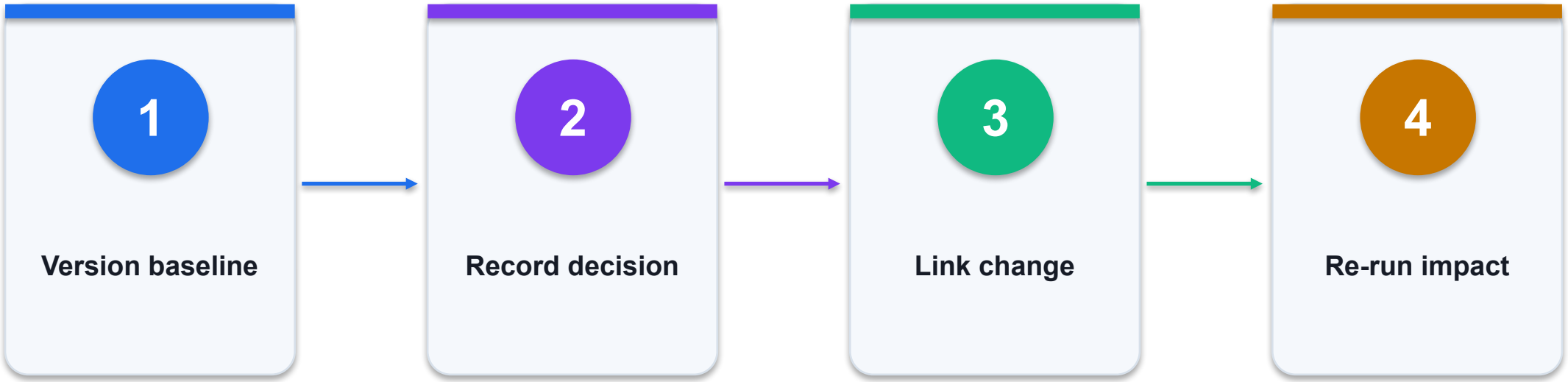
GOV-AI

DESIGN JUDGEMENT

Treat AI output as a proposal against a versioned baseline.

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Requirements baseline and change control · Mechanism



MECHANISM **Failure mode: A model-generated change silently alters an approved requirement. Control: Treat AI output as a proposal against a versioned baseline.**

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WORKED EXAMPLE

Requirements baseline and change control · Technical Anchor

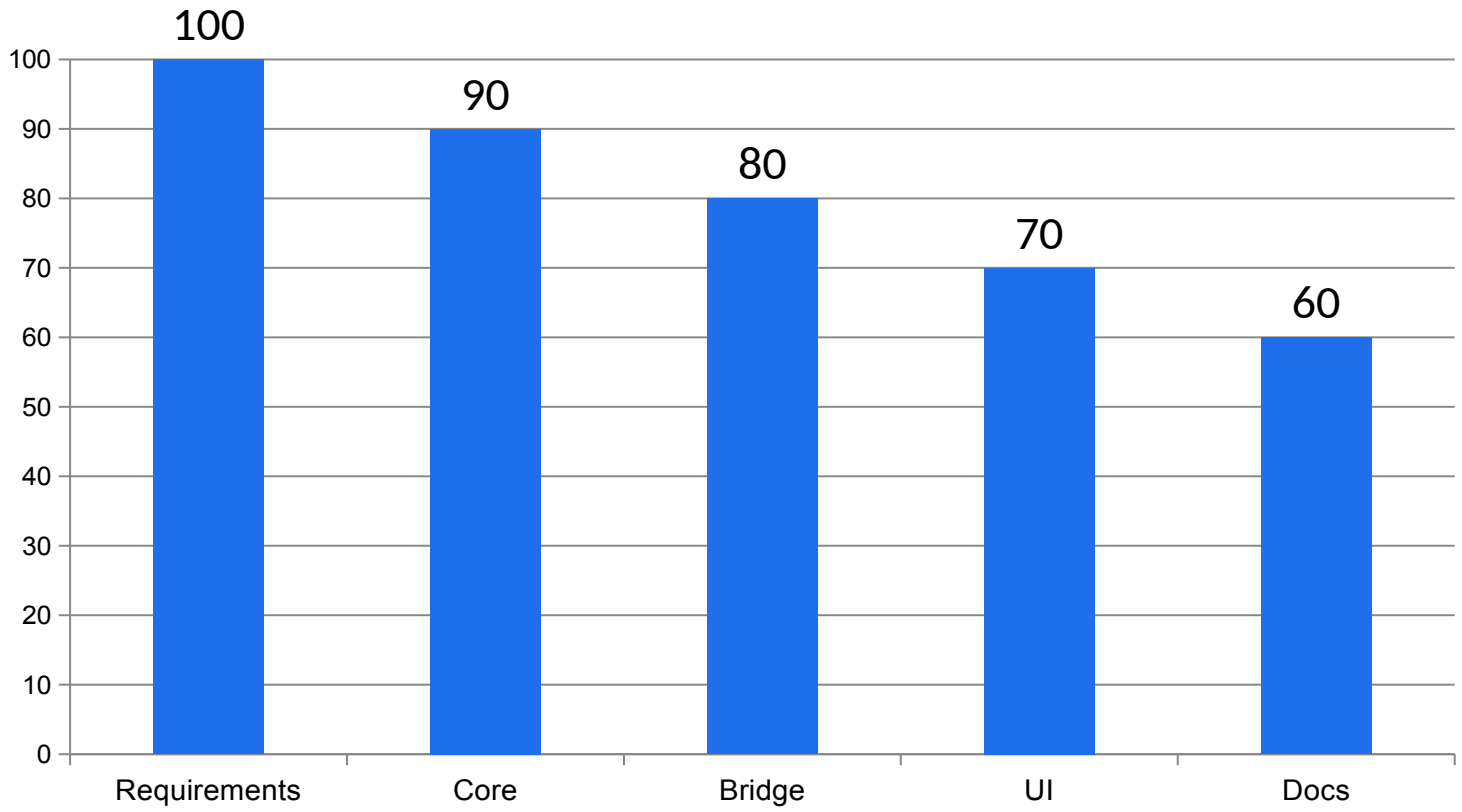
```
struct Change { std::string requirementId; std::string
rationale; };
```

1. requirement id
2. version
3. rationale
4. affected tests

EVIDENCE METRIC Changes with explicit downstream impact

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Topic 1 · Evidence Coverage by Layer



WHAT THE DATA SHOWS

Coverage should form a connected chain. A polished UI cannot compensate for untested invariants or an undocumented seam.

Editable chart · values are instructional thresholds, not production benchmarks

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HANDS-ON ACTIVITY · 70 MIN

Activity 1: Requirements and Component Boundary

CHALLENGE

Convert stakeholder needs into testable requirements and design the SwiftUI–Objective-C++–C++ boundary.

REGULATED MAPPING

K1, A1

YOU WILL PRODUCE

Requirement matrix, boundary diagram and acceptance criteria

Markdown, diagrams.net or PowerPoint, JSON fixture

Detailed procedures: Learner Guide and activities/01-requirements-component-boundary/README.md

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ACTIVITY MECHANISM

Activity 1 · Delivery Workflow

1 Frame

Inspect the BudgetBuddy scenario and identify actors, inputs, rules and outcomes.

K1, A1

2 Build

Write five uniquely identified functional requirements and three non-functional constraints.

K1, A1

3 Connect

Draw the SwiftUI shell, Objective-C++ adapter and portable C++ domain core as separate responsibilities.

K1, A1

4 Verify

Every requirement links to one interface operation and one observable acceptance criterion.

K1, A1

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 1 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
BudgetBuddy requirements only. Generate REQ-001..REQ-005 with actor,
input, integer-cent invariant, observable result, negative case and
evidence. Then create the inward dependency boundary from SwiftUI
through BudgetViewModel, BudgetEvaluating, BudgetBridge.h/.mm and
portable C++. Output each exact path plus the complete file. Do not
generate UI code. Verify all IDs, cent units and boundary layers with
rg.
```

PROTECTED FILE SCOPE

requirements-working.txt; boundary-working.txt; acceptance-review.txt

EXPECTED GREEN

`rg -n 'REQ-00[1-5]|SwiftUI|BudgetBridge|C\+\+|cents' requirements-working.txt boundary-working.txt` shows all five requirements and each boundary layer.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 1 · Evidence Contract

1 Artifact

Requirement matrix, boundary diagram and acceptance criteria

K1, A1

2 Acceptance

Every requirement links to one interface operation and one observable acceptance criterion.

K1, A1

3 Failure signal

If the boundary follows screen widgets, restate each operation as a business verb before redrawing.

K1, A1

4 Cleanup

Retain the versioned design artifacts; remove only disposable sketches.

K1, A1

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 1 · Verification Gate

PASS WHEN

Every requirement links to one interface operation and one observable acceptance criterion.

IF IT FAILS

If the boundary follows screen widgets, restate each operation as a business verb before redrawing.

LEAVE THE WORKSPACE

Retain the versioned design artifacts; remove only disposable sketches.

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HANDS-ON ACTIVITY · 85 MIN

Activity 2: Portable C++ Domain Core

CHALLENGE

Implement exact-cent value objects and a small budget policy as a standalone CMake library.

REGULATED MAPPING

K1, A1

YOU WILL PRODUCE

Compiling C++20 library and domain tests

C++20, CMake, CTest

Detailed procedures: Learner Guide and activities/02-portable-cpp-domain-core/README.md

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ACTIVITY MECHANISM

Activity 2 · Delivery Workflow

1 Frame

Configure the starter with CMake in an out-of-tree build folder.

K1, A1

2 Build

Implement Money and ExpenseDraft invariants using integer minor units.

K1, A1

3 Connect

Add tests for valid, zero, negative and over-budget cases.

K1, A1

4 Verify

CTest reports all domain tests passed and the public header contains no Objective-C or Swift constructs.

K1, A1

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 2 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate complete C++20 Money, ExpenseDraft, Decision, DomainError and
BudgetPolicy files plus CMake and tests. Preserve std::int64_t cents and
the approved evaluate signature. Reject invalid, overflow and overspend
cases without Apple imports. Output exact path then complete code, with
no TODOs. Verify a clean configure/build/CTest run and an empty Apple-
framework scan.
```

PROTECTED FILE SCOPE

**workspace/CMakeLists.txt;
workspace/include/BudgetPolicy.hpp;
workspace/src/BudgetPolicy.cpp;
workspace/tests/BudgetPolicyTests.cpp**

EXPECTED GREEN

`cmake -S workspace -B workspace/build && cmake --build workspace/build && ctest --test-dir workspace/build --output-on-failure`; all tests pass and Apple import scan is empty.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 2 · Evidence Contract

1 Artifact

Compiling C++20 library and domain tests

K1, A1

2 Acceptance

CTest reports all domain tests passed and the public header contains no Objective-C or Swift constructs.

K1, A1

3 Failure signal

Delete only the local build directory and reconfigure; inspect the first compiler diagnostic, not the last.

K1, A1

4 Cleanup

Remove build/; keep source, headers and tests.

K1, A1

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 2 · Verification Gate

PASS WHEN

CTest reports all domain tests passed and the public header contains no Objective-C or Swift constructs.

IF IT FAILS

Delete only the local build directory and reconfigure; inspect the first compiler diagnostic, not the last.

LEAVE THE WORKSPACE

Remove build/; keep source, headers and tests.

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RETRIEVAL PRACTICE

Topic 1 · Review

1 Explain

LO1: Determine basic software components using C++ methodologies to meet functional specifications.

LO1 · K1 · A1

2 Demonstrate

Show the Activity evidence that operationalises the outcome.

LO1 · K1 · A1

3 Diagnose

Name the failure signal and the layer where it originates.

LO1 · K1 · A1

4 Trace

Connect LO1 · K1 · A1 to the assessment and evidence pack.

LO1 · K1 · A1

DESIGN JUDGEMENT

Do not advance on confidence alone; point to the artifact or test result that supports the answer.

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02

TOPIC 2 · LO2 · K2 · A2

Vibe-Generated Portable Logic and Quality Gates

LO2: Apply C++ methodologies and tools for software creation.

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CMake defines a portable build graph

1 Declare target

library

CPP-CMAKE

2 Set standard

C++20

CPP-CMAKE

3 Attach tests

CTest

CPP-CMAKE

4 Build out-of-tree

build dir

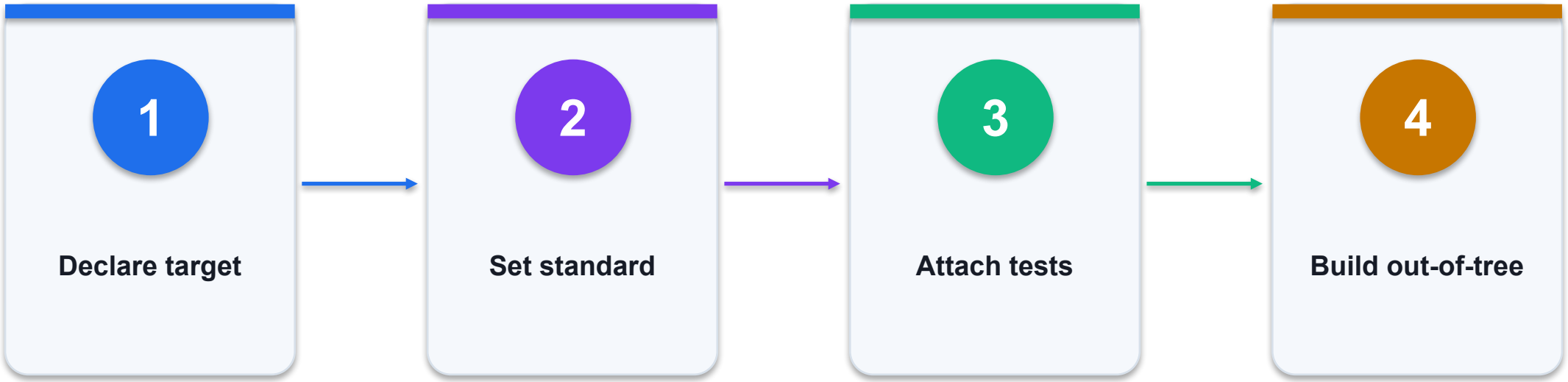
CPP-CMAKE

DESIGN JUDGEMENT

Make the core buildable with CMake and add it to Xcode as an explicit dependency.

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CMake defines a portable build graph · Mechanism



MECHANISM

Failure mode: Xcode-only configuration hides a broken standalone core. Control: Make the core buildable with CMake and add it to Xcode as an explicit dependency.

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WORKED EXAMPLE

CMake defines a portable build graph · Technical Anchor

```
add_library(DomainCore STATIC src/Budget.cpp)
```

1. library
2. C++20
3. CTest
4. build dir

EVIDENCE METRIC Reproducible build from clean checkout

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Use Red-Green-Refactor with AI

1 Write failing test

red

AI-TDD

2 Ask for minimal patch

minimal

AI-TDD

3 Run test

green

AI-TDD

4 Refactor with suite

cleanup

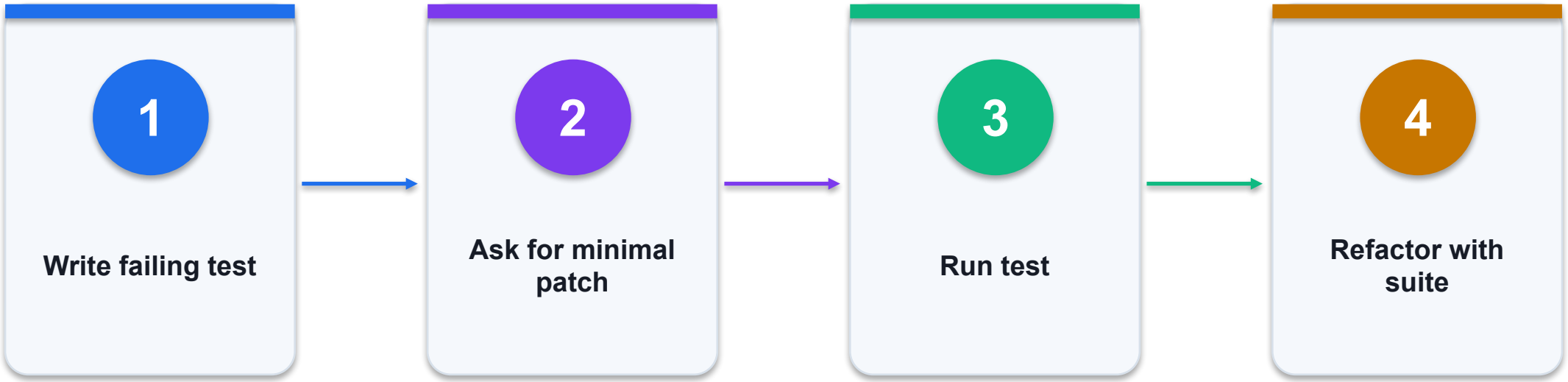
AI-TDD

DESIGN JUDGEMENT

Freeze expected behaviour first and demand the smallest causal patch.

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Use Red-Green-Refactor with AI · Mechanism



MECHANISM

Failure mode: The assistant writes implementation and tests that merely echo it. Control: Freeze expected behaviour first and demand the smallest causal patch.

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WORKED EXAMPLE

Use Red-Green-Refactor with AI · Technical Anchor

```
TEST(BudgetPolicy, RejectsNegativeLimit)
{ EXPECT_THROW(...); }
```

1. red
2. minimal
3. green
4. cleanup

EVIDENCE METRIC Every production change preceded by a meaningful failure

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Prompt context is an engineering input

1 **State contract**

goal

AI-PROMPT

2 **Attach relevant files**

context

AI-PROMPT

3 **Set constraints**

boundary

AI-PROMPT

4 **Define done**

checks

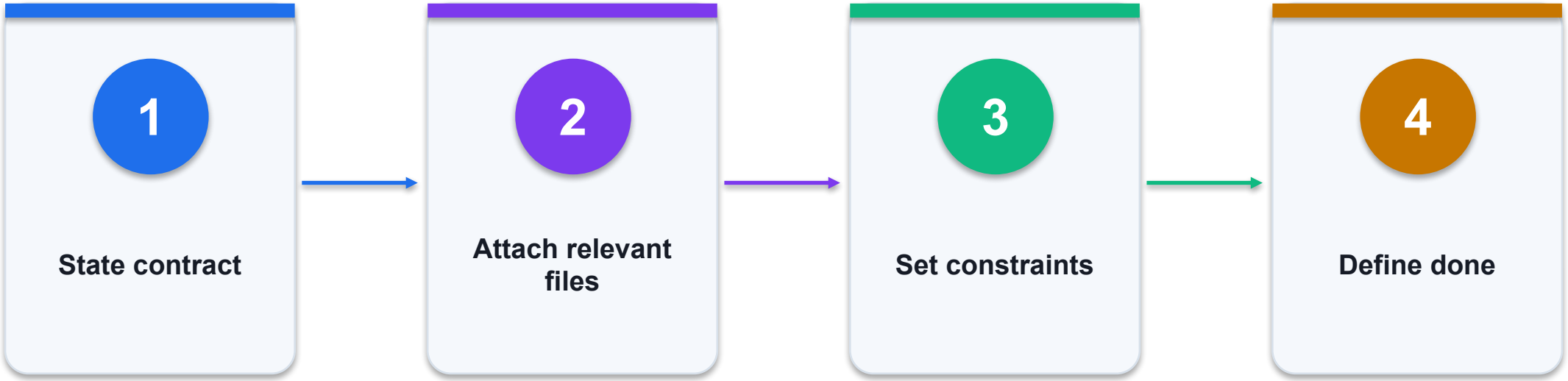
AI-PROMPT

DESIGN JUDGEMENT

Name protected interfaces and exact verification commands.

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Prompt context is an engineering input · Mechanism



MECHANISM **Failure mode: A broad prompt invites API churn and unrelated edits. Control: Name protected interfaces and exact verification commands.**

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WORKED EXAMPLE

Prompt context is an engineering input · Technical Anchor

```
Goal: implement evaluate(). Preserve BudgetPolicy.hpp.
Run ctest.
```

1. goal
2. context
3. boundary
4. checks

EVIDENCE METRIC Acceptance checks passed on first review

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Select types to encode invariants

1 Inspect range

int64_t

CPP-TYPES

2 Choose signedness

enum class

CPP-TYPES

3 Name unit

chrono

CPP-TYPES

4 Guard conversion

optional

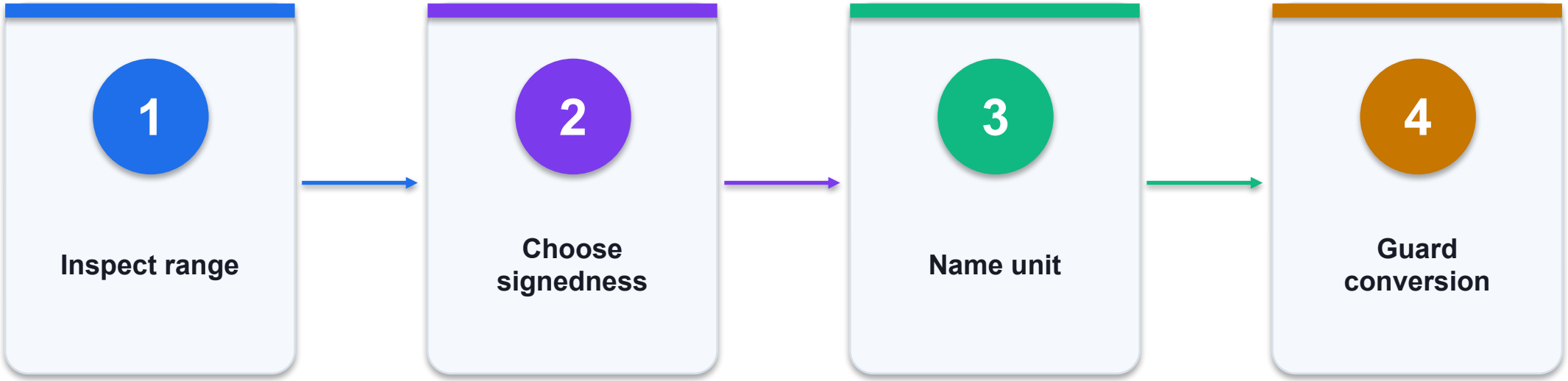
CPP-TYPES

DESIGN JUDGEMENT

Prefer domain types, scoped enumerations and explicit optional values.

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Select types to encode invariants · Mechanism



MECHANISM Failure mode: Primitive strings and doubles hide units and optionality. Control: Prefer domain types, scoped enumerations and explicit optional values.

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WORKED EXAMPLE

Select types to encode invariants · Technical Anchor

```
enum class ExpenseKind : std::uint8_t { food, travel,
other };
```

1. int64_t
2. enum class
3. chrono
4. optional

EVIDENCE METRIC Invalid conversions detected at boundaries

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Functions expose intention

1 Name verb

command

CPP-FUNCTION

2 Limit parameters

query

CPP-FUNCTION

3 Return value

const

CPP-FUNCTION

4 Keep pure

nodiscard

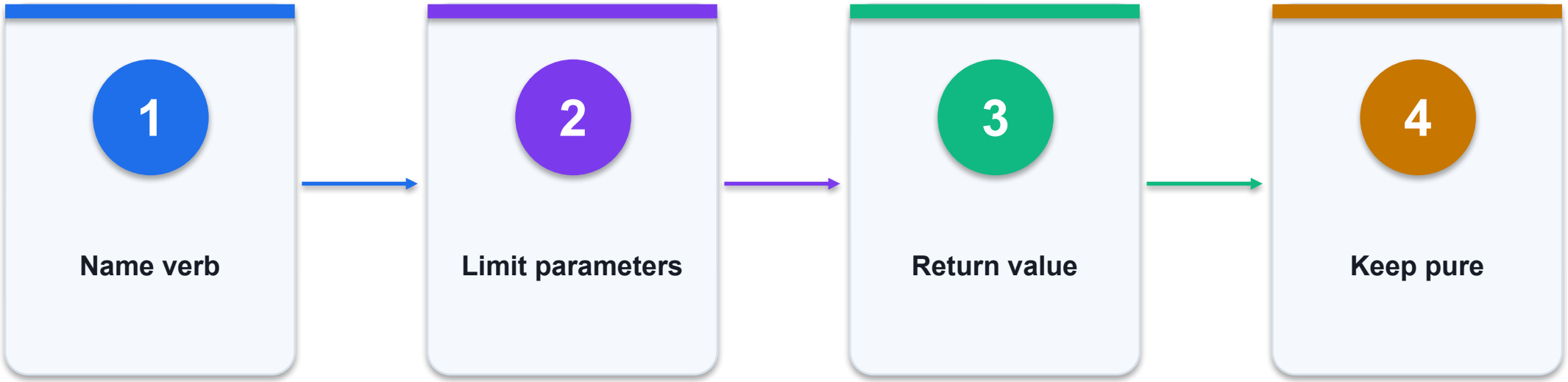
CPP-FUNCTION

DESIGN JUDGEMENT

Return data or typed errors; inject side effects through interfaces.

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Functions expose intention · Mechanism



MECHANISM

Failure mode: A function mutates globals and communicates failure by logging. Control: Return data or typed errors; inject side effects through interfaces.

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WORKED EXAMPLE

Functions expose intention · Technical Anchor

```
[[nodiscard]] Money remaining(Money limit, Money
spent);
```

1. command
2. query
3. const
4. nodiscard

EVIDENCE METRIC Pure functions covered by deterministic tests

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Control flow should reveal policy

1 **Guard invalid**

guard clause

CPP-CONTROL

2 **Select case**

switch

CPP-CONTROL

3 **Apply rule**

loop

CPP-CONTROL

4 **Return result**

result

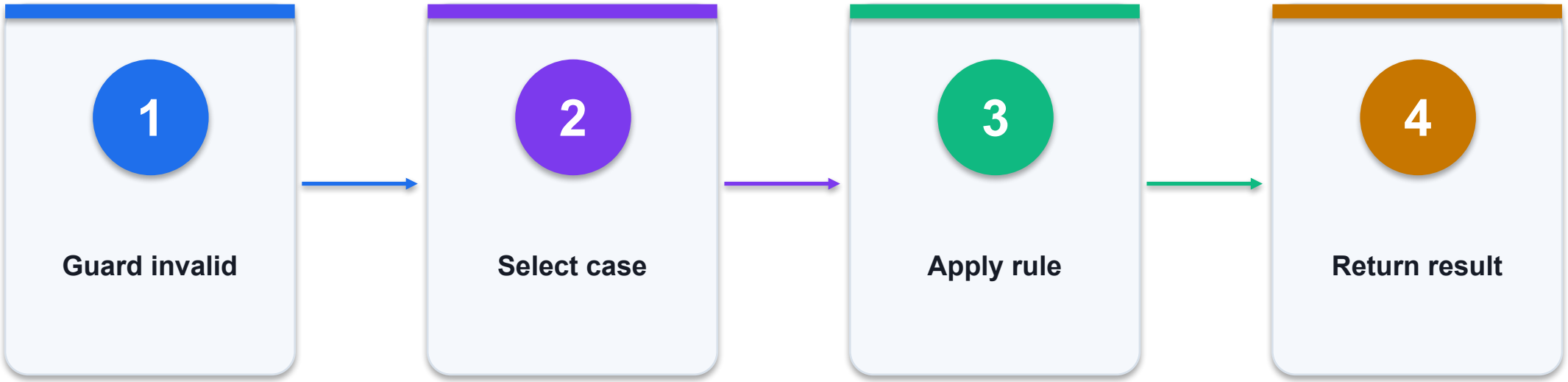
CPP-CONTROL

DESIGN JUDGEMENT

Use guards and small policy functions; reserve loops for collections.

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Control flow should reveal policy · Mechanism



MECHANISM Failure mode: Deep nesting obscures which condition owns a decision. Control: Use guards and small policy functions; reserve loops for collections.

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WORKED EXAMPLE

Control flow should reveal policy · Technical Anchor

```
if (draft.amount.cents <= 0) return
DomainError::invalidAmount;
```

1. guard clause
2. switch
3. loop
4. result

EVIDENCE METRIC Cyclomatic paths each have a named test

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RAII makes ownership visible

1 Acquire

unique_ptr

CPP-RAII

2 Bind to object

vector

CPP-RAII

3 Use safely

fstream

CPP-RAII

4 Release by scope

destructor

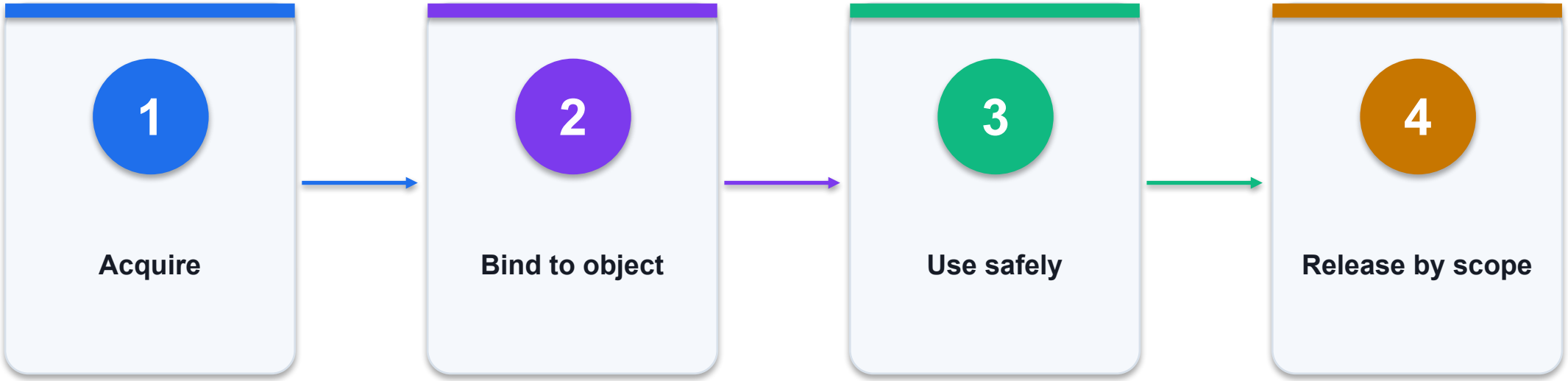
CPP-RAII

DESIGN JUDGEMENT

Keep RAII inside C++; cross the boundary using copied value snapshots.

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RAII makes ownership visible · Mechanism



MECHANISM

Failure mode: Raw ownership crosses the bridge and leaks on an error path. Control: Keep RAII inside C++; cross the boundary using copied value snapshots.

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WORKED EXAMPLE

RAII makes ownership visible · Technical Anchor

```
auto repo =
std::make_unique<JsonExpenseRepository>(path);
```

1. unique_ptr
2. vector
3. fstream
4. destructor

EVIDENCE METRIC Zero manual delete paths

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Dependency injection enables substitution

1 Define port

interface

ARCH-DI

2 Inject implementation

constructor

ARCH-DI

3 Exercise fake

fake

ARCH-DI

4 Swap adapter

production

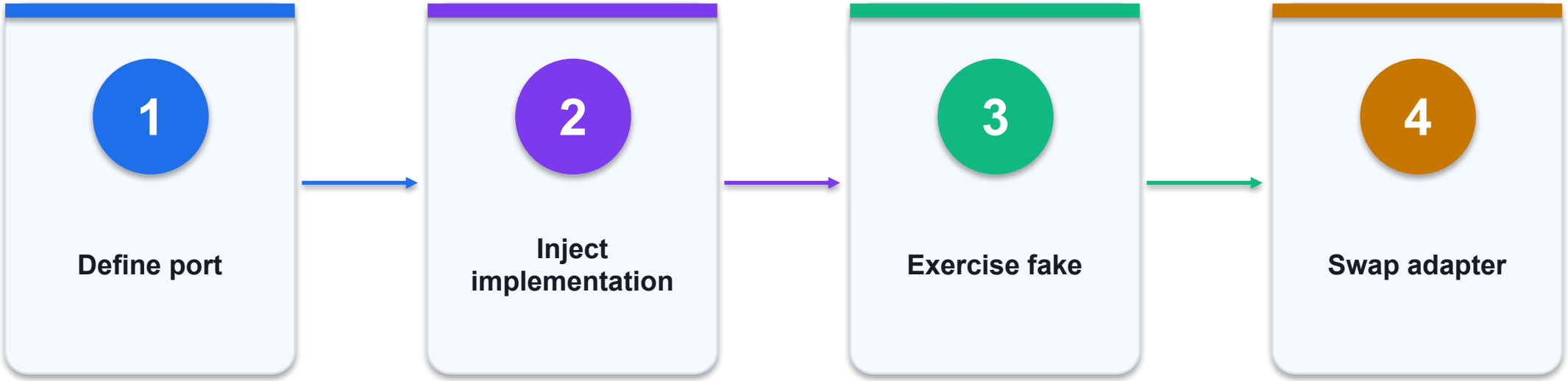
ARCH-DI

DESIGN JUDGEMENT

Inject repositories and clocks through narrow abstract ports.

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Dependency injection enables substitution · Mechanism



MECHANISM Failure mode: A singleton couples rules to persistence and time. Control: Inject repositories and clocks through narrow abstract ports.

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WORKED EXAMPLE

Dependency injection enables substitution · Technical Anchor

```
ExpenseService(ExpenseRepository& repo, Clock& clock);
```

1. interface
2. constructor
3. fake
4. production

EVIDENCE METRIC Core tests run without disk or iOS runtime

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Compiler diagnostics are design feedback

1 Enable warnings

-Wall

CPP-TOOLCHAIN

2 Treat as errors

-Wextra

CPP-TOOLCHAIN

3 Read locus

-Werror

CPP-TOOLCHAIN

4 Repair cause

sanitizer

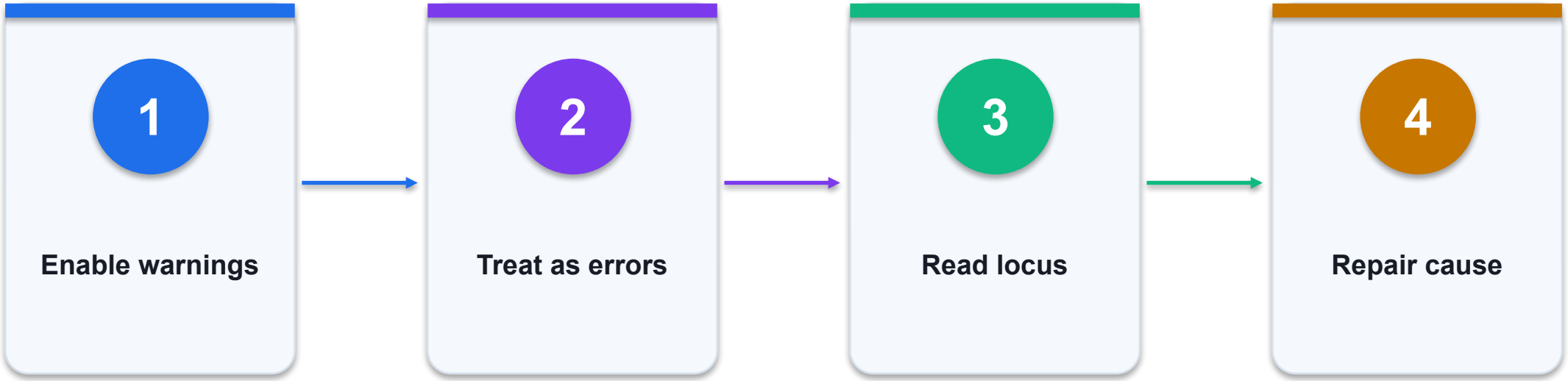
CPP-TOOLCHAIN

DESIGN JUDGEMENT

Use strict warnings and sanitizers during development; document platform differences.

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Compiler diagnostics are design feedback · Mechanism



MECHANISM

Failure mode: Generated code compiles only under permissive defaults. Control: Use strict warnings and sanitizers during development; document platform differences.

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WORKED EXAMPLE

Compiler diagnostics are design feedback · Technical Anchor

```
target_compile_options(DomainCore PRIVATE -Wall -Wextra
-Werror)
```

1. -Wall
2. -Wextra
3. -Werror
4. sanitizer

EVIDENCE METRIC Zero warnings in clean build

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Review AI diffs semantically

1 Check contract

API

AI-REVIEW

2 Trace data

invariant

AI-REVIEW

3 Inspect lifetime

ownership

AI-REVIEW

4 Run regression

tests

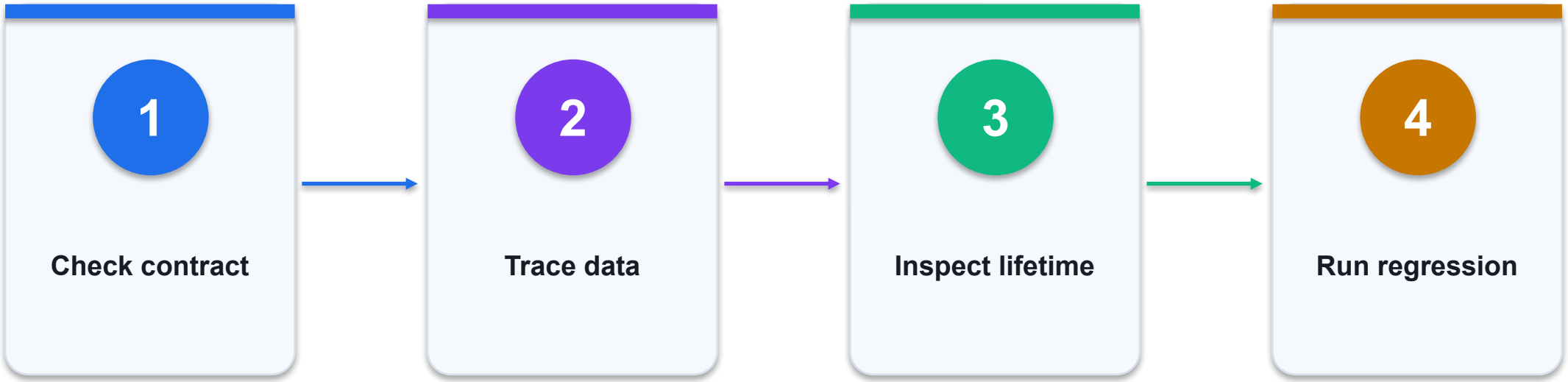
AI-REVIEW

DESIGN JUDGEMENT

Review data representation and lifetimes before style.

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Review AI diffs semantically · Mechanism



MECHANISM

Failure mode: A plausible patch introduces silent truncation or ownership ambiguity. Control: Review data representation and lifetimes before style.

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WORKED EXAMPLE

Review AI diffs semantically · Technical Anchor

```
// Review: conversion from Swift Int to int32_t may
overflow.
```

1. API
2. invariant
3. ownership
4. tests

EVIDENCE METRIC Review findings resolved before merge

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SCENARIO EVIDENCE

AI Code Review · Evidence before Acceptance



1. Locate the claim in code.
2. Reproduce material findings.
3. Classify severity and owner.
4. Run focused and regression checks.

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Organisational practice is executable

1 Write convention

format

TSC-A2

2 Automate check

lint

TSC-A2

3 Run locally

tests

TSC-A2

4 Enforce in CI

evidence

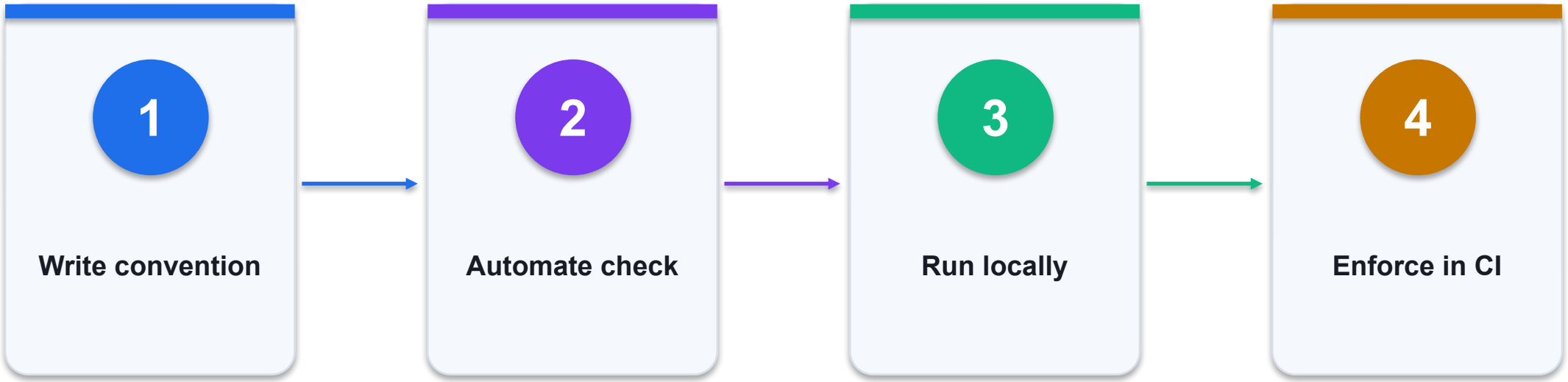
TSC-A2

DESIGN JUDGEMENT

Translate practices into formatter, compiler and test configuration.

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Organisational practice is executable · Mechanism



MECHANISM Failure mode: A design principle remains prose and is ignored under time pressure. Control: Translate practices into formatter, compiler and test configuration.

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WORKED EXAMPLE

Organisational practice is executable · Technical Anchor

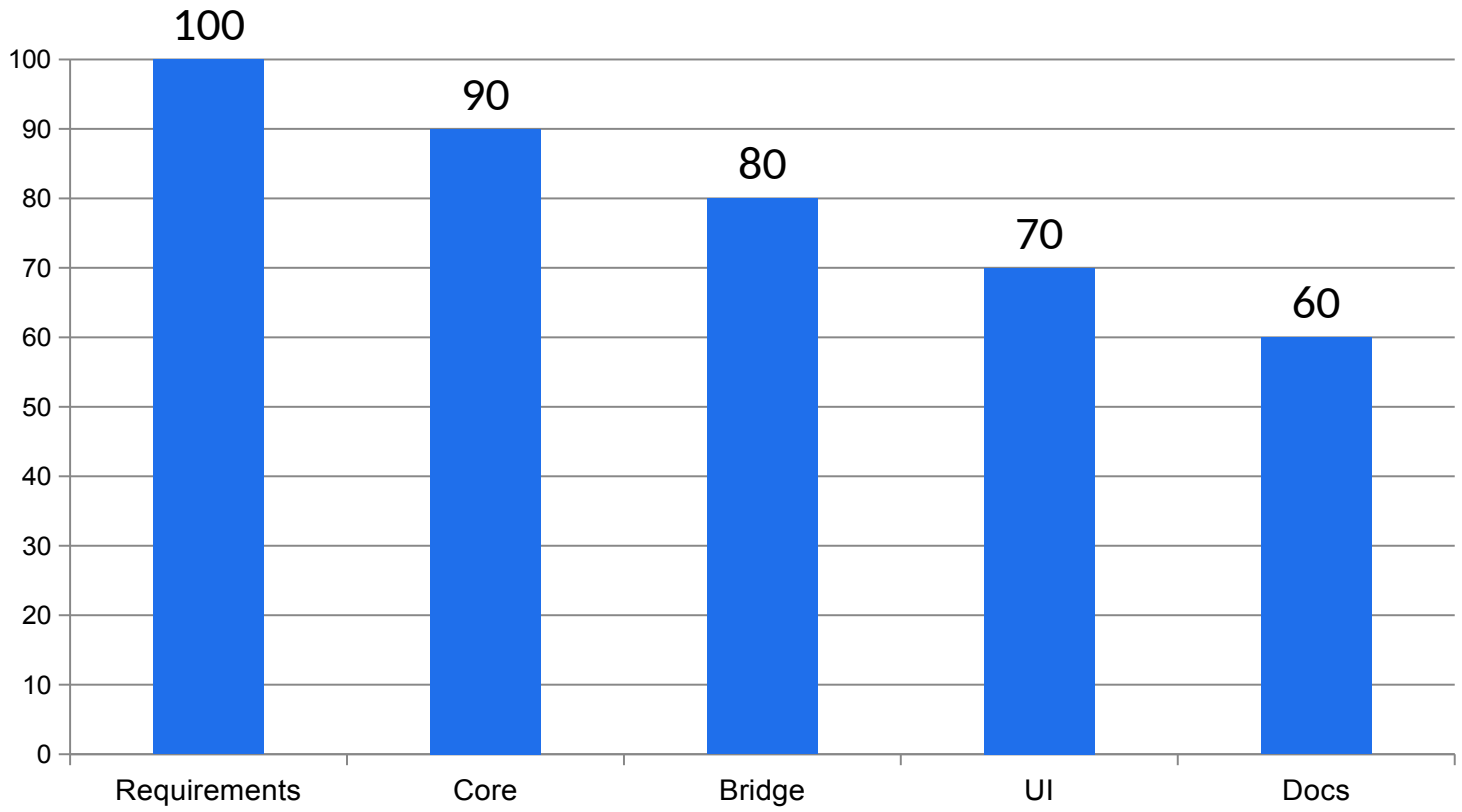
```
cmake --build build && ctest --test-dir build --output-on-failure
```

1. format
2. lint
3. tests
4. evidence

EVIDENCE METRIC Same checks locally and in CI

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Topic 2 · Evidence Coverage by Layer



WHAT THE DATA SHOWS

Coverage should form a connected chain. A polished UI cannot compensate for untested invariants or an undocumented seam.

Editable chart · values are instructional thresholds, not production benchmarks

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HANDS-ON ACTIVITY · 75 MIN

Activity 3: AI-Assisted TDD for C++

CHALLENGE

Use an evidence-bound AI prompt to implement a rule through Red-Green-Refactor.

REGULATED MAPPING

K2, A2

YOU WILL PRODUCE

Prompt contract, failing trace, minimal patch and passing trace

AI coding assistant, CTest, diff review

Detailed procedures: Learner Guide and activities/03-ai-assisted-tdd/README.md

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ACTIVITY MECHANISM

Activity 3 · Delivery Workflow

1 Frame

Run the supplied failing overspend test and save the red trace.

K2, A2

2 Build

Write a prompt naming the contract, protected files and exact finish checks.

K2, A2

3 Connect

Review types, ownership, error paths and unrelated changes.

K2, A2

4 Verify

Evidence includes a meaningful initial failure, a scoped diff and a fully passing suite.

K2, A2

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 3 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Use the recorded overspend RED trace and current BudgetPolicy files.
Produce the smallest causal patch that makes the missing rule pass.
Preserve the public signature, integer cents, existing tests and
unrelated formatting. Return the complete changed file, focused command
and full regression command. Do not weaken the failing assertion. Retain
red trace, scoped diff and GREEN trace.
```

PROTECTED FILE SCOPE

workspace/prompt-contract.txt; workspace/BudgetPolicy.hpp; workspace/BudgetPolicy.cpp; workspace/failing-test.cpp; workspace/red-trace.txt; workspace/green-trace.txt

EXPECTED GREEN

The initial test fails for the missing overspend rule; after the minimal patch the focused binary and full domain suite exit 0.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 3 · Evidence Contract

1 Artifact

Prompt contract, failing trace, minimal patch and passing trace

K2, A2

2 Acceptance

Evidence includes a meaningful initial failure, a scoped diff and a fully passing suite.

K2, A2

3 Failure signal

If the test passed before the change, strengthen the assertion before asking for implementation.

K2, A2

4 Cleanup

Keep prompt and evidence; revert experimental changes outside the accepted patch.

K2, A2

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 3 · Verification Gate

PASS WHEN

Evidence includes a meaningful initial failure, a scoped diff and a fully passing suite.

IF IT FAILS

If the test passed before the change, strengthen the assertion before asking for implementation.

LEAVE THE WORKSPACE

Keep prompt and evidence; revert experimental changes outside the accepted patch.

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HANDS-ON ACTIVITY · 65 MIN

Activity 4: Toolchain and Quality Gates

CHALLENGE

Turn organisational design practices into compiler, formatter and test gates.

REGULATED MAPPING

K2, A2

YOU WILL PRODUCE

Warning-clean build, quality script and evidence log

CMake, compiler warnings, sanitizers, shell

Detailed procedures: Learner Guide and activities/04-toolchain-quality-gates/README.md

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ACTIVITY MECHANISM

Activity 4 · Delivery Workflow

1 Frame

Enable C++20 plus -Wall, -Wextra and -Werror for the core target.

K2, A2

2 Build

Add a single verify script that configures, builds and tests.

K2, A2

3 Connect

Repair the causal line and run the complete verification again.

K2, A2

4 Verify

A clean checkout can run one command and receive a non-zero exit on the injected defect and zero after repair.

K2, A2

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 4 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate complete CMake and verify.sh files for C++20, guarded strict
warnings, optional AddressSanitizer and CTest. Demonstrate RED with one
controlled warning, then repair its causal line and demonstrate GREEN.
Do not ignore command failures or use an in-source build. Output
complete files and the expected non-zero then zero observations.
```

PROTECTED FILE SCOPE

workspace/CMakeLists.txt; workspace/verify.sh; workspace/quality-gates.txt; workspace/evidence-log.txt

EXPECTED GREEN

`sh workspace/verify.sh` fails on the injected warning and exits 0 after repair with all CTest cases passing.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 4 · Evidence Contract

1 Artifact

Warning-clean build, quality script and evidence log

K2, A2

2 Acceptance

A clean checkout can run one command and receive a non-zero exit on the injected defect and zero after repair.

K2, A2

3 Failure signal

If Apple Clang rejects a flag, gate it by compiler ID and document the equivalent.

K2, A2

4 Cleanup

Remove generated build and sanitizer output, not the verification script.

K2, A2

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 4 · Verification Gate

PASS WHEN

A clean checkout can run one command and receive a non-zero exit on the injected defect and zero after repair.

IF IT FAILS

If Apple Clang rejects a flag, gate it by compiler ID and document the equivalent.

LEAVE THE WORKSPACE

Remove generated build and sanitizer output, not the verification script.

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RETRIEVAL PRACTICE

Topic 2 · Review

1 Explain

LO2: Apply C++ methodologies and tools for software creation.

LO2 · K2 · A2

2 Demonstrate

Show the Activity evidence that operationalises the outcome.

LO2 · K2 · A2

3 Diagnose

Name the failure signal and the layer where it originates.

LO2 · K2 · A2

4 Trace

Connect LO2 · K2 · A2 to the assessment and evidence pack.

LO2 · K2 · A2

DESIGN JUDGEMENT

Do not advance on confidence alone; point to the artifact or test result that supports the answer.

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03

TOPIC 3 · LO3 · K3 · A3

AI-Generated SwiftUI App Shell and Controls

LO3: Select essential C++ controls and features to meet software design requirements.

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SwiftUI renders state as a function

1 **Own state**

@State

SWIFTUI-STATE

2 **Derive view**

@Observable

SWIFTUI-STATE

3 **Send intent**

body

SWIFTUI-STATE

4 **Render update**

action

SWIFTUI-STATE

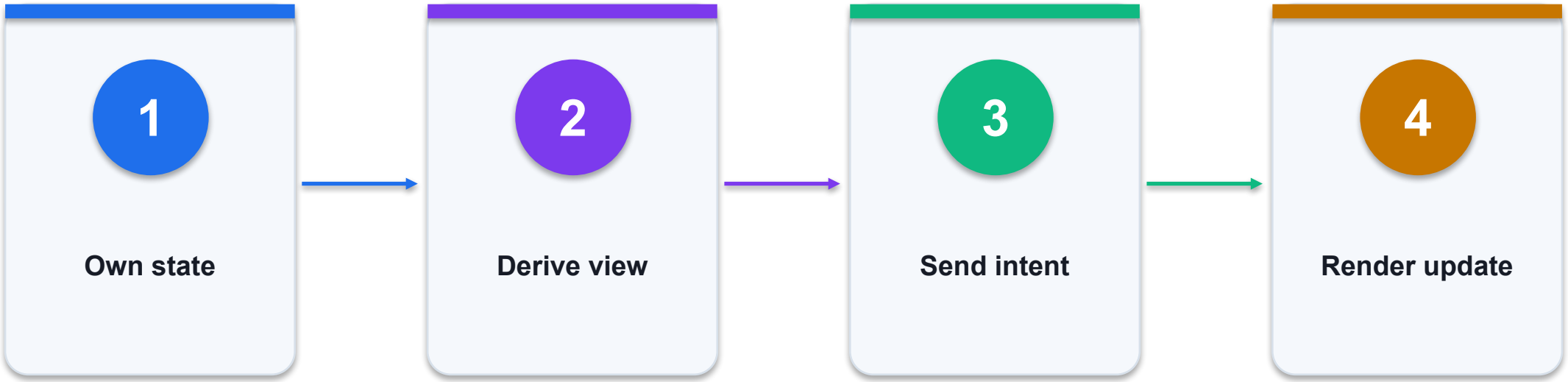
DESIGN JUDGEMENT

Keep View declarative; send intent to a state owner that calls the bridge.

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TOPIC 3 · PROCESS

SwiftUI renders state as a function · Mechanism



MECHANISM **Failure mode: Business rules execute inside body and repeat on every render. Control: Keep View declarative; send intent to a state owner that calls the bridge.**

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WORKED EXAMPLE

SwiftUI renders state as a function · Technical Anchor

```
@Observable final class BudgetViewModel { private(set)
var snapshot: Snapshot }
```

1. @State
2. @Observable
3. body
4. action

EVIDENCE METRIC UI states covered by previews or tests

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Select controls from the data contract

1 Identify type

TextField

TSC-A3

2 Choose affordance

Picker

TSC-A3

3 Constrain input

Toggle

TSC-A3

4 Expose feedback

Button

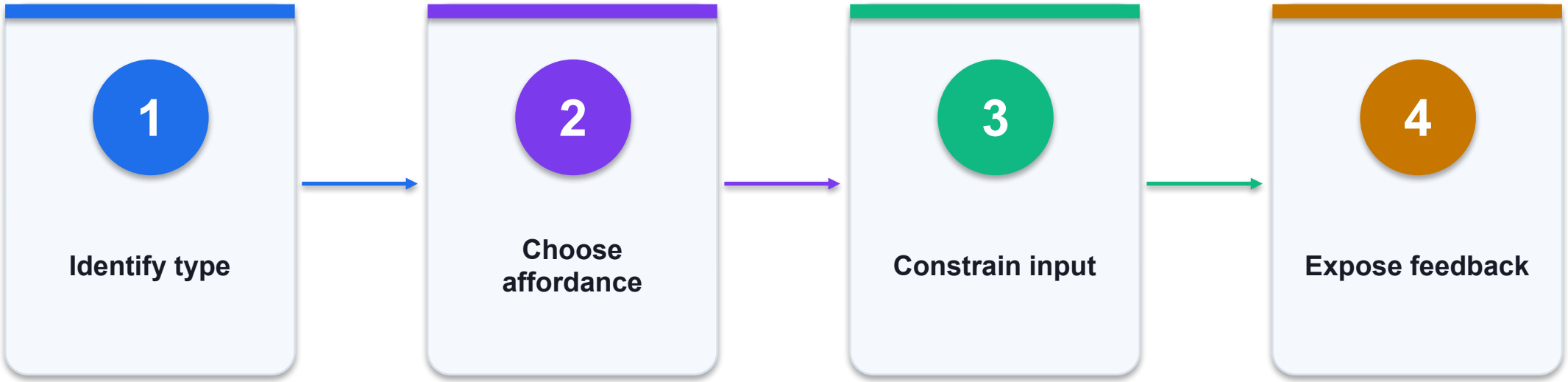
TSC-A3

DESIGN JUDGEMENT

Choose the control that communicates range, optionality and consequence.

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Select controls from the data contract · Mechanism



MECHANISM

Failure mode: A free-text field captures a value from a small controlled set. Control: Choose the control that communicates range, optionality and consequence.

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WORKED EXAMPLE

Select controls from the data contract · Technical Anchor

```
Picker("Category", selection: $draft.category) { ... }
```

1. TextField
2. Picker
3. Toggle
4. Button

EVIDENCE METRIC Invalid states prevented or explained

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Validation belongs at two boundaries

1 Guide in UI

format

VALIDATION

2 Validate domain

invariant

VALIDATION

3 Map error

message

VALIDATION

4 Preserve input

recovery

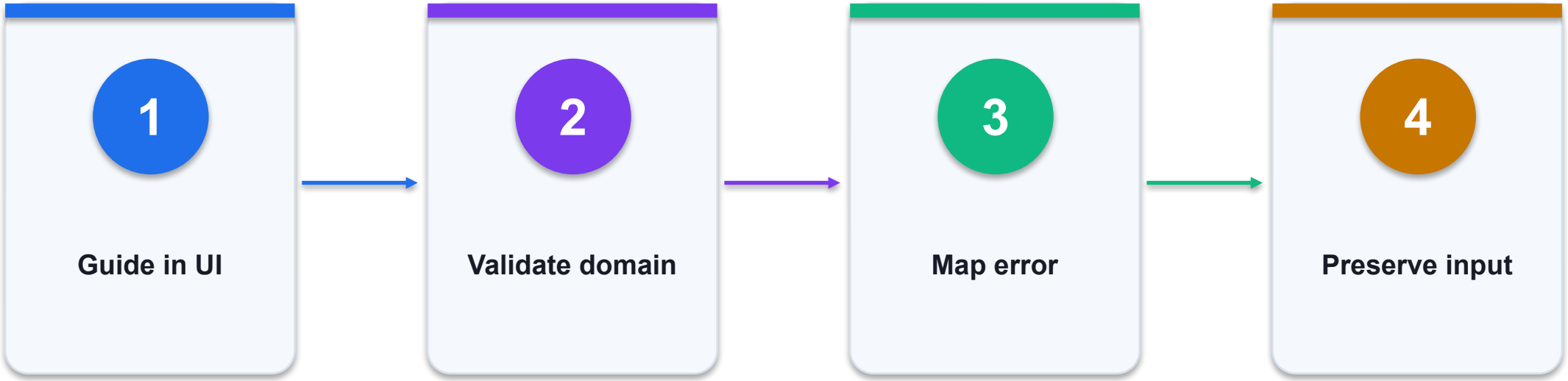
VALIDATION

DESIGN JUDGEMENT

Provide immediate SwiftUI guidance and enforce invariants again in C++.

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Validation belongs at two boundaries · Mechanism



MECHANISM Failure mode: UI validation is treated as sufficient security or correctness. Control: Provide immediate SwiftUI guidance and enforce invariants again in C++.

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WORKED EXAMPLE

Validation belongs at two boundaries · Technical Anchor

```
guard let cents = MoneyParser.cents(text) else
{ message = .invalidAmount }
```

1. format
2. invariant
3. message
4. recovery

EVIDENCE METRIC Rejected invalid inputs with recoverable state

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Navigation models workflow state

1 Define destination

NavigationStack

SWIFTUI-NAV

2 Bind path

NavigationPath

SWIFTUI-NAV

3 Pass identity

route

SWIFTUI-NAV

4 Handle return

selection

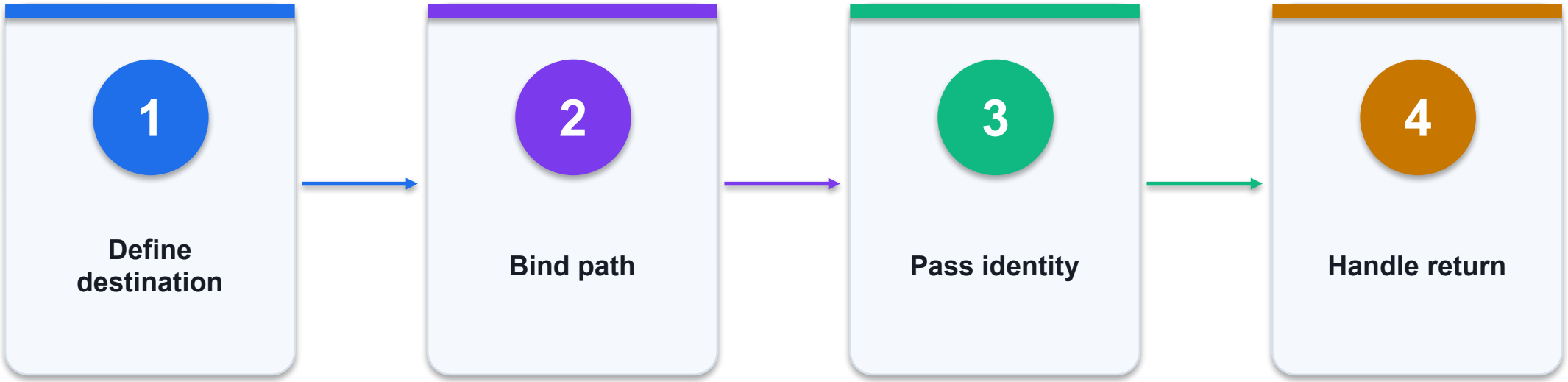
SWIFTUI-NAV

DESIGN JUDGEMENT

Route with identifiers and reload authoritative snapshots at boundaries.

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Navigation models workflow state · Mechanism



MECHANISM

Failure mode: Views pass mutable models through an implicit navigation chain. Control: Route with identifiers and reload authoritative snapshots at boundaries.

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WORKED EXAMPLE

Navigation models workflow state · Technical Anchor

```
enum Route: Hashable { case expenseDetail(UUID); case
settings }
```

1. NavigationStack
2. NavigationPath
3. route
4. selection

EVIDENCE METRIC Deep link and back path produce valid state

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Lists need stable identity

1 Fetch snapshot

Identifiable

SWIFTUI-LIST

2 Identify row

UUID

SWIFTUI-LIST

3 Render collection

ForEach

SWIFTUI-LIST

4 Apply intent

delete

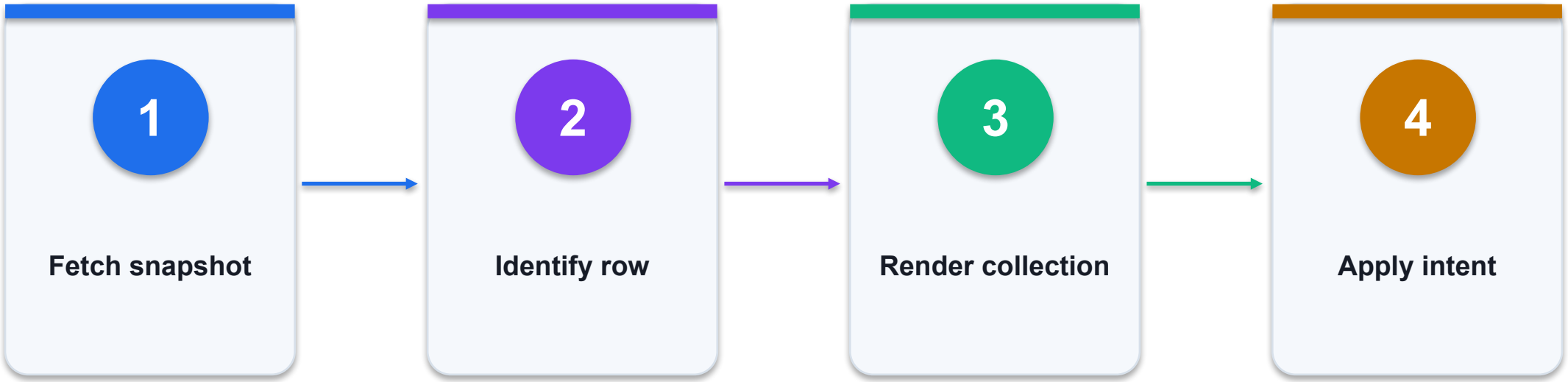
SWIFTUI-LIST

DESIGN JUDGEMENT

Expose stable IDs from the core and reconcile snapshots by identity.

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Lists need stable identity · Mechanism



MECHANISM Failure mode: Array indices are used as identity and shift after deletion. Control: Expose stable IDs from the core and reconcile snapshots by identity.

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WORKED EXAMPLE

Lists need stable identity · Technical Anchor

```
ForEach(model.snapshot.expenses, id: \.id)
{ ExpenseRow($0) }
```

1. Identifiable
2. UUID
3. ForEach
4. delete

EVIDENCE METRIC Row updates affect exactly one domain entity

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Swift concurrency guards responsiveness

1 **Start task**

Task

SWIFT-CONCURRENCY

2 **Leave main actor**

async

SWIFT-CONCURRENCY

3 **Await bridge work**

actor

SWIFT-CONCURRENCY

4 **Publish state**

MainActor

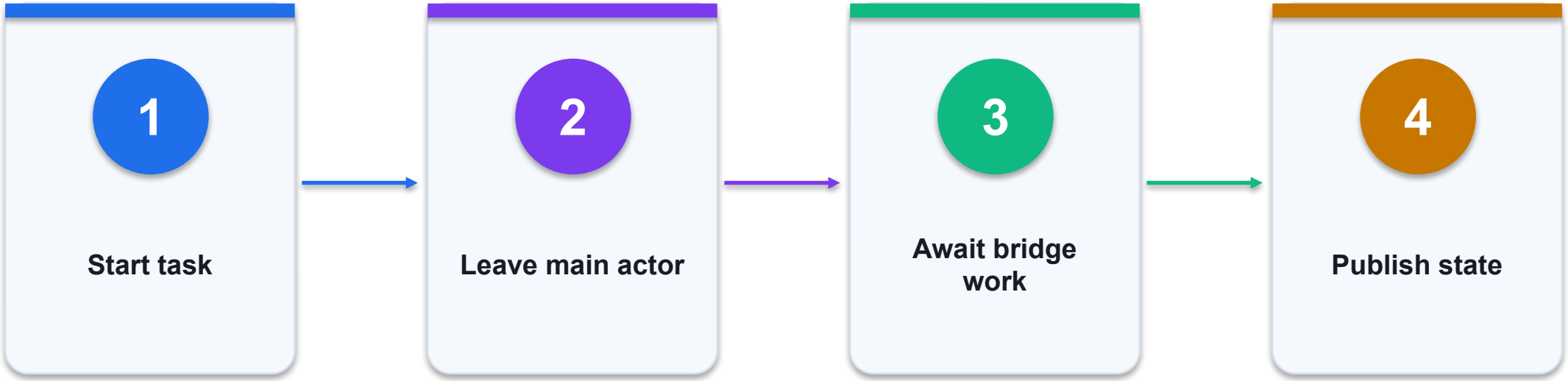
SWIFT-CONCURRENCY

DESIGN JUDGEMENT

Isolate expensive work and publish UI state on the main actor.

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Swift concurrency guards responsiveness · Mechanism



MECHANISM Failure mode: Disk or model inference runs synchronously from a button action. Control: Isolate expensive work and publish UI state on the main actor.

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WORKED EXAMPLE

Swift concurrency guards responsiveness · Technical Anchor

```
Task { await model.refresh() }
```

1. Task
2. async
3. actor
4. MainActor

EVIDENCE METRIC No blocking work on main thread

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Accessibility is part of control design

1 **Name purpose**

label

IOS-A11Y

2 **Expose value**

value

IOS-A11Y

3 **Support focus**

hint

IOS-A11Y

4 **Test scaling**

Dynamic Type

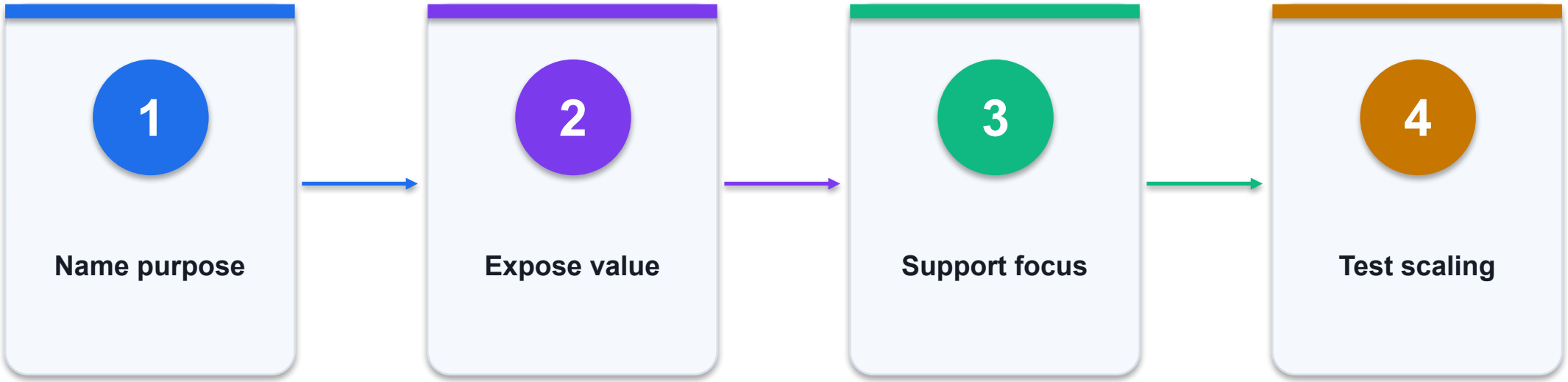
IOS-A11Y

DESIGN JUDGEMENT

Treat semantic labels and adaptable layout as acceptance criteria.

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Accessibility is part of control design · Mechanism



MECHANISM

Failure mode: An icon-only control has no accessible name. Control: Treat semantic labels and adaptable layout as acceptance criteria.

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WORKED EXAMPLE

Accessibility is part of control design · Technical Anchor

```
.accessibilityLabel("Delete expense")
```

1. label
2. value
3. hint
4. Dynamic Type

EVIDENCE METRIC Critical flow works with VoiceOver and large text

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Previews are executable design probes

1 **Create fixture**

empty

SWIFTUI-PREVIEW

2 **Inject fake**

loaded

SWIFTUI-PREVIEW

3 **Render states**

error

SWIFTUI-PREVIEW

4 **Inspect variants**

large text

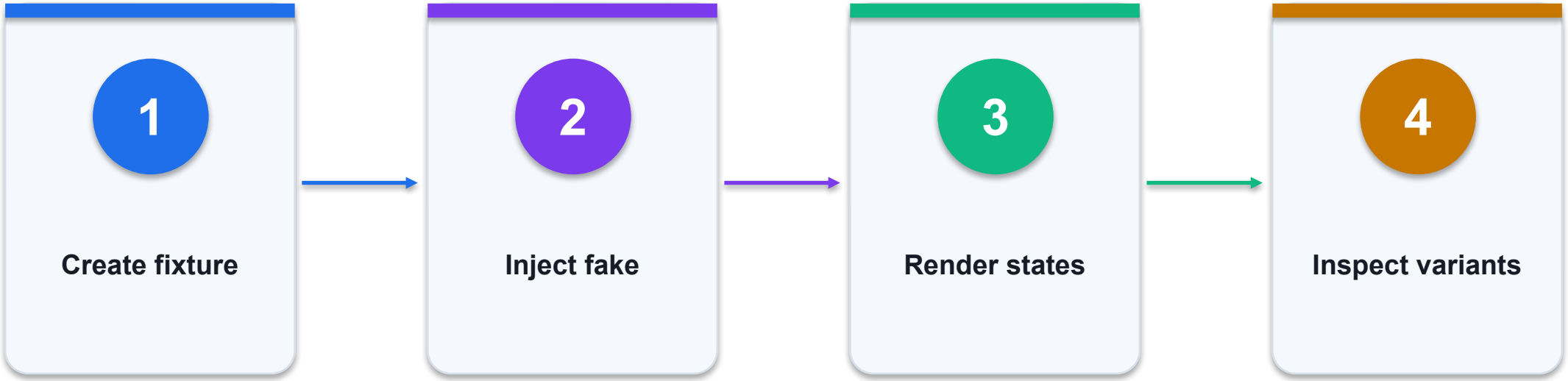
SWIFTUI-PREVIEW

DESIGN JUDGEMENT

Inject deterministic snapshots and display state extremes.

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Previews are executable design probes · Mechanism



MECHANISM Failure mode: A preview depends on production data and never reaches failure states. Control: Inject deterministic snapshots and display state extremes.

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WORKED EXAMPLE

Previews are executable design probes · Technical Anchor

```
#Preview("Error")
{ BudgetScreen(model: .fixture(.error)) }
```

1. empty
2. loaded
3. error
4. large text

EVIDENCE METRIC Representative states preview without network or storage

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Feature flags bound experimental AI

1 Define flag

identifier

GOV-FLAG

2 Default off

default

GOV-FLAG

3 Observe usage

telemetry

GOV-FLAG

4 Remove or graduate

owner

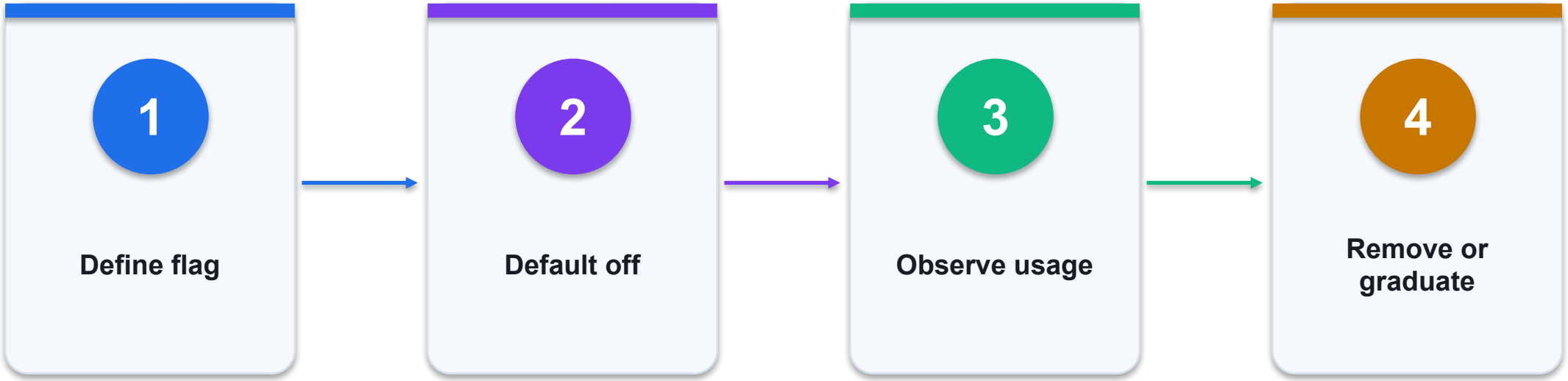
GOV-FLAG

DESIGN JUDGEMENT

Gate experiments, capture evidence and set a review date.

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Feature flags bound experimental AI · Mechanism



MECHANISM **Failure mode: An AI-generated feature becomes permanent hidden complexity. Control: Gate experiments, capture evidence and set a review date.**

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WORKED EXAMPLE

Feature flags bound experimental AI · Technical Anchor

```
if flags.isEnabled(.smartCategorySuggestion) { ... }
```

1. identifier
2. default
3. telemetry
4. owner

EVIDENCE METRIC Experimental path has owner and expiry

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Local Foundation Models stay supplemental

1 Check availability

availability

AI-ONDEVICE

2 Prepare bounded prompt

schema

AI-ONDEVICE

3 Validate output

validation

AI-ONDEVICE

4 Fall back

fallback

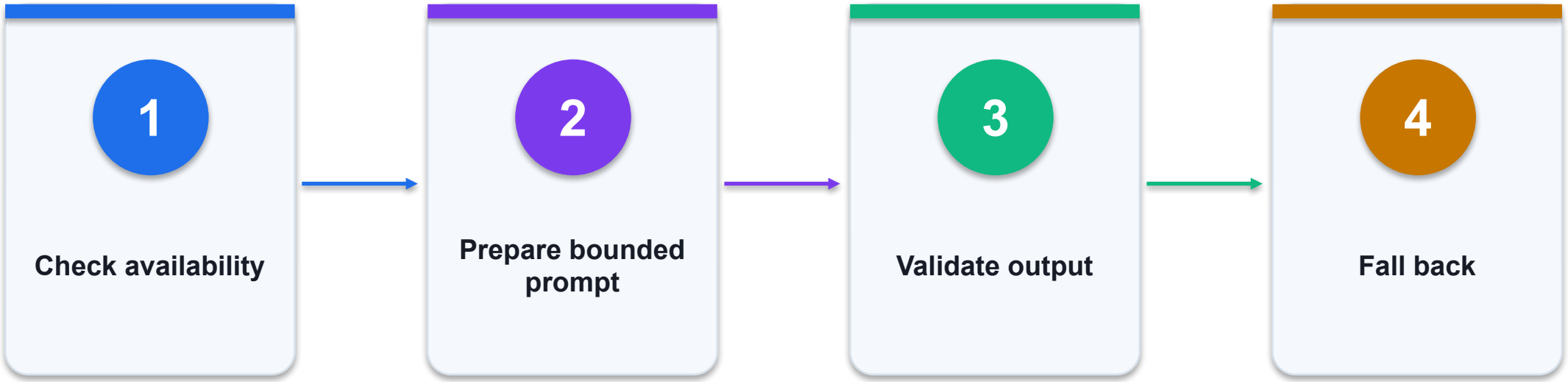
AI-ONDEVICE

DESIGN JUDGEMENT

Use model output as a suggestion; validate before committing to the core.

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Local Foundation Models stay supplemental · Mechanism



MECHANISM

Failure mode: A local model becomes a hidden source of business truth. Control: Use model output as a suggestion; validate before committing to the core.

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WORKED EXAMPLE

Local Foundation Models stay supplemental · Technical Anchor

```
guard modelSystem.isAvailable else
{ return .manualEntry }
```

1. availability
2. schema
3. validation
4. fallback

EVIDENCE METRIC Deterministic core works when model is unavailable

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Design selections require rationale

1 **List candidates**

control

TSC-K3

2 **Score fit**

criterion

TSC-K3

3 **Prototype risk**

trade-off

TSC-K3

4 **Record choice**

decision

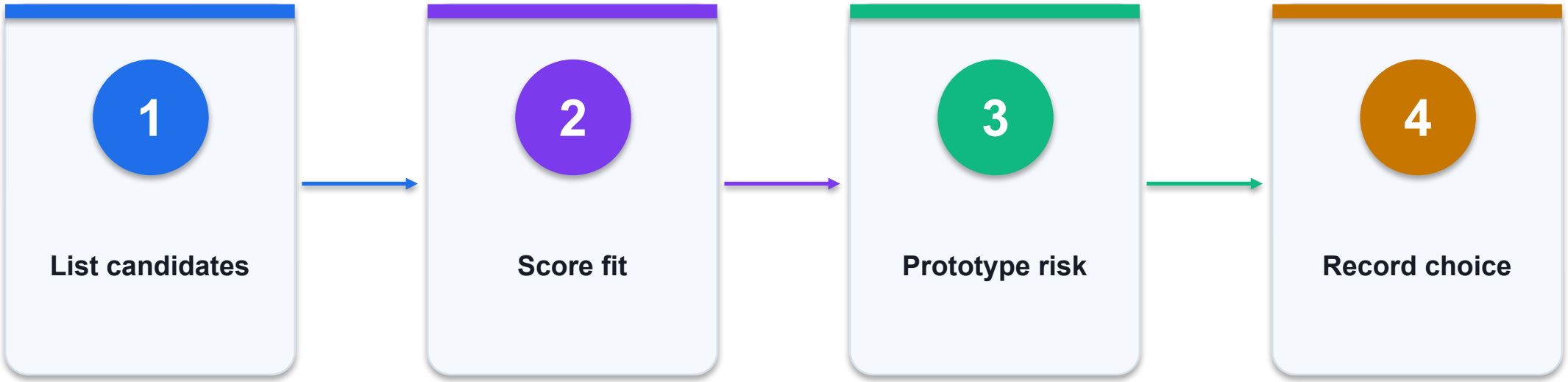
TSC-K3

DESIGN JUDGEMENT

Record why the chosen control better enforces the requirement.

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Design selections require rationale · Mechanism



MECHANISM

Failure mode: Controls are selected because they are familiar or visually fashionable. Control: Record why the chosen control better enforces the requirement.

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WORKED EXAMPLE

Design selections require rationale · Technical Anchor

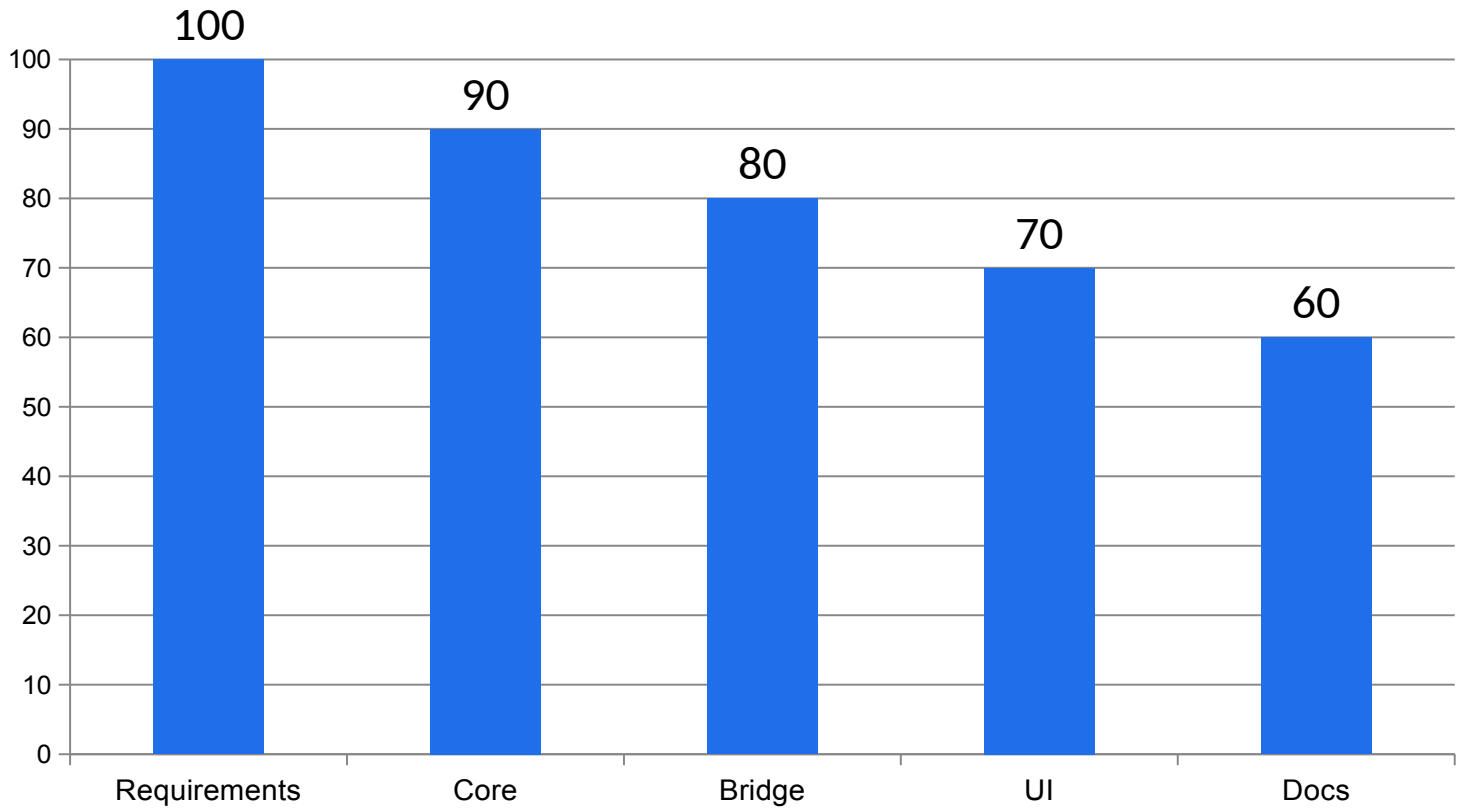
```
// Decision: Picker because category is closed and
required.
```

1. control
2. criterion
3. trade-off
4. decision

EVIDENCE METRIC Each material control linked to objective

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Topic 3 · Evidence Coverage by Layer



WHAT THE DATA SHOWS

Coverage should form a connected chain. A polished UI cannot compensate for untested invariants or an undocumented seam.

Editable chart · values are instructional thresholds, not production benchmarks

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HANDS-ON ACTIVITY · 85 MIN

Activity 5: Objective-C++ Adapter

CHALLENGE

Expose the portable domain core to Swift through a narrow Objective-C-compatible API.

REGULATED MAPPING

K3, A3

YOU WILL PRODUCE

Bridge header, .mm adapter and round-trip test

Xcode, Objective-C++, XCTest

Detailed procedures: Learner Guide and activities/05-objective-cpp-adapter/README.md

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ACTIVITY MECHANISM

Activity 5 · Delivery Workflow

1 Frame

Inspect the public bridge header and remove every C++-only type.

K3, A3

2 Build

Own ExpenseService with std::unique_ptr inside the .mm implementation.

K3, A3

3 Connect

Map typed domain failures to stable NSError codes.

K3, A3

4 Verify

Swift can call the bridge; cents, identity and error codes survive the round trip exactly.

K3, A3

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 5 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate BudgetBridge.h, BudgetBridge.mm and a bridge contract test.
Keep the header Objective-C compatible. Put C++ includes and
std::unique_ptr ownership in .mm. Preserve exact cents and map each
DomainError to a stable NSError code plus recovery description. Output
complete files. Verify iOS build, 10_000/2_500/1_250 -> 6_250 and every
error mapping.
```

PROTECTED FILE SCOPE

BudgetBridge.h; BudgetBridge.mm; bridge-test.swift; error-mapping.md

EXPECTED GREEN

iOS build and focused bridge tests exit 0; the public header contains no STL/template/reference/owning-pointer type and exact cents survive the round trip.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 5 · Evidence Contract

1 Artifact

Bridge header, .mm adapter and round-trip test

K3, A3

2 Acceptance

Swift can call the bridge; cents, identity and error codes survive the round trip exactly.

K3, A3

3 Failure signal

If Swift cannot import the header, check target membership and confirm C++ includes appear only in .mm.

K3, A3

4 Cleanup

Clean derived build data only if needed; retain tests and mapping table.

K3, A3

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 5 · Verification Gate

PASS WHEN

Swift can call the bridge; cents, identity and error codes survive the round trip exactly.

IF IT FAILS

If Swift cannot import the header, check target membership and confirm C++ includes appear only in .mm.

LEAVE THE WORKSPACE

Clean derived build data only if needed; retain tests and mapping table.

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HANDS-ON ACTIVITY · 80 MIN

Activity 6: SwiftUI Controls and State

CHALLENGE

Build an accessible SwiftUI expense-entry shell whose controls reflect the domain contract.

REGULATED MAPPING

K3, A3

YOU WILL PRODUCE

SwiftUI entry flow, deterministic previews and validation states

SwiftUI, Observation, Xcode previews

Detailed procedures: Learner Guide and activities/06-swiftui-controls-state/README.md

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ACTIVITY MECHANISM

Activity 6 · Delivery Workflow

1 Frame

Create an observable view model that owns a bridge-facing snapshot.

K3, A3

2 Build

Use TextField for amount, Picker for closed category and Button for explicit commit.

K3, A3

3 Connect

Render empty, loaded, validation-error and large-text previews.

K3, A3

4 Verify

The view remains declarative, invalid input is recoverable, and previews cover four states without production data.

K3, A3

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 6 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate BudgetEvaluating, BudgetBridgeClient, FakeBudgetBridge,
BudgetModel and @Observable BudgetViewModel. Then generate
BudgetBuddyApp, AppRoot, BudgetScreen, summary/form/list/row views and
deterministic previews. Use accessible TextField, Picker and Button
identifiers. Keep policy in C++. Output complete files. Verify Swift
Testing state tests, iOS 17+ build and empty/loaded/error/large-text
previews.
```

PROTECTED FILE SCOPE

BudgetEvaluating.swift; BudgetBridgeClient.swift; FakeBudgetBridge.swift; BudgetModel.swift; BudgetViewModel.swift; BudgetBuddyApp.swift; AppRoot.swift; BudgetScreen.swift; supporting views and previews

EXPECTED GREEN

Swift Testing proves parsing/state/error/retry with a fake; iOS 17+ build succeeds; four deterministic previews construct; required accessibility identifiers exist.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 6 · Evidence Contract

1 Artifact

SwiftUI entry flow, deterministic previews and validation states

K3, A3

2 Acceptance

The view remains declarative, invalid input is recoverable, and previews cover four states without production data.

K3, A3

3 Failure signal

If previews fail, inject the fixture model and remove filesystem work from initialisation.

K3, A3

4 Cleanup

Retain preview fixtures; remove screenshots containing personal data.

K3, A3

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 6 · Verification Gate

PASS WHEN

The view remains declarative, invalid input is recoverable, and previews cover four states without production data.

IF IT FAILS

If previews fail, inject the fixture model and remove filesystem work from initialisation.

LEAVE THE WORKSPACE

Retain preview fixtures; remove screenshots containing personal data.

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RETRIEVAL PRACTICE

Topic 3 · Review

1 Explain

LO3: Select essential C++ controls and features to meet software design requirements.

LO3 · K3 · A3

2 Demonstrate

Show the Activity evidence that operationalises the outcome.

LO3 · K3 · A3

3 Diagnose

Name the failure signal and the layer where it originates.

LO3 · K3 · A3

4 Trace

Connect LO3 · K3 · A3 to the assessment and evidence pack.

LO3 · K3 · A3

DESIGN JUDGEMENT

Do not advance on confidence alone; point to the artifact or test result that supports the answer.

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04

TOPIC 4 · LO4 · K4 · A4

iOS Interoperability, Persistence and Validation

LO4: Examine the interoperability and functionality of C++ software components.

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Objective-C++ owns language translation

1 Receive Swift value

.h

OBJCXX-BRIDGE

2 Convert to C++

.mm

OBJCXX-BRIDGE

3 Call domain

namespace

OBJCXX-BRIDGE

4 Return Objective-C value

NSError

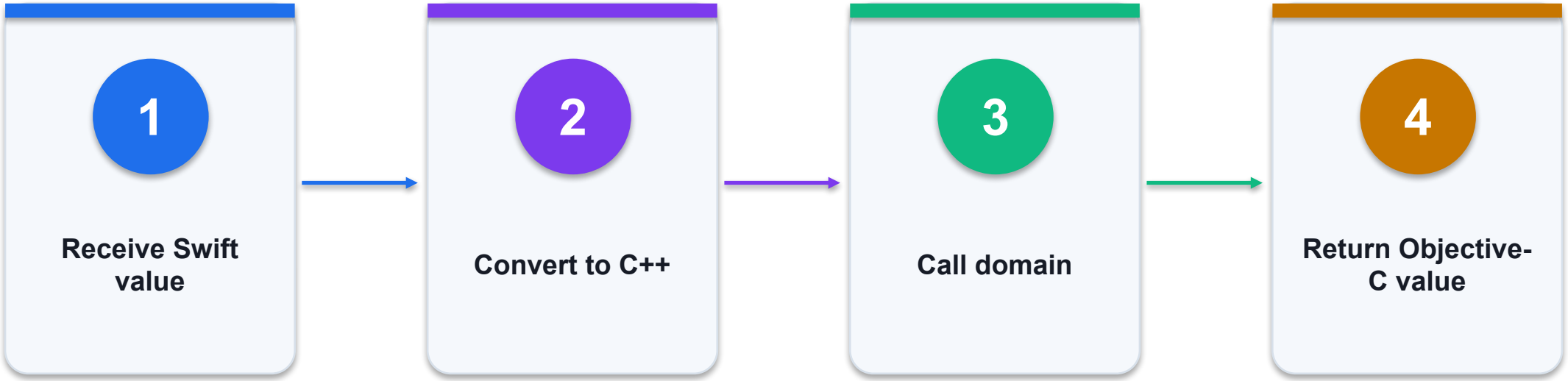
OBJCXX-BRIDGE

DESIGN JUDGEMENT

Keep the header Objective-C compatible and put C++ includes in .mm.

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Objective-C++ owns language translation · Mechanism



MECHANISM Failure mode: C++ types leak through the public Objective-C header. Control: Keep the header Objective-C compatible and put C++ includes in .mm.

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WORKED EXAMPLE

Objective-C++ owns language translation · Technical Anchor

```
// BudgetBridge.mm
auto result = core_->add(toCpp(draft));
```

1. .h
2. .mm
3. namespace
4. NSError

EVIDENCE METRIC One translation point per operation

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Ownership across ARC and RAII

1 Swift retains bridge

strong

OBJCXX-OWNERSHIP

2 Bridge owns unique_ptr

unique_ptr

OBJCXX-OWNERSHIP

3 Core owns values

value

OBJCXX-OWNERSHIP

4 Scope releases

deinit

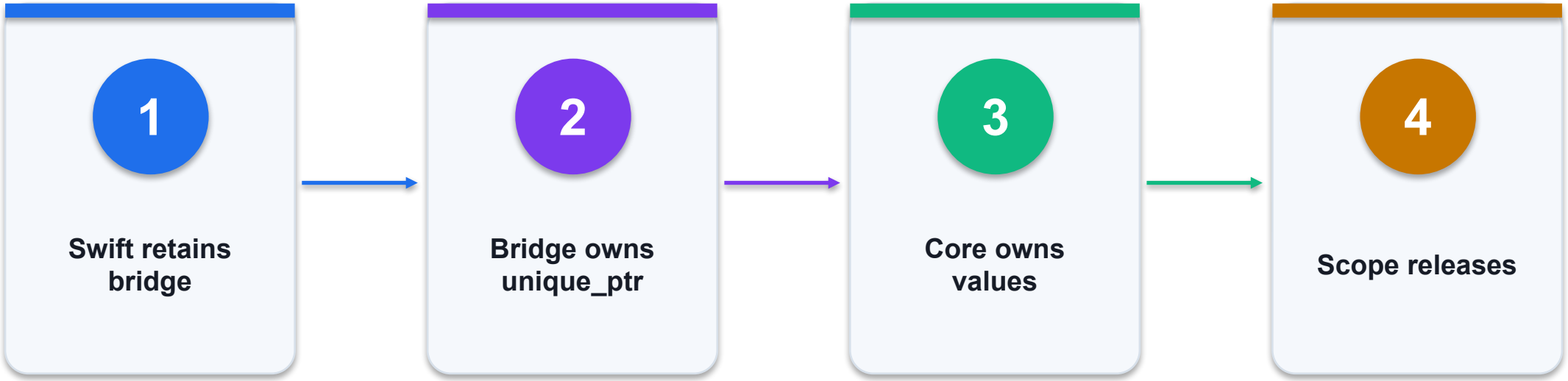
OBJCXX-OWNERSHIP

DESIGN JUDGEMENT

Let the adapter own long-lived C++ services; return copied immutable DTOs.

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Ownership across ARC and RAII · Mechanism



MECHANISM Failure mode: Swift stores a pointer to a temporary C++ object. Control: Let the adapter own long-lived C++ services; return copied immutable DTOs.

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WORKED EXAMPLE

Ownership across ARC and RAII · Technical Anchor

```
std::unique_ptr<ExpenseService> service_;
```

1. strong
2. unique_ptr
3. value
4. deinit

EVIDENCE METRIC No leaks or dangling pointers under repeated use

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Map typed errors once

1 **Receive variant**

domain code

OBJCXX-ERROR

2 **Extract error**

domain

OBJCXX-ERROR

3 **Assign code**

message

OBJCXX-ERROR

4 **Create NSError**

recovery

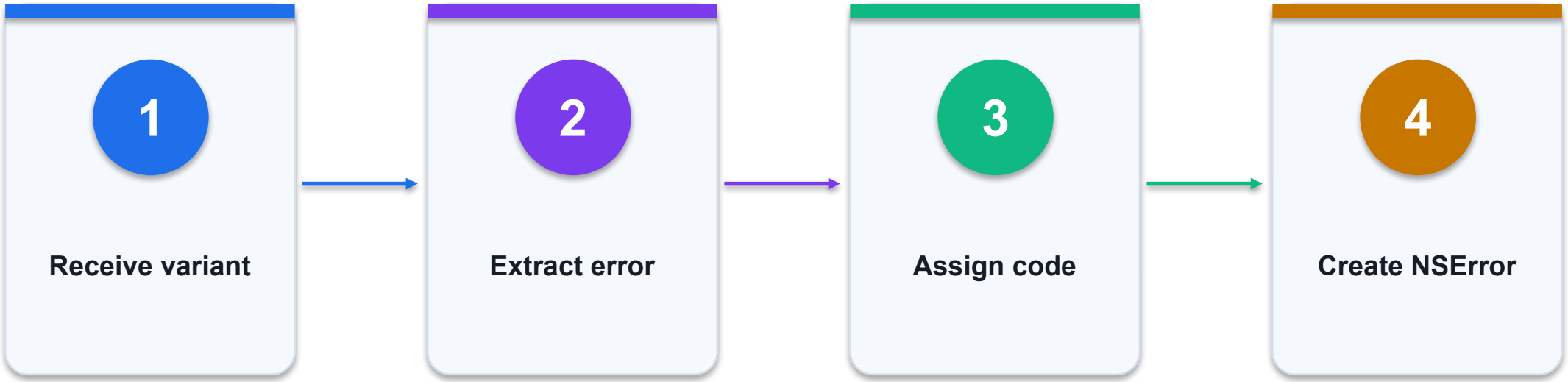
OBJCXX-ERROR

DESIGN JUDGEMENT

Centralise DomainError-to-NSError mapping in the adapter.

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Map typed errors once · Mechanism



MECHANISM

Failure mode: Every screen invents a different string for the same domain failure. Control: Centralise DomainError-to-NSError mapping in the adapter.

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WORKED EXAMPLE

Map typed errors once · Technical Anchor

```
NSError *error = [NSError errorWithDomain:BudgetDomain
code:code userInfo:info];
```

1. domain code
2. domain
3. message
4. recovery

EVIDENCE METRIC Stable error codes reach Swift tests

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DTOs freeze the boundary shape

1 **Read domain**

identity

OBJCXX-DTO

2 **Copy fields**

amount

OBJCXX-DTO

3 **Create DTO**

category

OBJCXX-DTO

4 **Consume in Swift**

timestamp

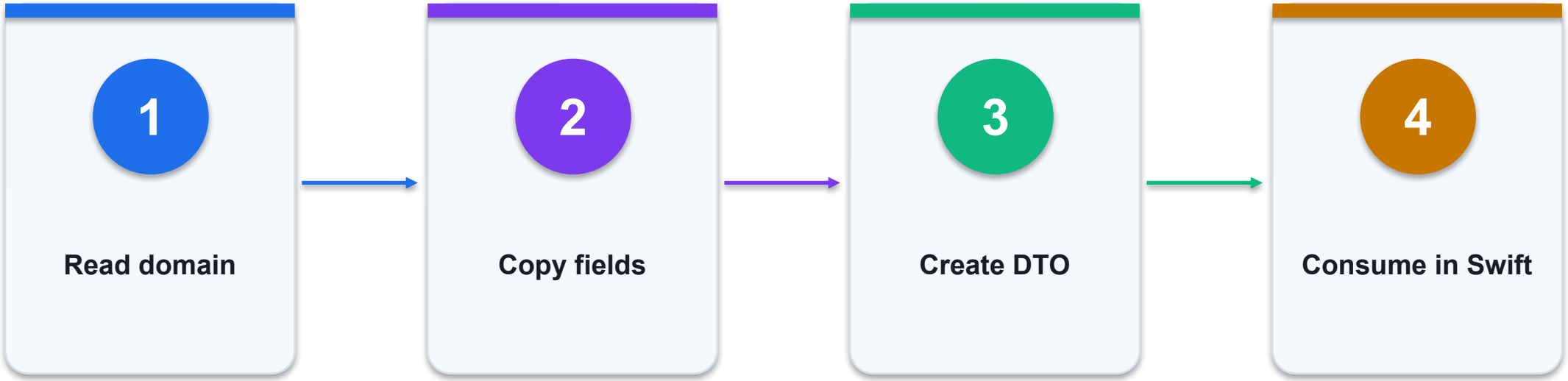
OBJCXX-DTO

DESIGN JUDGEMENT

Use explicit DTO objects with stable identity and documented units.

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DTOs freeze the boundary shape · Mechanism



MECHANISM Failure mode: Swift receives multiple arrays whose indices must stay aligned. Control: Use explicit DTO objects with stable identity and documented units.

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WORKED EXAMPLE

DTOs freeze the boundary shape · Technical Anchor

```
@interface ExpenseDTO : NSObject @property(readonly)
int64_t cents; @end
```

1. identity
2. amount
3. category
4. timestamp

EVIDENCE METRIC Swift sees immutable, complete records

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JSON persistence stays behind a port

1 Serialise snapshot

encoder

CPP-PERSIST

2 Write atomically

temp file

CPP-PERSIST

3 Read data

rename

CPP-PERSIST

4 Validate schema

schemaVersion

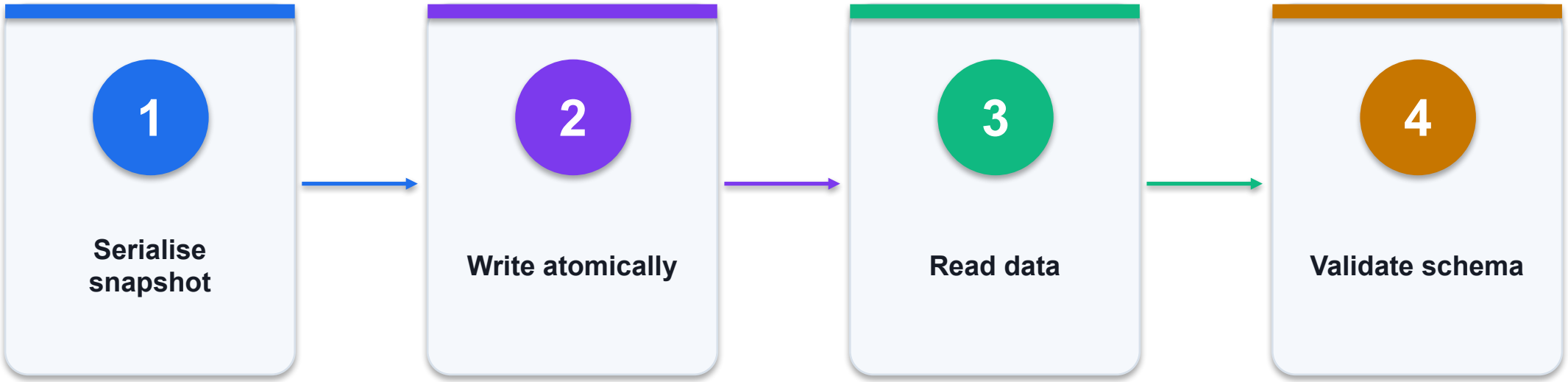
CPP-PERSIST

DESIGN JUDGEMENT

Write to a temporary file, fsync where relevant, then replace atomically.

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JSON persistence stays behind a port · Mechanism



MECHANISM **Failure mode: A partial write corrupts the only saved file. Control: Write to a temporary file, fsync where relevant, then replace atomically.**

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WORKED EXAMPLE

JSON persistence stays behind a port · Technical Anchor

```
repository.save(snapshot); // port hides JSON and
filesystem
```

1. encoder
2. temp file
3. rename
4. schemaVersion

EVIDENCE METRIC Restart reproduces last accepted state

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Schema evolution needs a version

1 **Read version**

v1

CPP-SCHEMA

2 **Select migration**

migration

CPP-SCHEMA

3 **Transform**

default

CPP-SCHEMA

4 **Persist current**

current

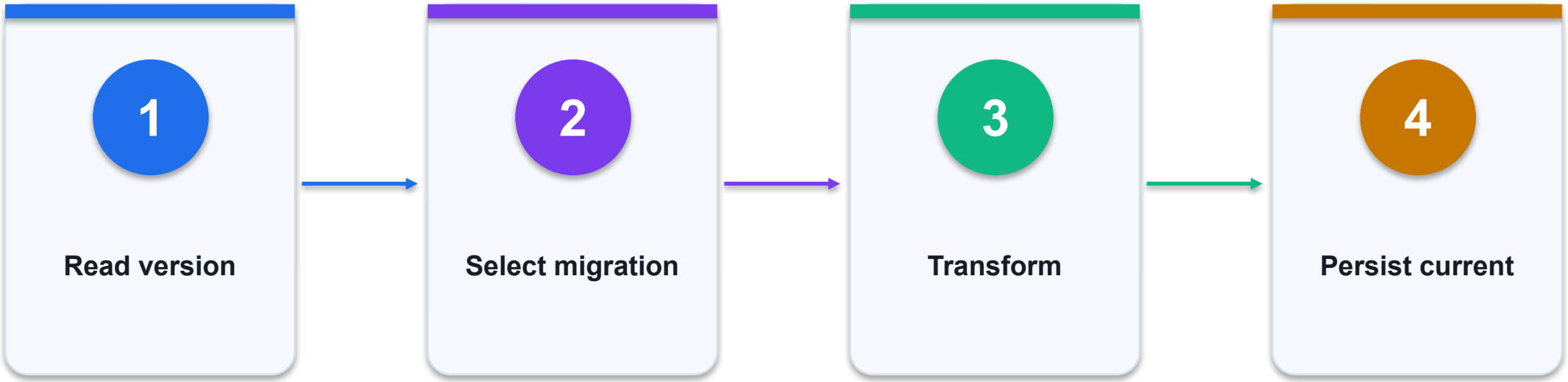
CPP-SCHEMA

DESIGN JUDGEMENT

Version persisted documents and test every supported migration path.

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Schema evolution needs a version · Mechanism



MECHANISM Failure mode: A new field makes existing user data unreadable. Control: Version persisted documents and test every supported migration path.

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WORKED EXAMPLE

Schema evolution needs a version · Technical Anchor

```
if (doc.schemaVersion == 1) return migrateV1(doc);
```

1. v1
2. migration
3. default
4. current

EVIDENCE METRIC Legacy fixtures load into current model

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Interoperability has observable indicators

1 Build both targets

compile

TSC-K4

2 Call boundary

round-trip

TSC-K4

3 Compare values

identity

TSC-K4

4 Inspect failures

error

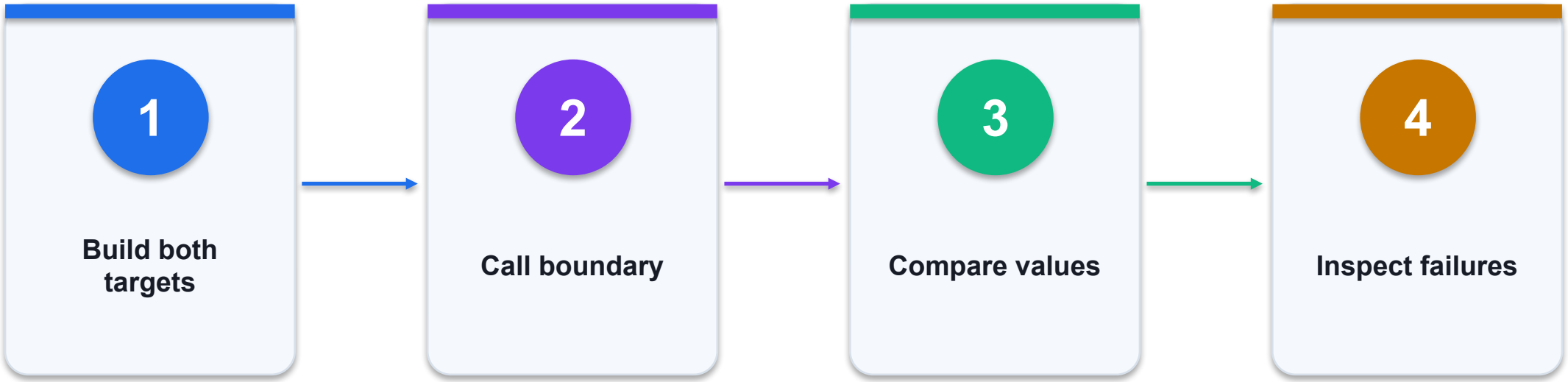
TSC-K4

DESIGN JUDGEMENT

Measure compile compatibility, value fidelity, error fidelity and lifecycle safety.

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Interoperability has observable indicators · Mechanism



MECHANISM

Failure mode: Only a happy-path simulator demo is used as evidence. Control: Measure compile compatibility, value fidelity, error fidelity and lifecycle safety.

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WORKED EXAMPLE

Interoperability has observable indicators · Technical Anchor

```
EXPECT_EQ(swiftRoundTrip.amountCents,
cppInput.amount.cents);
```

1. compile
2. round-trip
3. identity
4. error

EVIDENCE METRIC Contract tests pass for success and failure paths

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Contract tests isolate the seam

1 Arrange adapter

fixture

TEST-INTEROP

2 Send DTO

input

TEST-INTEROP

3 Invoke core

call

TEST-INTEROP

4 Assert mapped result

output

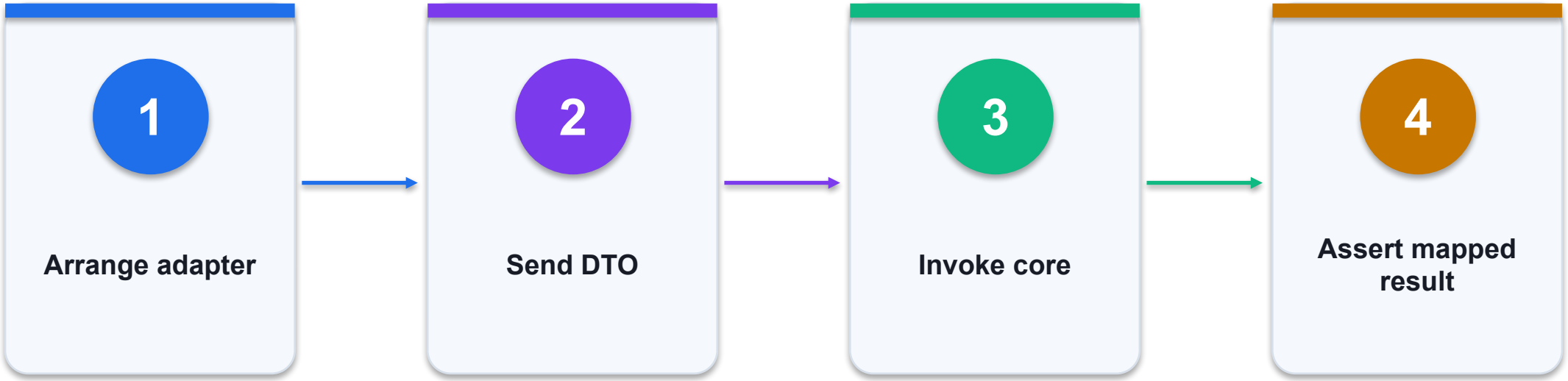
TEST-INTEROP

DESIGN JUDGEMENT

Test the translation seam directly with deterministic core doubles.

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Contract tests isolate the seam · Mechanism



MECHANISM

Failure mode: Bridge tests are postponed until full end-to-end testing. Control: Test the translation seam directly with deterministic core doubles.

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WORKED EXAMPLE

Contract tests isolate the seam · Technical Anchor

```
func testNegativeAmountMapsToStableErrorCode() throws {
... }
```

1. fixture
2. input
3. call
4. output

EVIDENCE METRIC Adapter regression detected without UI automation

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End-to-end tests prove user value

1 Launch fixture

launch arg

XCTEST-E2E

2 Enter expense

accessibility id

XCTEST-E2E

3 Save

persistence

XCTEST-E2E

4 Relaunch and verify

assertion

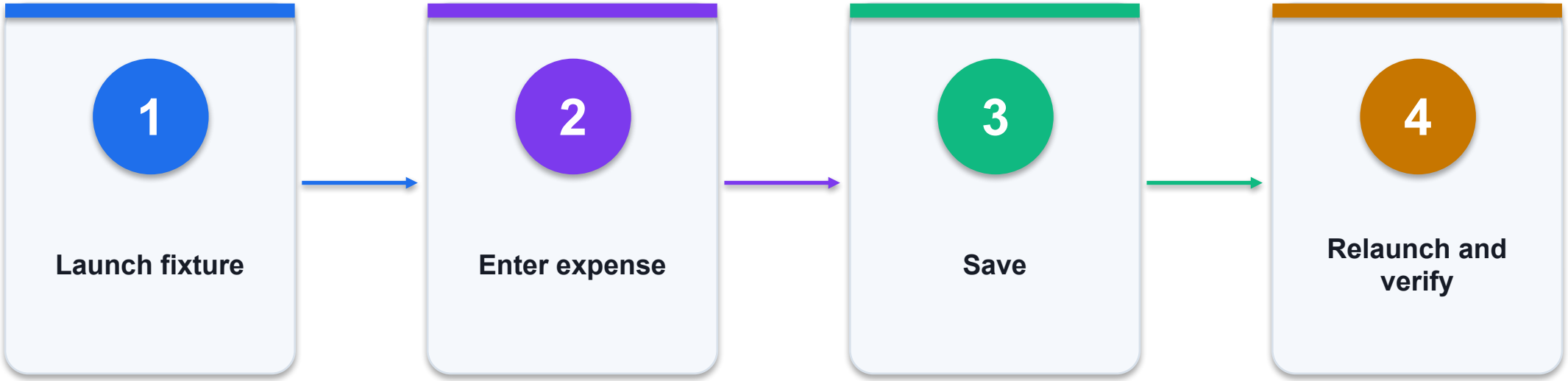
XCTEST-E2E

DESIGN JUDGEMENT

Choose a thin golden path that crosses every architectural boundary.

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End-to-end tests prove user value · Mechanism



MECHANISM Failure mode: A UI test checks labels but not the C++ decision or persistence result. Control: Choose a thin golden path that crosses every architectural boundary.

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WORKED EXAMPLE

End-to-end tests prove user value · Technical Anchor

```
app.buttons["saveExpense"].tap();
XCTAssertTrue(app.staticTexts["$12.50"].exists)
```

1. launch arg
2. accessibility id
3. persistence
4. assertion

EVIDENCE METRIC Critical journey passes from control to persisted core state

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Failure injection tests recovery

1 **Inject fault**

disk full

TEST-FAILURE

2 **Observe mapping**

invalid JSON

TEST-FAILURE

3 **Show recovery**

bridge error

TEST-FAILURE

4 **Retry safely**

retry

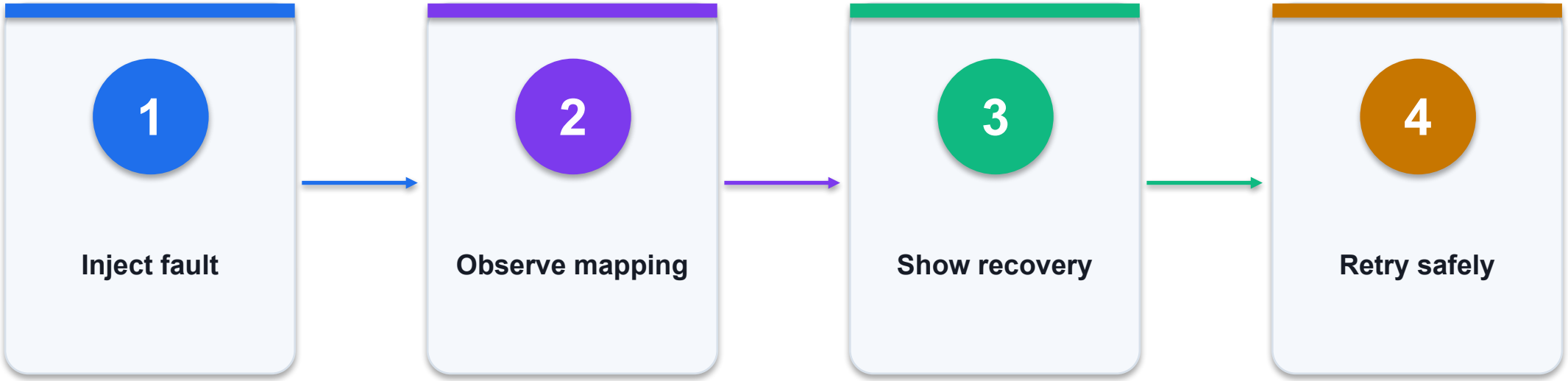
TEST-FAILURE

DESIGN JUDGEMENT

Inject controlled faults and assert state transitions plus recovery action.

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Failure injection tests recovery · Mechanism



MECHANISM

Failure mode: Errors are logged but the UI remains stuck in a loading state. Control: Inject controlled faults and assert state transitions plus recovery action.

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WORKED EXAMPLE

Failure injection tests recovery · Technical Anchor

```
repo.failNextSave = true // deterministic test double
```

1. disk full
2. invalid JSON
3. bridge error
4. retry

EVIDENCE METRIC Failure states preserve user input and offer a valid next action

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Functionality needs a scorecard

1 **Choose criteria**

correctness

TSC-A4

2 **Set threshold**

latency

TSC-A4

3 **Run suite**

accessibility

TSC-A4

4 **Review evidence**

recovery

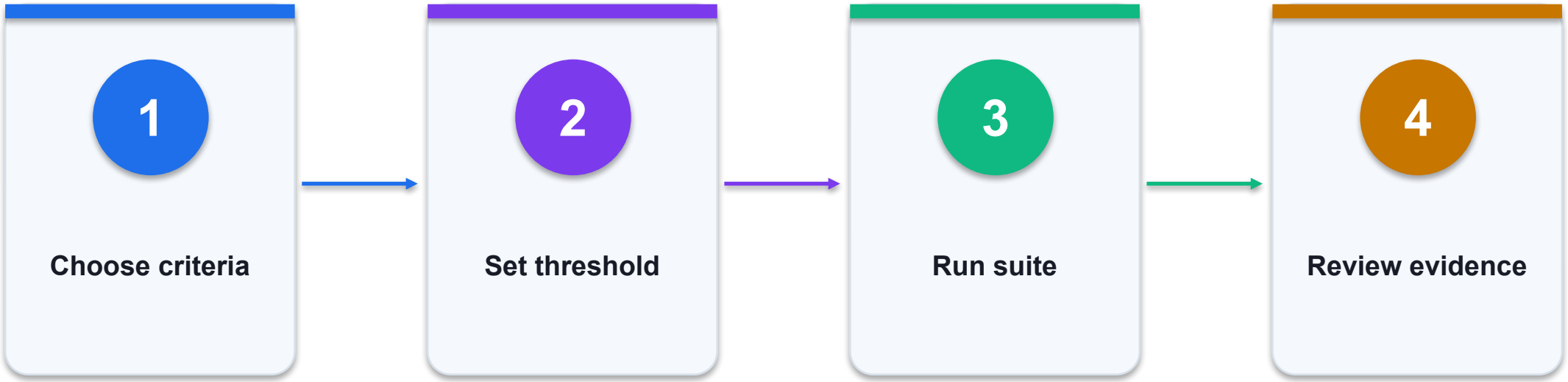
TSC-A4

DESIGN JUDGEMENT

Combine domain, bridge, UI and quality indicators into one release decision.

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Functionality needs a scorecard · Mechanism



MECHANISM **Failure mode: A build is declared ready because it compiles. Control: Combine domain, bridge, UI and quality indicators into one release decision.**

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WORKED EXAMPLE

Functionality needs a scorecard · Technical Anchor

```
ReleaseGate{.domain=1.0, .bridge=1.0, .ui=0.95, .crashes=0};
```

1. correctness
2. latency
3. accessibility
4. recovery

EVIDENCE METRIC All release gates meet declared thresholds

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SCENARIO EVIDENCE

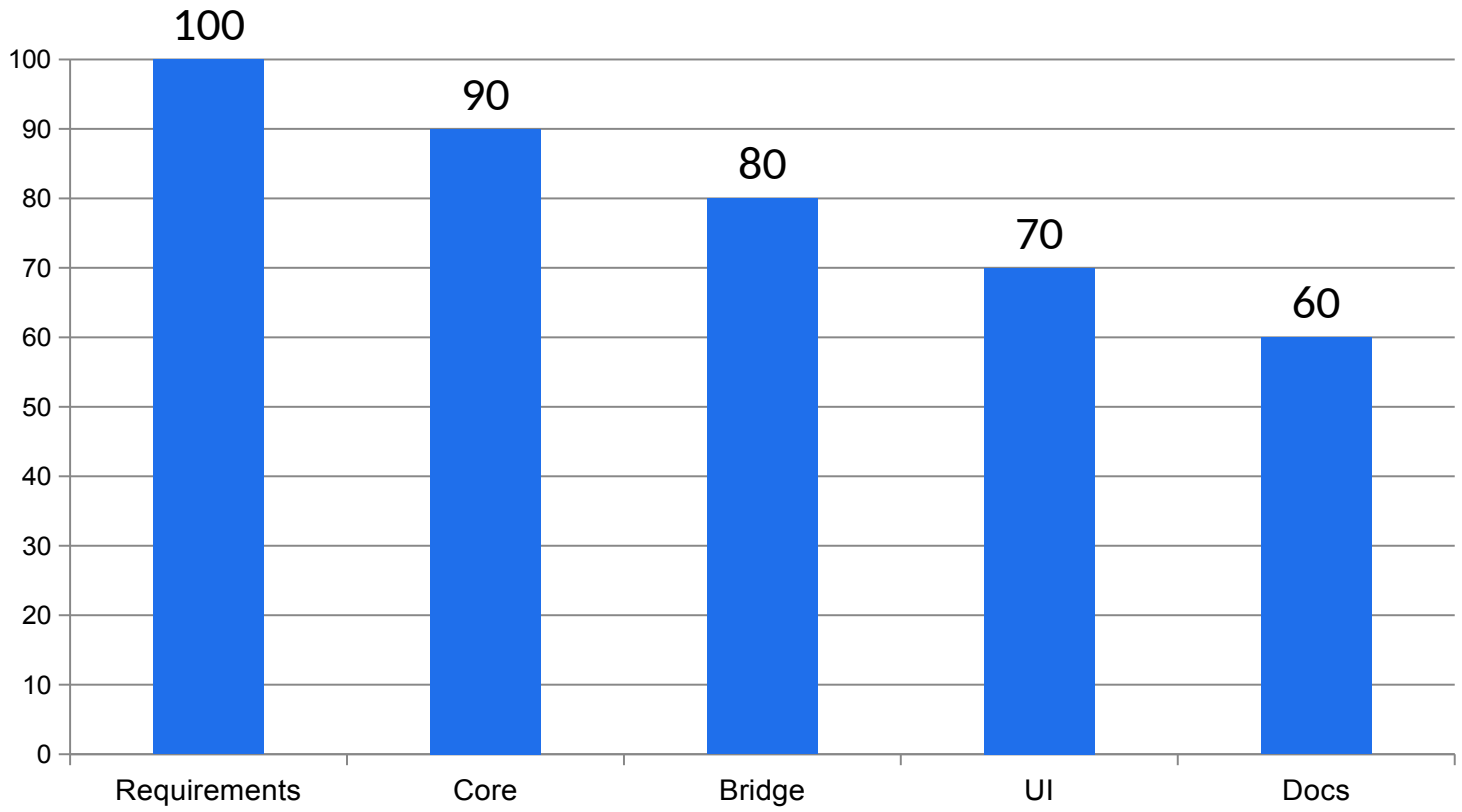
Application Validation · Layered Evidence



1. Domain rules first.
2. Bridge contract second.
3. Golden UI journey third.
4. Release decision last.

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Topic 4 · Evidence Coverage by Layer



WHAT THE DATA SHOWS

Coverage should form a connected chain. A polished UI cannot compensate for untested invariants or an undocumented seam.

Editable chart · values are instructional thresholds, not production benchmarks

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HANDS-ON ACTIVITY · 75 MIN

Activity 7: Persistence and Error Recovery

CHALLENGE

Persist domain snapshots safely and prove recovery from invalid data and write failure.

REGULATED MAPPING

K4, A4

YOU WILL PRODUCE

Versioned JSON repository, migration fixture and recovery tests

C++ filesystem, JSON fixture, CTest

Detailed procedures: Learner Guide and activities/07-persistence-error-recovery/README.md

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ACTIVITY MECHANISM

Activity 7 · Delivery Workflow

1 Frame

Implement repository save behind the ExpenseRepository port.

K4, A4

2 Build

Write to a temporary file before replacing the current snapshot.

K4, A4

3 Connect

Inject invalid JSON and a failed save through deterministic doubles.

K4, A4

4 Verify

Restart restores accepted data; corrupt input and failed writes become stable recoverable errors.

K4, A4

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 7 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate versioned atomic JSON persistence and v1-to-v2 migration with
exact Int64 cents and stable IDs. Write a temporary file before
replacement, keep tests inside a supplied temporary directory and map
corrupt/read/write failures to typed recovery. Output complete
repository, fixture, schema and test files. Verify migration,
save/reload, failed replacement and retry with CTest.
```

PROTECTED FILE SCOPE

JsonExpenseRepository.hpp/.cpp; snapshot-v1.json; persistence-schema.md; JsonExpenseRepositoryTests.cpp

EXPECTED GREEN

Clean CTest passes migration, round-trip and injected-failure tests; the previous snapshot remains readable after a failed replacement.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 7 · Evidence Contract

1 Artifact

Versioned JSON repository, migration fixture and recovery tests

K4, A4

2 Acceptance

Restart restores accepted data; corrupt input and failed writes become stable recoverable errors.

K4, A4

3 Failure signal

If tests touch the real home directory, inject a temporary path and fail the test on escape.

K4, A4

4 Cleanup

Delete only temporary test stores; retain migration fixtures.

K4, A4

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 7 · Verification Gate

PASS WHEN

Restart restores accepted data; corrupt input and failed writes become stable recoverable errors.

IF IT FAILS

If tests touch the real home directory, inject a temporary path and fail the test on escape.

LEAVE THE WORKSPACE

Delete only temporary test stores; retain migration fixtures.

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HANDS-ON ACTIVITY · 80 MIN

Activity 8: Interoperability Validation Scorecard

CHALLENGE

Assess the C++/Objective-C++/SwiftUI system with domain, bridge and end-to-end evidence.

REGULATED MAPPING

K4, A4

YOU WILL PRODUCE

Test matrix, execution evidence and release scorecard

CTest, XCTest, XCUITest, Instruments

Detailed procedures: Learner Guide and activities/08-interop-validation-scorecard/README.md

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ACTIVITY MECHANISM

Activity 8 · Delivery Workflow

1 Frame

Map each requirement to a domain, bridge or UI test level.

K4, A4

2 Build

Run exact-cent and typed-error domain tests.

K4, A4

3 Connect

Exercise the golden SwiftUI journey through save and relaunch.

K4, A4

4 Verify

Every release-gate score links to reproducible evidence and no criterion relies only on a visual demo.

K4, A4

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 8 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate the layered validation matrix. Use CTest for portable code,
Swift Testing for Swift unit/integration behaviour and XCTest with
XCUIAutomation for the visible journey. Map REQ-001..REQ-005 to exact
test names, commands, thresholds and evidence. Reject screenshot-only
proof or PASS with an empty path. Output complete matrix, scorecard and
test files.
```

PROTECTED FILE SCOPE

test-matrix.csv; scorecard.json; Swift Testing test files; BudgetBuddyUITests.swift; evidence logs

EXPECTED GREEN

Every requirement has domain, bridge/integration, or UI evidence as appropriate; all mandatory thresholds resolve to reproducible passing commands.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 8 · Evidence Contract

1 Artifact

Test matrix, execution evidence and release scorecard

K4, A4

2 Acceptance

Every release-gate score links to reproducible evidence and no criterion relies only on a visual demo.

K4, A4

3 Failure signal

If a UI failure is ambiguous, reproduce it one layer lower before changing the architecture.

K4, A4

4 Cleanup

Remove disposable run artifacts; retain the scorecard and final evidence links.

K4, A4

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 8 · Verification Gate

PASS WHEN

Every release-gate score links to reproducible evidence and no criterion relies only on a visual demo.

IF IT FAILS

If a UI failure is ambiguous, reproduce it one layer lower before changing the architecture.

LEAVE THE WORKSPACE

Remove disposable run artifacts; retain the scorecard and final evidence links.

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RETRIEVAL PRACTICE

Topic 4 · Review

1 Explain

LO4: Examine the interoperability and functionality of C++ software components.

LO4 · K4 · A4

2 Demonstrate

Show the Activity evidence that operationalises the outcome.

LO4 · K4 · A4

3 Diagnose

Name the failure signal and the layer where it originates.

LO4 · K4 · A4

4 Trace

Connect LO4 · K4 · A4 to the assessment and evidence pack.

LO4 · K4 · A4

DESIGN JUDGEMENT

Do not advance on confidence alone; point to the artifact or test result that supports the answer.

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05

TOPIC 5 · LO5 · K5 · A5

Architecture Evidence, Documentation and Deployment

LO5: Generate C++ design documentation aligned with user specifications.

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C4 views explain different questions

1 Context

people

DOC-C4

2 Container

deployables

DOC-C4

3 Component

responsibilities

DOC-C4

4 Code

interfaces

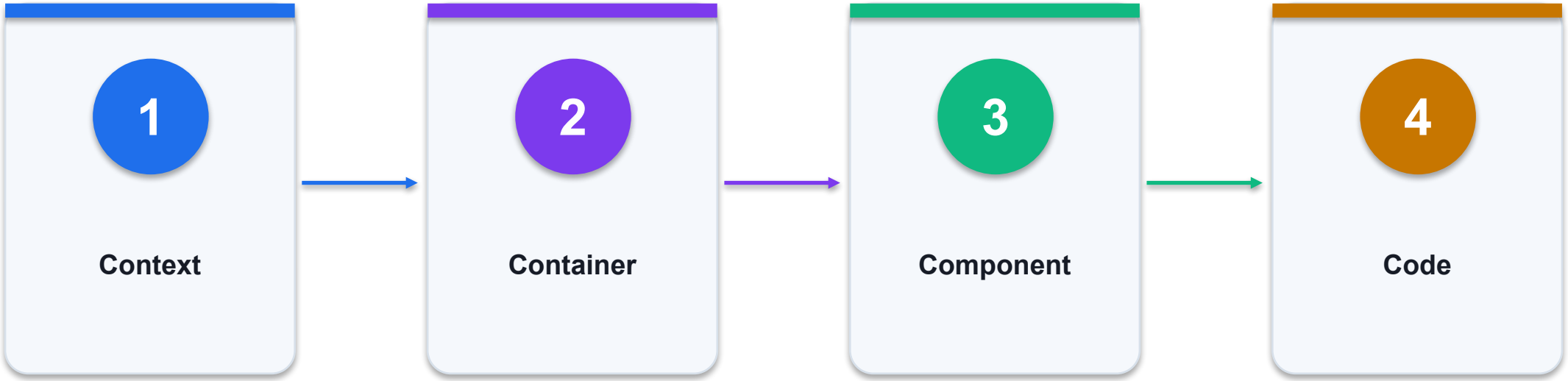
DOC-C4

DESIGN JUDGEMENT

Use layered diagrams and state the audience and decision for each.

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C4 views explain different questions · Mechanism



MECHANISM **Failure mode: A single dense diagram mixes business actors with classes. Control: Use layered diagrams and state the audience and decision for each.**

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WORKED EXAMPLE

C4 views explain different questions · Technical Anchor

```
Context -> iOS App -> SwiftUI | Objective-C++ | C++ Core
```

1. people
2. deployables
3. responsibilities
4. interfaces

EVIDENCE METRIC Stakeholder question answered at correct zoom

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ADRs preserve design reasoning

1 **Name decision**

status

DOC-ADR

2 **Record context**

context

DOC-ADR

3 **Compare options**

decision

DOC-ADR

4 **State consequences**

consequence

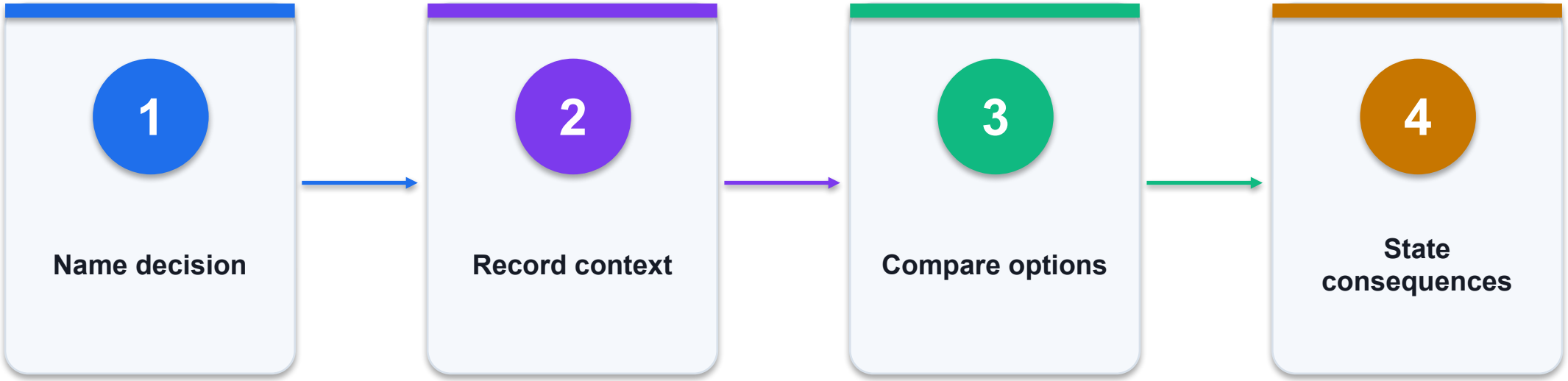
DOC-ADR

DESIGN JUDGEMENT

Write a short ADR for boundaries, persistence and AI usage.

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ADRs preserve design reasoning · Mechanism



MECHANISM

Failure mode: Documentation states what exists but not why alternatives were rejected. Control: Write a short ADR for boundaries, persistence and AI usage.

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WORKED EXAMPLE

ADRs preserve design reasoning · Technical Anchor

```
# ADR-002: Use Objective-C++ as the sole C++
translation boundary
```

1. **status**
2. **context**
3. **decision**
4. **consequence**

EVIDENCE METRIC Material choices have retrievable rationale

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Traceability closes the specification loop

1 **Assign IDs**

REQ

TSC-A5

2 **Link design**

component

TSC-A5

3 **Link test**

test

TSC-A5

4 **Report status**

evidence

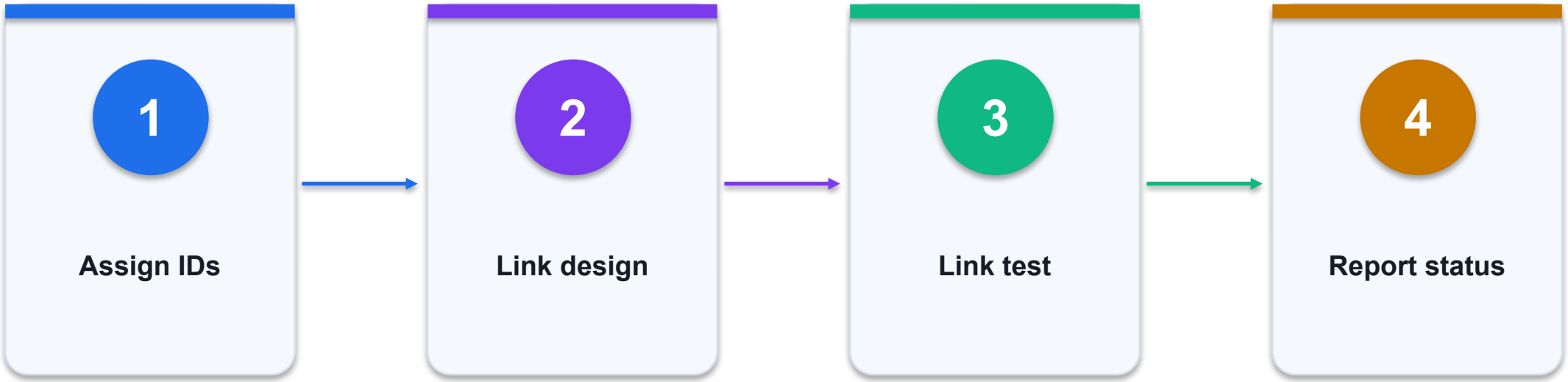
TSC-A5

DESIGN JUDGEMENT

Maintain a bidirectional matrix from requirement to implementation and test.

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Traceability closes the specification loop · Mechanism



MECHANISM

Failure mode: A feature is marked done without proving the approved requirement. Control: Maintain a bidirectional matrix from requirement to implementation and test.

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WORKED EXAMPLE

Traceability closes the specification loop · Technical Anchor

```
REQ-004 | BudgetPolicy |
BudgetPolicyTests/RejectsNegative | PASS
```

1. REQ
2. component
3. test
4. evidence

EVIDENCE METRIC Requirements with design and passing evidence

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API documentation states contracts

1 **Describe purpose**

precondition

DOC-API

2 **Name parameters**

unit

DOC-API

3 **Define errors**

ownership

DOC-API

4 **Give example**

error

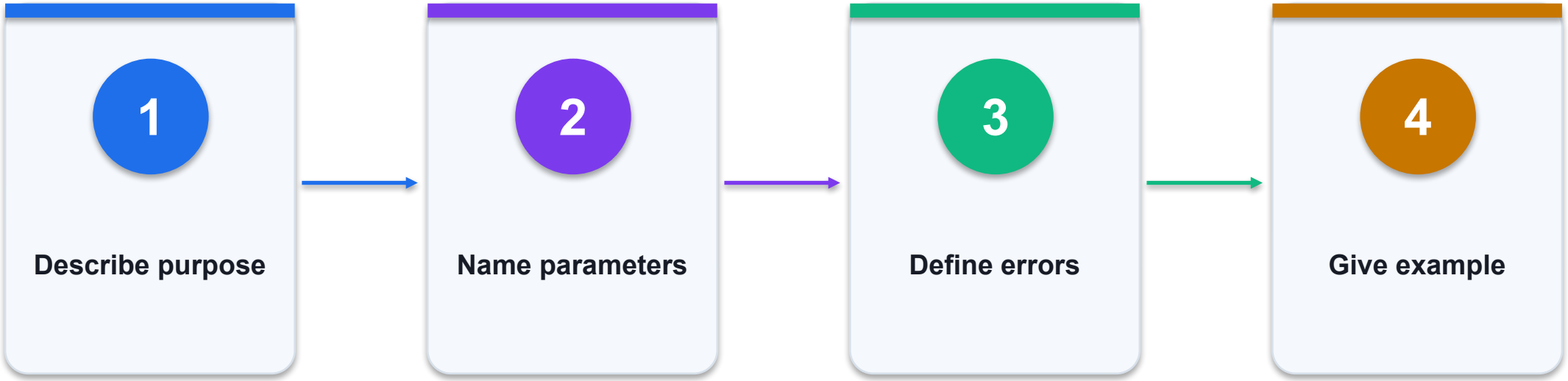
DOC-API

DESIGN JUDGEMENT

Document invariants, units, lifetime and failure semantics.

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API documentation states contracts · Mechanism



MECHANISM

Failure mode: Generated comments repeat the method name and omit units or ownership. Control: Document invariants, units, lifetime and failure semantics.

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WORKED EXAMPLE

API documentation states contracts · Technical Anchor

```
/// Evaluates a draft. Amount is cents; returns typed
error; no mutation.
```

1. precondition
2. unit
3. ownership
4. error

EVIDENCE METRIC Consumers implement without reading source

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Architecture diagrams must match code

1 Generate inventory

target

DOC-VERIFY

2 Compare dependencies

module

DOC-VERIFY

3 Update diagram

edge

DOC-VERIFY

4 Review diff

owner

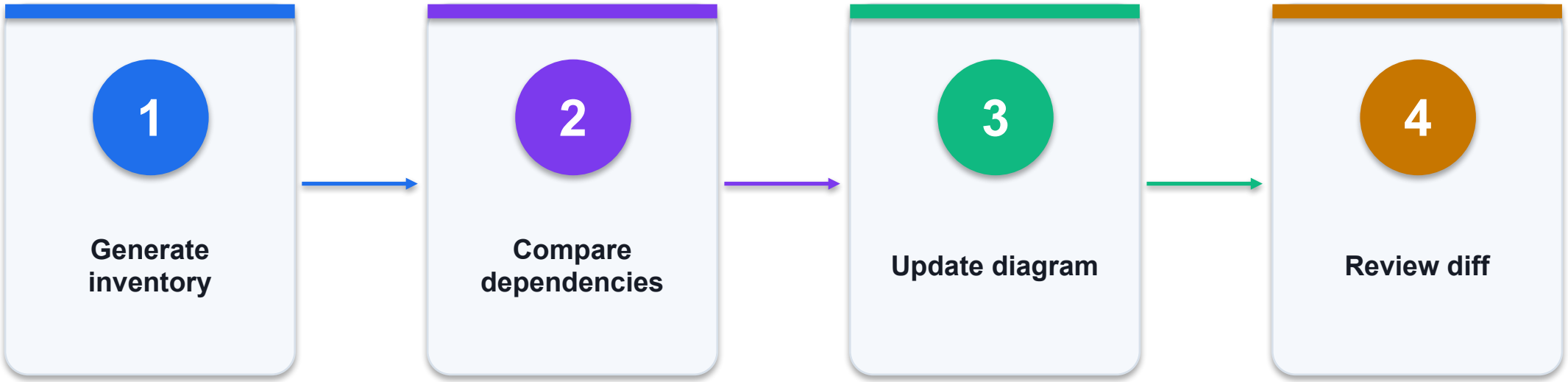
DOC-VERIFY

DESIGN JUDGEMENT

Validate diagrams against target membership and imports before release.

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Architecture diagrams must match code · Mechanism



MECHANISM **Failure mode: A polished diagram describes an architecture the build no longer uses. Control: Validate diagrams against target membership and imports before release.**

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WORKED EXAMPLE

Architecture diagrams must match code · Technical Anchor

```
SwiftUI -> BudgetBridge -> DomainCore; no reverse
dependency
```

1. target
2. module
3. edge
4. owner

EVIDENCE METRIC Zero undocumented or impossible dependency arrows

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Documentation is a tested deliverable

1 **Build docs**

README

DOC-QA

2 **Check links**

links

DOC-QA

3 **Run examples**

commands

DOC-QA

4 **Review audience**

roles

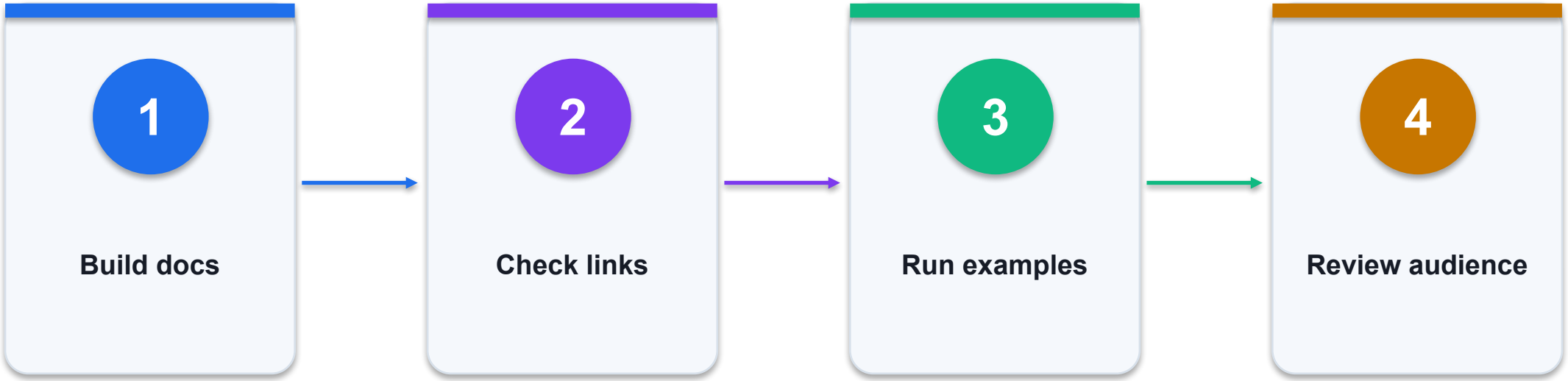
DOC-QA

DESIGN JUDGEMENT

Exercise documentation in a clean environment and capture the transcript.

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Documentation is a tested deliverable · Mechanism



MECHANISM

Failure mode: Setup steps depend on an unstated local path or secret. Control: Exercise documentation in a clean environment and capture the transcript.

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WORKED EXAMPLE

Documentation is a tested deliverable · Technical Anchor

```
cmake -S DomainCore -B build && cmake --build build &&
ctest --test-dir build
```

1. README
2. links
3. commands
4. roles

EVIDENCE METRIC New developer reaches a passing test from clean checkout

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Deployment separates signing from design

1 **Choose identity**

bundle id

IOS-DEPLOY

2 **Create archive**

certificate

IOS-DEPLOY

3 **Validate build**

archive

IOS-DEPLOY

4 **Prepare release**

metadata

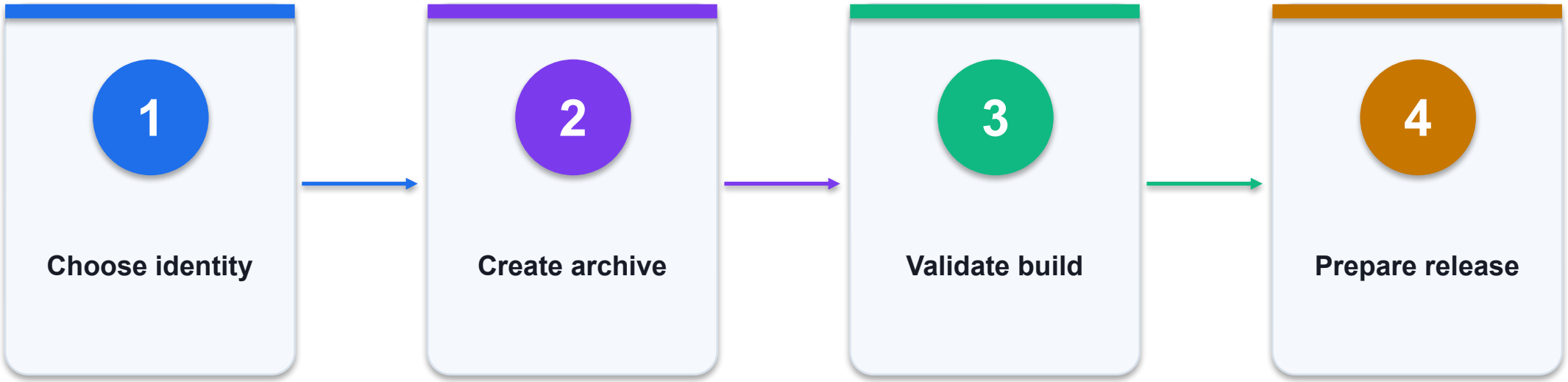
IOS-DEPLOY

DESIGN JUDGEMENT

Treat signing, archive validation and store submission as separate evidence gates.

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Deployment separates signing from design · Mechanism



MECHANISM

Failure mode: A local simulator run is described as App Store readiness. Control: Treat signing, archive validation and store submission as separate evidence gates.

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WORKED EXAMPLE

Deployment separates signing from design · Technical Anchor

```
xcodebuild -scheme BudgetBridgeApp -configuration
Release archive
```

1. bundle id
2. certificate
3. archive
4. metadata

EVIDENCE METRIC Validated archive with reproducible design evidence

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Privacy manifests document data use

1 **Inventory APIs**

API

IOS-PRIVACY

2 **Classify data**

data

IOS-PRIVACY

3 **Declare reasons**

purpose

IOS-PRIVACY

4 **Verify package**

manifest

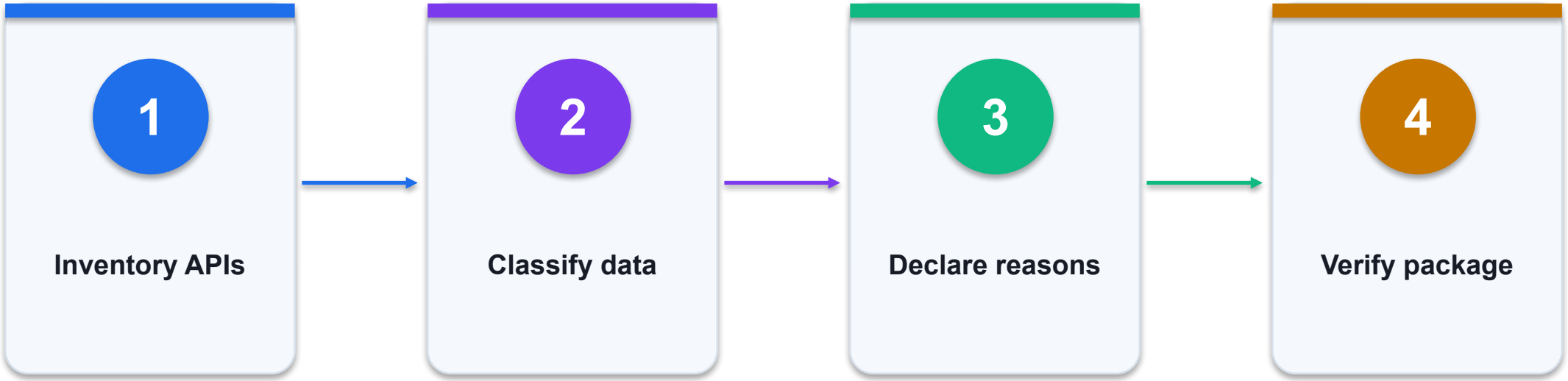
IOS-PRIVACY

DESIGN JUDGEMENT

Re-run privacy inventory whenever dependencies or telemetry change.

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Privacy manifests document data use · Mechanism



MECHANISM **Failure mode: An AI-added SDK expands collection without documentation review. Control: Re-run privacy inventory whenever dependencies or telemetry change.**

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WORKED EXAMPLE

Privacy manifests document data use · Technical Anchor

```
PrivacyInfo.xcprivacy // reviewed against linked
frameworks
```

1. API
2. data
3. purpose
4. manifest

EVIDENCE METRIC Declared use matches code and third-party packages

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AI review needs accountable evidence

1 **Request review**

prompt

AI-REVIEW

2 **Verify claim**

evidence

AI-REVIEW

3 **Classify finding**

severity

AI-REVIEW

4 **Resolve or accept**

owner

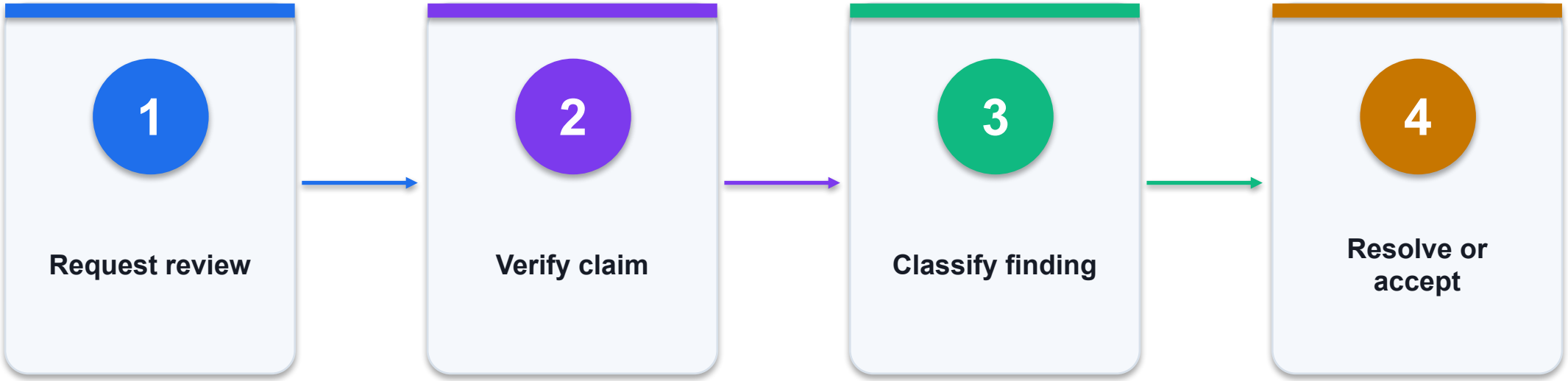
AI-REVIEW

DESIGN JUDGEMENT

Require code locus and a repeatable test for material findings.

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AI review needs accountable evidence · Mechanism



MECHANISM

Failure mode: AI review comments are copied into approval without reproduction. Control: Require code locus and a repeatable test for material findings.

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WORKED EXAMPLE

AI review needs accountable evidence · Technical Anchor

```
Finding: cents overflow. Evidence: boundary test with
Int.max. Owner: developer.
```

1. prompt
2. evidence
3. severity
4. owner

EVIDENCE METRIC Accepted findings have owner and rationale

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Release notes map value to change

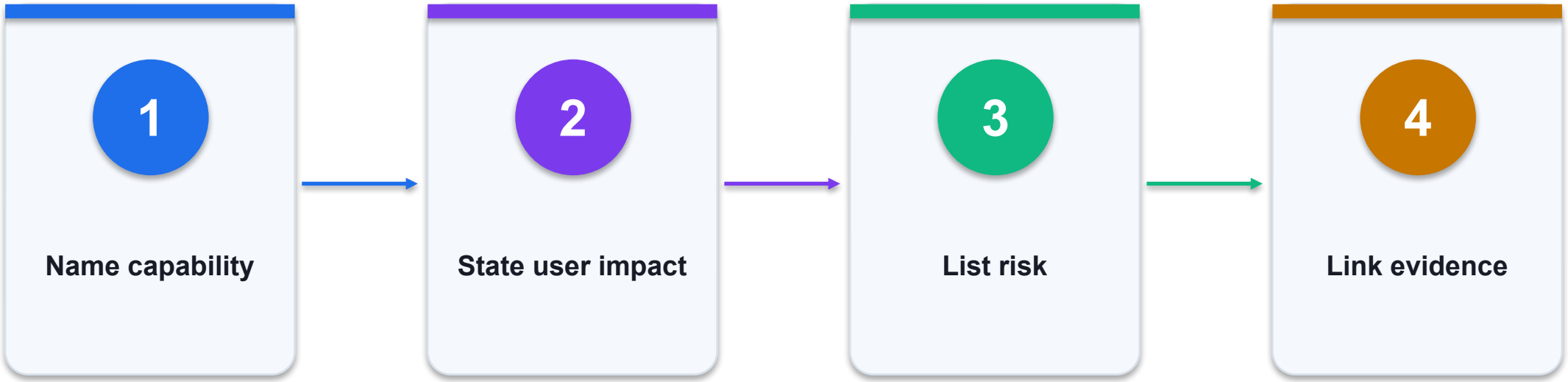
1. **Name capability**
   feature
   DOC-RELEASE
2. **State user impact**
   impact
   DOC-RELEASE
3. **List risk**
   known issue
   DOC-RELEASE
4. **Link evidence**
   verification
   DOC-RELEASE

DESIGN JUDGEMENT

Write outcome-oriented notes and disclose known limitations.

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Release notes map value to change · Mechanism



MECHANISM

Failure mode: Release notes list commits but not observable user outcomes. Control: Write outcome-oriented notes and disclose known limitations.

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WORKED EXAMPLE

Release notes map value to change · Technical Anchor

```
Added exact-cent budgeting; validated restart
persistence and VoiceOver flow.
```

1. feature
2. impact
3. known issue
4. verification

EVIDENCE METRIC Every release claim points to a checked artifact

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The evidence pack supports handover

1 **Index artifacts**

design

TSC-K5

2 **Record versions**

tests

TSC-K5

3 **Assign owners**

deployment

TSC-K5

4 **Set review date**

operations

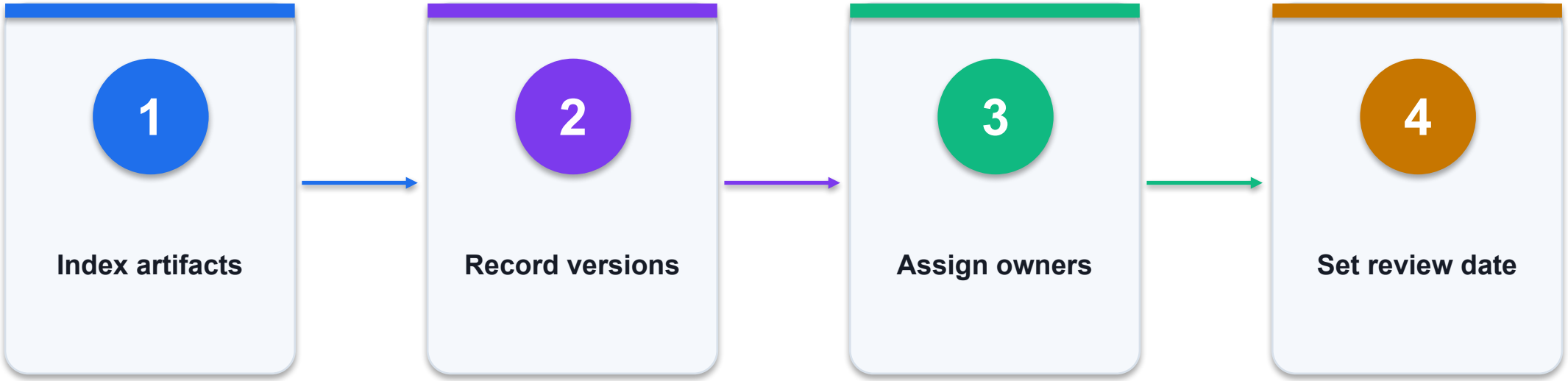
TSC-K5

DESIGN JUDGEMENT

Package requirements, design, tests, risks and deployment evidence with provenance.

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The evidence pack supports handover · Mechanism



MECHANISM

Failure mode: Evidence is scattered across local screenshots and chat history. Control: Package requirements, design, tests, risks and deployment evidence with provenance.

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WORKED EXAMPLE

The evidence pack supports handover · Technical Anchor

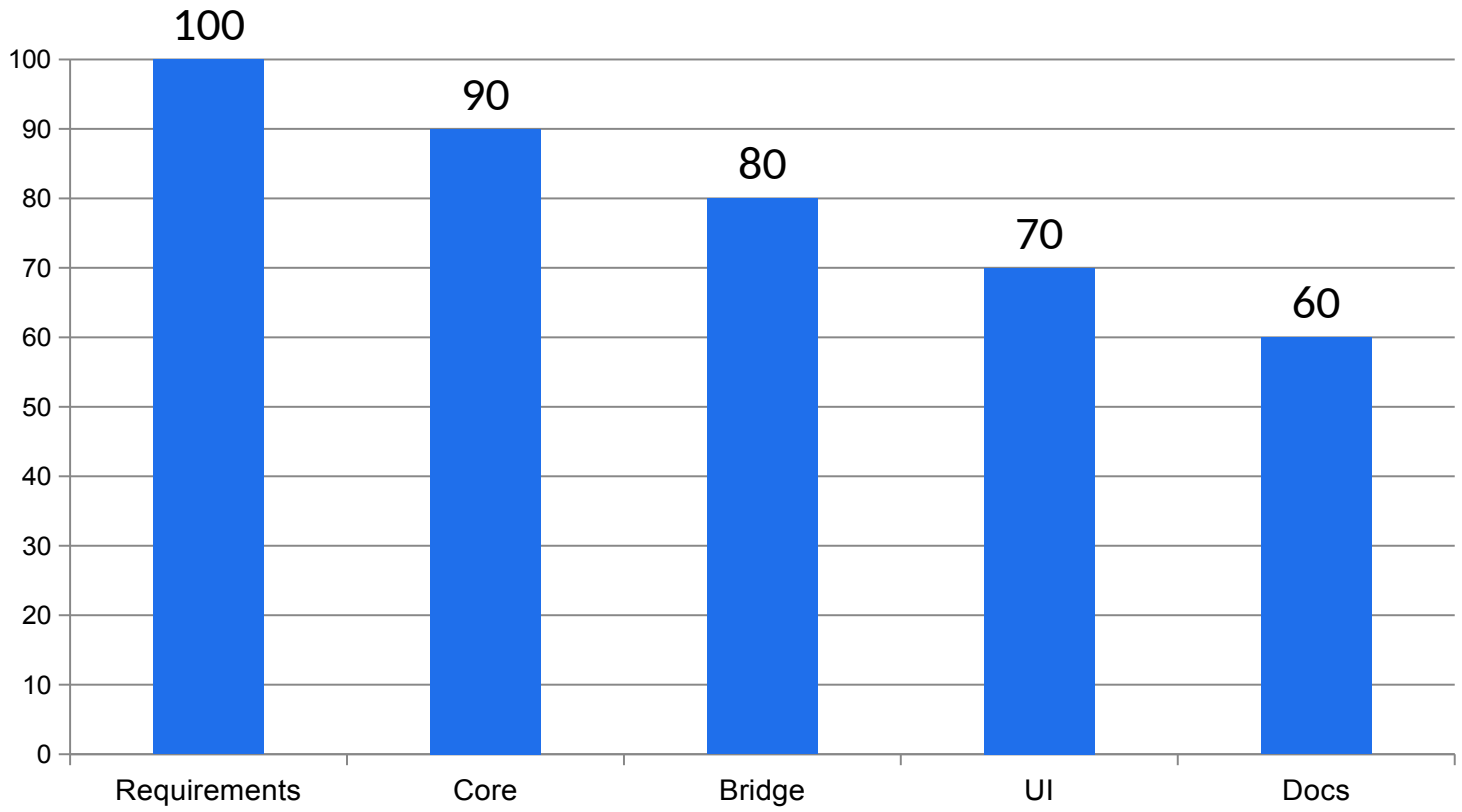
```
EvidencePack v1.2: requirements, ADRs, tests,
validation, release checklist
```

1. design
2. tests
3. deployment
4. operations

EVIDENCE METRIC A reviewer can reproduce the release decision

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Topic 5 · Evidence Coverage by Layer



WHAT THE DATA SHOWS

Coverage should form a connected chain. A polished UI cannot compensate for untested invariants or an undocumented seam.

Editable chart · values are instructional thresholds, not production benchmarks

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HANDS-ON ACTIVITY · 75 MIN

Activity 9: Architecture and Traceability Documentation

CHALLENGE

Create design documentation that maps user specifications to architecture, interfaces and tests.

REGULATED MAPPING

K5, A5

YOU WILL PRODUCE

C4 diagrams, two ADRs, API contract and traceability matrix

Markdown, Mermaid or editable shapes, Doxygen-style comments

Detailed procedures: Learner Guide and activities/09-architecture-traceability-docs/README.md

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ACTIVITY MECHANISM

Activity 9 · Delivery Workflow

1 Frame

Create context, container and component views at distinct levels of detail.

K5, A5

2 Build

Write ADRs for the Objective-C++ boundary and JSON persistence.

K5, A5

3 Connect

Build a requirement-to-component-to-test-to-evidence matrix.

K5, A5

4 Verify

All five regulated outcomes are traceable and no documented dependency contradicts the build graph.

K5, A5

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 9 · Vibe Coding Generation Prompt

GENERATION PROMPT

```
Generate C4 views, ADR-001 bridge boundary, ADR-002 persistence, bridge
API contract and traceability.csv from the verified build graph.
Document units, ownership, nullability and typed errors. Do not invent a
module, test or evidence path. Output complete documents. Reconcile
every dependency arrow and require implementation, test, evidence and
status for REQ-001..REQ-005.
```

PROTECTED FILE SCOPE

architecture.md; ADR-001-objective-cpp-boundary.md; ADR-002-json-persistence.md; bridge-contract.md; traceability.csv; README.md

EXPECTED GREEN

All REQ-001..REQ-005 rows have real implementation, test, evidence, and status; architecture imports match the inward dependency rule.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 9 · Evidence Contract

1 Artifact

C4 diagrams, two ADRs, API contract and traceability matrix

K5, A5

2 Acceptance

All five regulated outcomes are traceable and no documented dependency contradicts the build graph.

K5, A5

3 Failure signal

If the diagram is crowded, split by stakeholder question instead of shrinking labels.

K5, A5

4 Cleanup

Retain source diagrams and Markdown; remove obsolete exported drafts.

K5, A5

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 9 · Verification Gate

PASS WHEN

All five regulated outcomes are traceable and no documented dependency contradicts the build graph.

IF IT FAILS

If the diagram is crowded, split by stakeholder question instead of shrinking labels.

LEAVE THE WORKSPACE

Retain source diagrams and Markdown; remove obsolete exported drafts.

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HANDS-ON ACTIVITY · 105 MIN

Activity 10: Capstone: BudgetBuddy Release Evidence

CHALLENGE

Integrate the portable C++ core, Objective-C++ adapter and SwiftUI shell into a verified release evidence pack.

REGULATED MAPPING

A1, A2, A3, A4, A5

YOU WILL PRODUCE

Working app, complete tests, architecture pack and deployment checklist

C++20, CMake, Xcode, SwiftUI, XCTest, documentation templates

Detailed procedures: Learner Guide and activities/10-capstone-release-evidence/README.md

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ACTIVITY MECHANISM

Activity 10 · Delivery Workflow

1 Frame

Confirm the approved requirement baseline and component boundary.

A1, A2, A3, A4, A5

2 Build

Build and test the portable C++ domain core independently.

A1, A2, A3, A4, A5

3 Connect

Complete the accessible SwiftUI flow and persistence recovery path.

A1, A2, A3, A4, A5

4 Verify

A clean evidence index proves requirements, implementation, interoperability, documentation and deployment readiness without claiming App Store submission.

A1, A2, A3, A4, A5

DESIGN JUDGEMENT

The slide communicates the workflow; the Learner Guide carries the detailed procedure and commands.

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COPY / PASTE PROMPT · BUDGETBUDDY

Activity 10 · Final Project Audit Prompt

GENERATION PROMPT

```
Verify BudgetBuddy, FocusCards and HabitPulse from project.yml: build,
Swift Testing, XCUIAutomation, launch and inspect each AppIcon. Prove
BudgetBuddy exact cents through Objective-C++. Prepare ASC identity,
archive/export/upload, metadata, screenshots, privacy, age,
availability, attach-build, submit and readback gates. Never expose
secrets or claim external execution; leave trainer-only credentialled
gates HOLD.
```

PROTECTED FILE SCOPE

verify-local.sh; CI workflow; privacy-security-review.md; evidence-index.csv; release-scorecard.json; release-notes.md; deployment-position.md; file-inventory.txt; final-audit.md

EXPECTED GREEN

All three generated Xcode projects build, test and launch on a resolved iPhone Simulator; BudgetBuddy proves exact cents through Objective-C++; icons and privacy manifests are present; local evidence is complete; every unexecuted archive/upload/metadata/privacy/age/availability/attach/submit/readback gate remains HOLD.

REVIEW / REPAIR

Use the paired review prompt in the Activity guide. Fix the first causal failure, preserve tests, then rerun focused and full checks.

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VERIFY BEFORE MOVING ON

Activity 10 · Evidence Contract

1 Artifact

Working app, complete tests, architecture pack and deployment checklist

A1, A2, A3, A4, A5

2 Acceptance

A clean evidence index proves requirements, implementation, interoperability, documentation and deployment readiness without claiming App Store submission.

A1, A2, A3, A4, A5

3 Failure signal

When a system test fails, localise the fault by rerunning domain then bridge tests before editing UI code.

A1, A2, A3, A4, A5

4 Cleanup

Remove local build caches and secrets; retain source, fixtures, tests and evidence pack.

A1, A2, A3, A4, A5

DESIGN JUDGEMENT

Evidence is a retrievable artifact or test result, not a statement that the activity was completed.

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TEST IT

Activity 10 · Verification Gate

PASS WHEN

A clean evidence index proves requirements, implementation, interoperability, documentation and deployment readiness without claiming App Store submission.

IF IT FAILS

When a system test fails, localise the fault by rerunning domain then bridge tests before editing UI code.

LEAVE THE WORKSPACE

Remove local build caches and secrets; retain source, fixtures, tests and evidence pack.

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RETRIEVAL PRACTICE

Topic 5 · Review

1 Explain

LO5: Generate C++ design documentation aligned with user specifications.

LO5 · K5 · A5

2 Demonstrate

Show the Activity evidence that operationalises the outcome.

LO5 · K5 · A5

3 Diagnose

Name the failure signal and the layer where it originates.

LO5 · K5 · A5

4 Trace

Connect LO5 · K5 · A5 to the assessment and evidence pack.

LO5 · K5 · A5

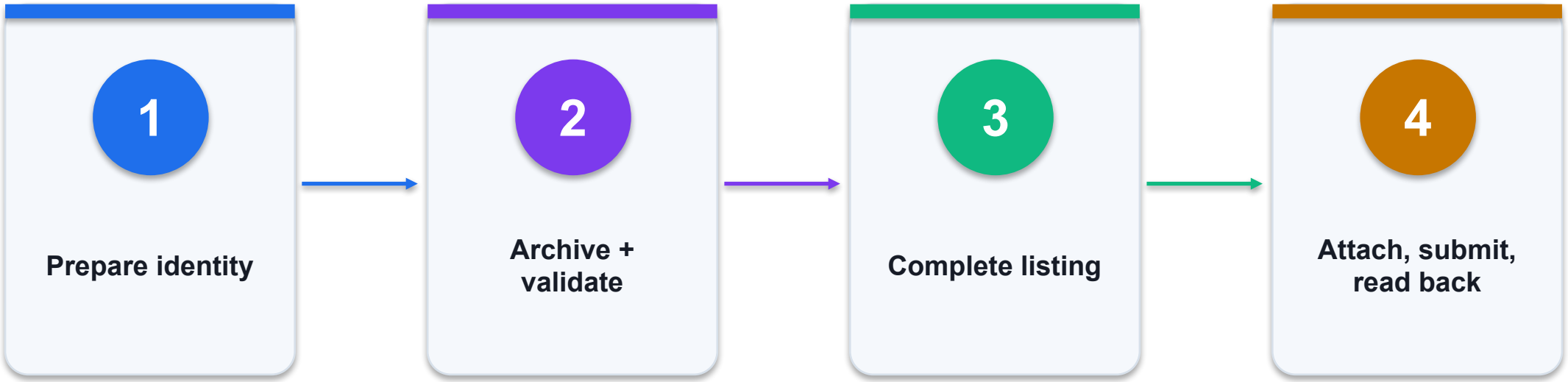
DESIGN JUDGEMENT

Do not advance on confidence alone; point to the artifact or test result that supports the answer.

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TRAINER DEMO / FUTURE CREDENTIALLED GATE

App Store Connect · Evidence Gates, Not a Submission Claim



MECHANISM Local PASS covers XcodeGen, build, tests, Simulator and reviewed assets. Signing, upload, metadata, screenshots, privacy, age rating, availability, build attachment and submission remain HOLD until executed and read back by an authorised trainer.

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COPY / ADAPT · KEEP EXTERNAL GATES HOLD

App Store Connect · Bounded Release Prompt

```
Audit the selected app identity, version/build, Release settings,
opaque 1024px icon, privacy manifest, required-reason APIs and test
evidence. Prepare complete archive/export/upload commands and
checklists for metadata, real-app screenshots, App Privacy, age
rating, rights, price and explicit availability. Then define
processed-build attachment, submission and Apple-side readback
evidence. Never invent identifiers or credentials; never claim a
gated action ran. Mark signing, upload and real ASC mutations
TRAINER DEMO / FUTURE CREDENTIALLED GATE and HOLD until read back.
```

LOCAL PASS

Generate · build · Swift Testing · XCTest UI · Simulator · icon/privacy inspection

EXTERNAL HOLD

Sign · archive · upload · listing · privacy · age · availability · submit

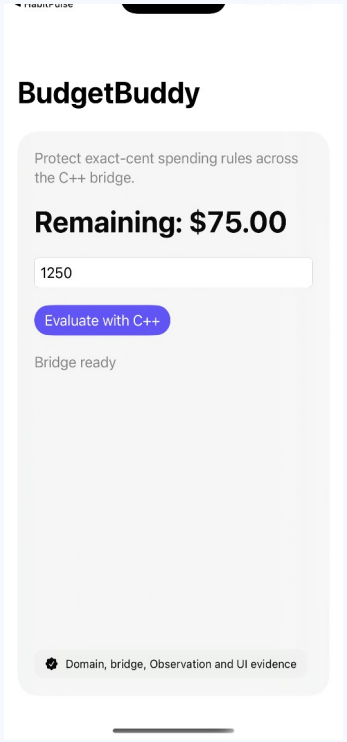
DONE ONLY WHEN

Identity, build, availability and review state are read back from Apple

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REAL RUNTIME EVIDENCE · XCODE 26.6

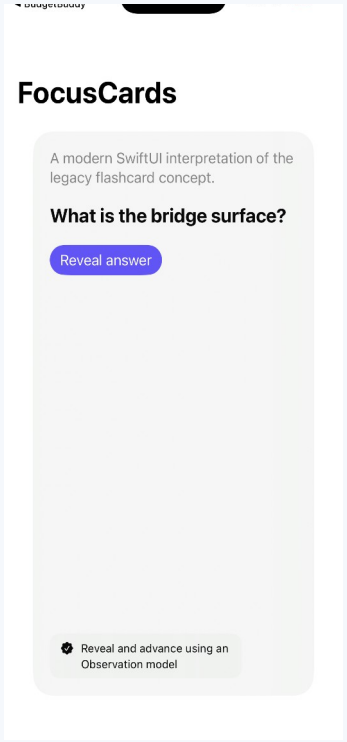
Three Apps · Proven in the iPhone Simulator



BudgetBuddy

Exact-cent save routes through Objective-C++ into the C++ policy.

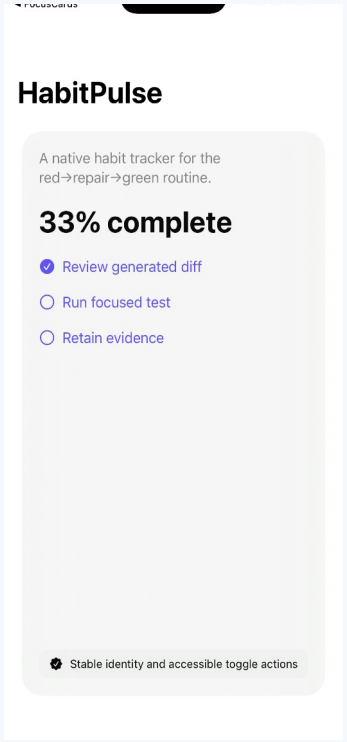
BUILD + TEST + LAUNCH



FocusCards

Reveal/advance state is generated with Observation and verified by tests.

BUILD + TEST + LAUNCH



HabitPulse

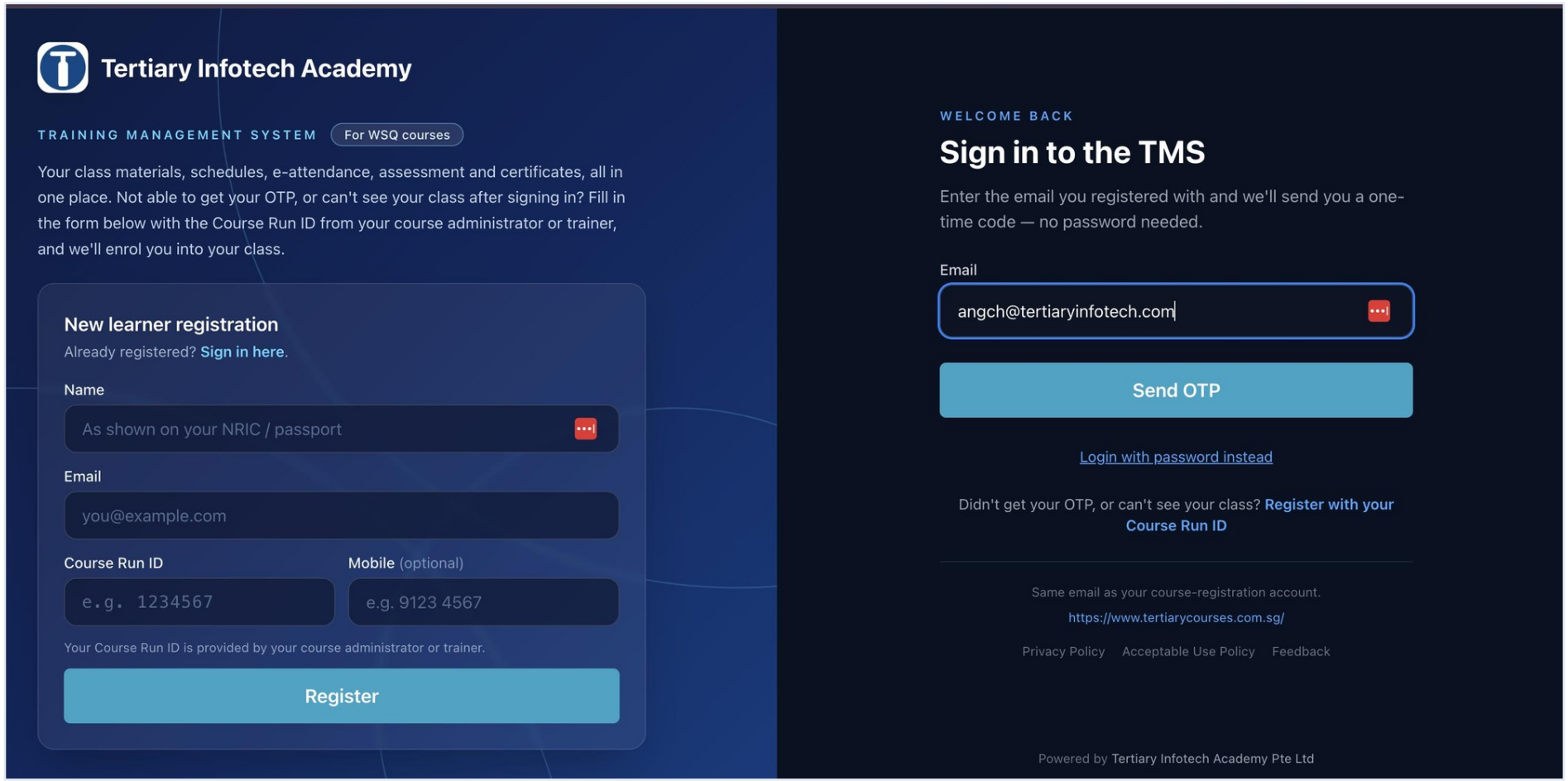
Stable-identity habit actions expose the red → repair → green routine.

BUILD + TEST + LAUNCH

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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LMS WORKFLOW

Download Course Material

1 · Sign in

Open the LMS using your assigned learner account.

LMS

2 · Open course

Confirm TGS-2023037545 and the full course title.

LMS

3 · Download

Use the Learner Guide PDF and Activity folders for detailed procedures.

LMS

4 · Submit

Upload completed assessment documents to the same course record.

LMS

DESIGN JUDGEMENT

Always confirm course identity before downloading or submitting.

https://lms-tms.tertiaryinfotech.com/

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CLOSING ADMINISTRATION

Assessment

1 **WA**

Five open-ended questions covering K1–K5.

ASSESSMENT

2 **PP**

Three practical tasks covering A1–A5.

ASSESSMENT

3 **Time**

One hour for each instrument.

ASSESSMENT

4 **Submit**

Upload candidate papers to the LMS and retain confirmation.

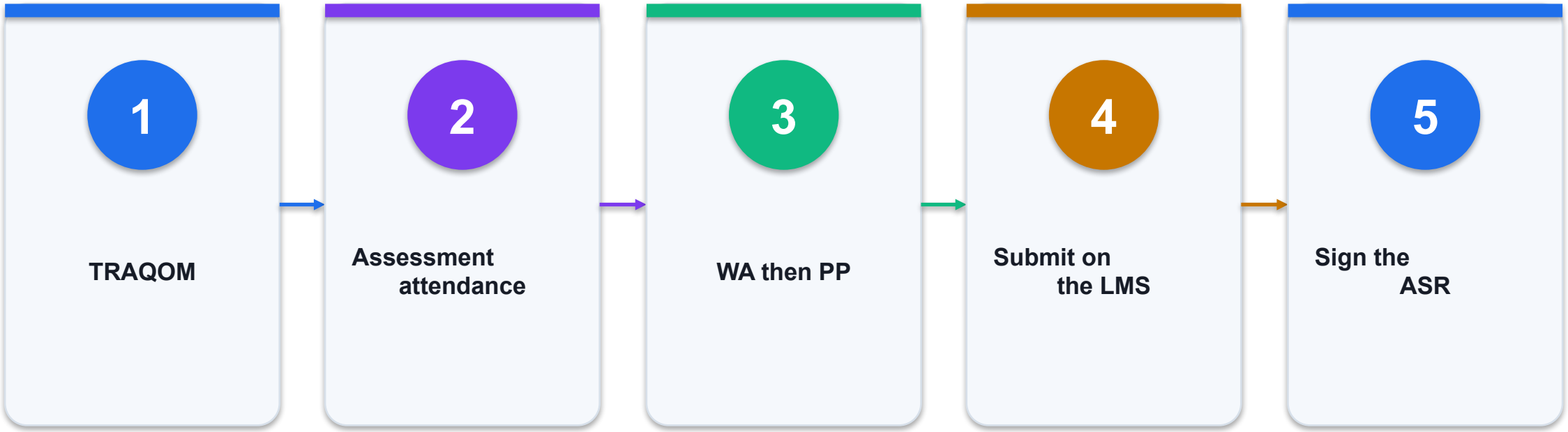
ASSESSMENT

DESIGN JUDGEMENT

Answer keys remain trainer-only.

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Assessment Flow



ASR = Assessment Summary Record · complete the sign-off only after both instruments are submitted.

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MANDATORY · SSG DIGITAL ATTENDANCE

Digital Attendance · Mandatory Close

1 01

Open the SSG TRAQOM attendance request shown by the trainer.

TRAQOM

2 02

Verify course, session and identity details before confirming.

TRAQOM

3 03

Complete attendance at the required start and end checkpoints.

TRAQOM

4 04

Tell the trainer immediately if the status is not confirmed.

TRAQOM

DESIGN JUDGEMENT

TRAQOM records SSG digital attendance. It is not a course survey.

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THANK YOU

Build the boundary.
Prove the behaviour.
Document the decision.

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